# Algebra II

Spring 2026

# 1 Modules

## 1.1 Basics

### 1.1.1 Modules and Homomorphisms

**Definition.** Let $R$ be a ring ($R$ not necessarily commutative but $1 \in R$). Let $(M, +)$ be an abelian group. We say $M$ is a **(left) $R$-module** if there is a map $R \times M \to M, (a, x) \mapsto a \cdot x$ with the following properties:

1. $a \cdot (x + y) = a \cdot x + a \cdot y$
2. $(a + b) \cdot x = a \cdot x + b \cdot x$
3. $(ab) \cdot x = a \cdot (b \cdot x)$
4. $1 \cdot x = x$

for all $a, b \in R, x, y \in M$.

**Example.**

1. $R$ ring $\implies$ $R$ is an $R$-module with respect to multiplication in $R$
2. $R$ field $\implies$ every $R$-vector space is an $R$-module
3. $R$ ring $\implies$ $R^n$ is an $R$-module
4. $R$ commutative ring $\implies$ every ideal in $R$ is an $R$-module
5. $R = \mathbb{Z}, M\, R$-module $\implies$ $M$ is an abelian group. Conversely, every abelian group is a $\mathbb{Z}$-module: if $r \in \mathbb{N}$,
$$r \cdot x := \underbrace{x + \cdots + x}_{r \text{ times}} \text{ and } (-1) \cdot x := -x)$$
6. $R$ commutative ring, $M$ left $R$-module $\implies$ $M$ is a right $R$-module via $x * a := a \cdot x$ (this only works because $R$ is a commutative ring)
7. We saw other modules last semester, which were special because they were over $R$ where $R$ is a $k$-algebra

**Proposition 1.** *Let $R$ be a ring and $M$ be an $R$-module. Say $x, y \in M$ and $a \in R$. Then*

1. $a \cdot (x - y) = a \cdot x - a \cdot y$
2. $(a - b) \cdot x = a \cdot x - b \cdot x$
3. $0_R \cdot x = 0_M$
4. $(-a) \cdot x = -(a \cdot x) = a \cdot (-x)$
5. $(-a) \cdot (-x) = a \cdot x$

**Definition.** Let $R$ be a ring and $M$ be an $R$-module. If $N \subseteq M$ such that

- $(N, +|_N)$ is an abelian group, and
- $N$ is an $R$-module via $\cdot|_{R \times N}$

then $N$ is a **submodule** of $M$. (Notation $N \leq M$)

$M$ is **simple** if 0 and $M$ are the only submodules of $M$

**Note.** If $K$ is a field, then the simple $K$-modules are the $K$-vector spaces of dimension 0 and 1.

**Proposition 2.** $N \leq M \iff x + y \in N, a \cdot x \in N$ *for all* $x, y \in N, a \in R$

**Example.** The $R$-submodules of $R$ (as an $R$-module) are exactly the ideals of $R$.

**Proposition 3.** *Let $R$ be a ring, $M$ an $R$-module, and $N_i \leq N$ for $i \in I$. Then*

1. $$\bigcap_{i \in I} N_i \leq M$$
2. $$\sum_{i \in I} N_i := \left\{ {\sum_{i \in I}}' x_i | x_i \in N_i \right\} \leq M$$

**Definition.** $\sum_{i \in I} N_i$ is called the **sum** of the $N_i$ (in $M$)

It is called the **direct sum** (internal direct sum) if $N_i \cap \sum_{j \in I \setminus \{i\}} N_j = 0$ for all $i \in I$ (Notation: $\oplus_{i \in I} N_i$)

**Definition.** If $R$ is a ring and $M$ is an $R$-module, a **congruence relation** on $M$ is an equivalence relation such that $x_1 \sim x_2$ and $y_1 \sim y_2 \implies x_1 + y_1 \sim x_2 + y_2$ and $a \cdot x_1 \sim a \cdot x_2$ for all $a \in R$.

**Proposition 4.** 1. *If $\sim$ is a congruence relation on $M$, then $\overline{M} := M/\sim$ is an $R$-module via induced structure*

2. *If $\sim$ is a congruence relation on $M$, then $N := \{m \in M | m \sim 0\} \leq M$ with $x \sim y \iff x - y \in N$*

3. *If $N \leq M$, $x \sim y \iff x - y \in N$ defines a congruence relation*

**Definition.** $\overline{M} := M/N = M/\sim$ (where $\sim$ is the congruence relation associated to $N$) is the **quotient of $M$ by $N$ (or by $\sim$)**

**Definition.** If $M, N$ are $R$-modules, $\varphi : M \to N$ is an **$R$-module homomorphism** if for all $x, y \in M, a \in R$:

- $\varphi(x + y) = \varphi(x) + \varphi(y)$
- $\varphi(a \cdot x) = a \cdot \varphi(x)$

$\operatorname{Ker} \varphi := \varphi^{-1}(\{0\})$ is called the **kernel**

$\operatorname{Hom}_R(M, N)$ = set of $R$-module homomorphisms $M \to N$

We say $\varphi \in \operatorname{Hom}_R(M, N)$ is a **monomorphism** if $\varphi$ is injective ( $\iff$ $\operatorname{Ker} \varphi = \{0\}$), an **epimorphism** if $\varphi$ is surjective, and an **isomorphism** if $\varphi$ is bijective.

**Proposition 5.** *1. $\operatorname{Hom}_R(M, N)$ is an abelian group under addition*

*2. $R$ commutative $\implies$ $\operatorname{Hom}_R(M, N)$ is an $R$-module via $(a \cdot \varphi)(x) := a \cdot \varphi(x)$ (note: this does not work if $R$ is not commutative)*

**Theorem 6.** *(Homomorphism Theorem)*
*If $\varphi \in \operatorname{Hom}_R(M, N)$,*

1. *$\varphi(M) \leq N$*
2. *$\operatorname{Ker} \varphi \leq M$*
3. *$\varphi$ induces an isomorphism $\varphi(M) \cong M/\operatorname{Ker} \varphi$*

$$\begin{tikzcd} M \arrow[r, "\varphi"] \arrow[d, "\text{epi}"] & N \\ M/\operatorname{Ker}\varphi \arrow[r, "\overline{\varphi}"] & \varphi(M) \arrow[u, "\text{mono}"'] \end{tikzcd}$$

**Theorem 7.** *(Isomorphism Theorems)*
*The isomorphism theorems are the exact same as for groups (and omitted here)*

**Definition.** Let $R$ be a ring, $M$ an $R$-module, and $S := \{s_i | i \in I\} \subseteq M$. We say

- $S$ is a **generating set** if $M = \{\sum_{i \in I}' a_i s_i | a_i \in R\}$
- $M$ is **finitely generated** if it has a finite generating set

- $M$ is **cyclic** if it is generated by one element
- $S$ is **free** if $\sum_{i\in I}{}' a_i s_i = 0$ forces $a_i = 0$ for all $i \in I$ (in this case representations in terms of the $s_i$ are unique)
- $S$ is a **basis** for $M$ if $S$ is a free generating set
- If $M$ has a basis, $M$ is a **free** $R$-module

**Example.**

1. $R$ field $\implies$ all $R$-modules are free
2. $R = \mathbb{Z}, M = \mathbb{Z}/n\mathbb{Z}$ is cyclic as a module but not free since $n \cdot x = 0 \forall x \in M$

**Proposition 8.** *If $N \leq M$ such that both $N$ and $M/N$ are finitely generated, then $M$ is finitely generated.*

**Theorem 9.** *If $M$ is a finitely generated free $R$-module where $R$ is commutative, then every basis of $M$ has the same number of elements.*

*Proof.* Without loss of generality, assume $R \neq 0$. Then there exists a maximal ideal $P$ in $R$. $R/P$ is a field, call it $K$.

Take a finite basis $S = \{s_1, \dots, s_m\}$ of $M$.

$$N := P \cdot M = \sum_{i=1}^{n} P s_i \leq M \text{ (easy to check)}$$

$V := M/N$ is an $R/P$-module (check), hence a $K$-vector space with generating set $\{\overline{s_1}, \dots, \overline{s_n}\}$, where $\overline{s_i} := s_i + N$. Then it is easy to see that $\overline{s_1}, \dots, \overline{s_n}$ are linearly independent, so $n = \dim_K(V)$ and hence is independent of choice of $S$, by linear algebra. $\square$

**Definition.** Let $M$ be a finitely generated free $R$-module. The number of elements in a basis for $M$ is called the **rank** of $M$ (Notation: $\mathrm{rk}_R(M)$) or **dimension** of $M$ (Notation: $\dim_R(M)$)

**Example.** If an $R$-module $M$ is finitely generated free of rank $n$, then $M \cong R^n$.

### 1.1.2 Direct product and direct sum

**Definition.** Let $R$ be a ring, $\varnothing \neq I$ an index set, and $M_i R$-modules ($i \in I$). An $R$-module $M$ together with $\pi_i \in \mathrm{Hom}_R(M, M_i)$ is the **direct product** of the $M_i$ if for all $N, \varphi_i \in \mathrm{Hom}_R(N, M_i)$ there exists a unique $\varphi \in \mathrm{Hom}_R(N, M)$ such that

$$\begin{tikzcd} M \arrow[dr, "\pi_i"'] & & N \arrow[ll, dashed, "\exists!\varphi"'] \arrow[dl, "\varphi_i"] \\ & M_i & \end{tikzcd}$$

commutes. (Notation: $M = \prod_{i\in I} M_i$)

**Exercise.** Direct product exists and is unique up to isomorphism.

**Proposition 10.** *Let $M$ be an $R$-module, $\varphi_i \in \operatorname{Hom}_R(M,M) = \operatorname{End}_R(M)$ $(i = 1, \dots, n)$ such that $\sum_{i=1}^n \varphi_i = \mathrm{id}_M$ and $\varphi_i \circ \varphi_j = 0$ for $i \neq j$. Then*

$$\varphi_i^2 = \varphi_i \text{ for all } i \text{ and } M \cong \prod_{i=1}^n \varphi_i(M)$$

*Proof.* **Exercise** □

**Definition.** Let $\varnothing \neq I$ be an index set, and for $i \in I$ let $M_i$ be an $R$-module:

An $R$-module $M$ together with $\varepsilon_i \in \operatorname{Hom}_R(M_i, M)$ is the **direct sum** (or **coproduct**) of the $M_i$ if for all $N, \varphi_i \in \operatorname{Hom}_R(M_i, N)$ there exists a unique $\varphi \in \operatorname{Hom}_R(N, M)$ such that

$$\begin{tikzcd} M \arrow[rr, dashed, "\exists!\varphi"] & & N \\ & M_i \arrow[ul, "\varepsilon_i"] \arrow[ur, "\varphi_i"'] & \end{tikzcd}$$

commutes. (Notation: $\bigoplus_{i \in I} M_i$)

**Example.** $R^n = \bigoplus_{i=1}^n R$

**Corollary 12.**

$$\operatorname{Hom}_R(\bigoplus_{i \in I} N_i, M) \cong \prod_{i \in I} \operatorname{Hom}_R(N_i, M)$$

*as abelian groups.*

### 1.1.3 Free and projective modules

**Definition.** Let $I$ be a set. An $R$-module $F_I$ with $i : I \to F_I$ is a **free $R$-module on $I$** if for all $R$-modules $N$ and maps $\varepsilon : I \to N$ we have the following UMP:

$$\begin{tikzcd} I \arrow[r, "i"] \arrow[d, "\varepsilon"] & F_I \arrow[dl, dashed, "\exists!\varphi \in \operatorname{Hom}_R(F_I, N)"] \\ N & \end{tikzcd}$$

**Note.** The elements of $i(I)$ form a basis of $F_I$.

**Corollary 13.** *Every $R$-module is a quotient of a free $R$-module.*

**Note.** $M$ is finitely generated $\iff$ $M$ is a quotient of a finitely generated free $R$-module (i.e. $R^k \twoheadrightarrow M$ some $k$)

**Definition.** $M$ is **finitely related** if there exists $\varphi : F \twoheadrightarrow M$ a homomorphism with $F$ free such that $\operatorname{Ker}\varphi$ is finitely generated.

$M$ is **finitely presented** if $M$ is finitely generated and finitely related.

$M$ is **coherent** if $M$ is finitely generated and all its finitely generated submodules are finitely presented.

**Theorem 14.** *Let $R$ be a ring, $M$ an $R$-module, $F$ a free $R$-module, and $\varphi : M \twoheadrightarrow F$ an epimorphism: then there exists $\psi \in \operatorname{Hom}_R(F, M)$ such that $\varphi \circ \psi = \mathrm{id}_F$ and $M = \operatorname{Ker}\varphi \oplus \psi(F)$. In particular, every short exact sequence*

$$0 \to \operatorname{Ker}\varphi \to M \xrightarrow{\varphi} F \to 0$$

*with $F$ free splits (this second part is an **exercise**).*

**Example.**

1. Not free: $\mathbb{Z}/m\mathbb{Z}$ as a $\mathbb{Z}$-module
2. $R = \mathbb{C}[x_1, \dots]$ is finitely generated as an $R$-module (free of rank 1) but the submodule $N = (x_1, x_2, \dots)$ is not finitely generated as an $R$-module; this is the set of polynomials with constant term 0.

   To prove $N$ is not finitely generated as an $R$-module, we will show each finitely generated ideal $Rf_1 + Rf_2 + \cdots + Rf_k$ in $R$ doesn't contain $x_i$ for all large $i$, so this ideal is not $N$.

   Since each of the polynomials $f_1, \dots, f_k$ involves only a finite number of variables, there's a large $n$ such that all $x_i$ appearing in one of $f_1, \dots, f_k$ have $i < n$. The substitution homomorphism $R \to R$ that sends $x_i$ to 0 for $i < n$ and $x_i$ to 1 for $i \geq n$ sends $f_1, \dots, f_k$ to 0 and therefore it sends every $R$-linear combination of $f_1, \dots, f_k$ to 0. Since this homomorphism sends $x_i$ to 1 for $i \geq n$, such $x_i$ do not lie in $Rf_1 + \cdots + Rfk$.
3. $R = \mathbb{C}[x_1, \dots]/(x_{n+1}x_1 - x_n x_2 | n \geq 1)$

   $\overline{x_i} :=$ image of $x_i$ in $R$

   $I := (\overline{x_1}, \overline{x_2}) \trianglelefteq R$ is finitely generated but not finitely related

   e.g. $\varphi : R^2 = Re_1 \oplus Re_2 \twoheadrightarrow I, e_i \mapsto \overline{x_i}$ does not have finitely generated kernel (note: to fully prove we would have to show why you cannot write $I$ some other way)

   $\operatorname{Ker}\varphi = (\overline{x_{n+1}}e_1 - \overline{x_n}e_2 | n \geq 1)$

**Definition.** An $R$-module $P$ is **projective** if for every epimorphism $\varphi : M \twoheadrightarrow P$, $M \cong \operatorname{Ker}\varphi \oplus U$ for some $U \leq M$. (In this case, we say $\operatorname{Ker}\varphi$ is a direct summand)

Note: By Theorem 14, free $\implies$ projective.

**Proposition 15.** *Let $R$ be a ring and $P$ be an $R$-module. Then the following are equivalent:*

1. *$P$ is projective*
2. *$\exists$ an $R$-module $M$ such that $M \oplus P$ is free*
3. *(Lifting property) For all $R$-modules $M, M'$: for all $\varphi \in \operatorname{Hom}_R(P, M')$, epimorphisms $\pi \in \operatorname{Hom}_R(M, M'), \exists \psi \in \operatorname{Hom}_R(P, M)$ so that the following diagram commutes:*

$$\begin{tikzcd} & P \arrow[dl, dashed, "\exists\psi"'] \arrow[dr, "\varphi"] & \\ M \arrow[rr, "\pi"] & & M' \end{tikzcd}$$

**Example.** $\mathbb{Z}/2\mathbb{Z}$ is not free as a $\mathbb{Z}/6\mathbb{Z}$-module, because there's no element $x \in \mathbb{Z}/2\mathbb{Z}$ that can generate the module freely without annihilation by a nonzero ring element (e.g. if $x$ is nonzero in $\mathbb{Z}/2\mathbb{Z}$ then $2 \in \mathbb{Z}/6\mathbb{Z}$ is nonzero but annihilates $x$).

But we claim that $\mathbb{Z}/2\mathbb{Z}$ is a projective $\mathbb{Z}/6\mathbb{Z}$-module.
To prove this, we see that $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z} = \mathbb{Z}/6\mathbb{Z}$, which is indeed free as a $\mathbb{Z}/6\mathbb{Z}$-module.

## 1.2 Modules over PIDs

### 1.2.1 Torsion and torsion-free modules

**Definition.** Let $R$ be a ring and $M$ an $R$-module.
$x \in M$ is a **torsion element** if $a \cdot x = 0$ for some $a \in R \setminus \{0\}$.

$\operatorname{Ann}(x) := \operatorname{Ann}_R(x) := \{b \in R | b \cdot x = 0\} \trianglelefteq R$ is called the **annihilator of** $x$

$\operatorname{Ann}(M) := \operatorname{Ann}_R(M) := \cap_{x \in M} \operatorname{Ann}_R(x)$

$M_{\text{tor}} := \{x \in M | \operatorname{Ann}_R(x) \neq 0\}$ is called the **torsion** of $M$

$M$ is a **torsion module** if $M_{\text{tor}} = M$

$M$ is **torsion-free** if $M_{\text{tor}} = 0$

**Example.**

1. $R = \mathbb{Z}, M = \mathbb{Z}/m\mathbb{Z}$ is torsion
2. $R = \mathbb{Z}/m\mathbb{Z}, M = \mathbb{Z}/m\mathbb{Z} \implies$
   $M_{\text{tor}} = \{\bar{a} \in \mathbb{Z}/m\mathbb{Z} | \gcd(a, m) \neq 1\}$, so $M$ is torsion-free if $m$ is prime
   Note: In domains, free $\implies$ torsion-free (not true in general e.g. for non-domains)

**Proposition 1.** *Let $R$ be an integral domain, $M$ an $R$-module. Then:*

1. $M_{tor} \leq M$

2. $\overline{M} := M/M_{tor}$ *is torsion-free*

*Proof.* 1. Let $x, y \in M_{\text{tor}}$. Then for some $a, b \in R \setminus \{0\}$, $a \cdot x = 0, b \cdot y = 0$. Hence $(ab) \cdot (xy) = 0$ and $ab \neq 0$ since $R$ is a domain, so $xy \in M_{\text{tor}}$. Also, easy to see that $RM_{\text{tor}} \subseteq M_{\text{tor}}$.

2. Let $\overline{x} \in \overline{M}$ such that $a \cdot \overline{x} = 0$ for some $a \in R \setminus \{0\}$. Then $ax \in M_{\text{tor}}$ for some representative $x$ of $\overline{x} \implies b \cdot (ax) = 0$ for some $b \in R \setminus \{0\}$. Now $ab \neq 0$ as before, so this means $x \in M_{\text{tor}}$ hence $\overline{x} = 0$.

□

**Theorem 2.** *Let $R$ be a PID. Then:*

1. *$M$ finitely generated $R$-module and $N \leq M \implies N$ is a finitely generated $R$-module.*

2. *$M$ finitely generated free $R$-module and $N \leq M \implies N$ is a finitely generated free $R$-module.*

*Proof.* Let $s_1, \ldots, s_m$ be generators for $M$. Induct on $m$.

$m = 1$:

1. If $M = R \cdot s_1$ and $N \leq M$, let $A := \{a \in R | as_1 \in N\} \trianglelefteq R$. Since $R$ is a PID, $A = (d)$ for some $d \in R$. So $N = R(ds_1)$ i.e. $ds_1$ generates $N$, meaning $N$ is finitely generated as an $R$-module

2. Notice that if $M$ is free, then $s_1 \notin M_{\text{tor}} \implies M_{\text{tor}} = 0$ (since $R$ is a domain) $\implies N_{\text{tor}} = 0 \implies ds_1 \notin N_{\text{tor}} \implies N$ is free.

Inductive step ($m - 1 \to m$):

1. Let
$$A_1 := \{a_1 \in R | \exists a_2, \ldots, a_m \in R \text{ s.t.} \sum_{i=1}^{m} a_i s_i \in N\}$$
$A_1$ is an ideal in $R$. Since $R$ is a PID, say $A_1 = (d_1)$.

   Let $t_1 \in N$ such that $t_1 - d_1 s_1 \in M_1 := \sum_{i=2}^{m} Rs_i$.

   **Case 1:** If $d_1 = 0, N \leq M_1 \leq M$. So by induction, we have $N$ finitely generated.
   **Case 2:** If $d_1 \neq 0$, let $y \in N$. Write $y = \sum_{i=1}^{m} b_i s_i$ and $b_1 = c_1 d_1$ some $c_1 \in R$.
   Let $u := t_1 - d_1 s_1$. Then $y - c_1 t_1 = y - c_1 u - c_1 d_1 s_1 = y - c_1 u - b_1 s_1 \in M_1$. Thus $y - c_1 t_1 \in M_1 \cap N =: N_1 \leq M_1$. Now $N_1$ is generated by finitely many elements $t_2, \ldots, t_r$.

2. We still need to show that if $s_1, \dots, s_m$ is a basis for $M$, then $\{t_1, \dots, t_r\}$ can be chosen freely.

   By induction, we can assume $\{t_2, \dots, t_r\}$ are free. Now if

$$b_1 t_1 + \sum_{i=2}^{m} b_i t_i = 0, \quad b_1 \neq 0,$$

   then $b_1 t_1 \in N_1 \leq M_1$ and $b_1 t_1 - b_1 d_1 s_1 \in M_1 \implies b_1 d_1 s_1 \in M_1 \rightarrow\leftarrow$. Hence $M$ is free on $s_1, \dots, s_m$.

$\square$

**Note.** 1. If $R$ is a PID and $M$ is a free $R$-module, then (we will see later that) every submodule of $M$ is free.

2. In the above proof, we used the fact that for cyclic modules over PIDS, $M$ torsion free $\iff$ $M$ free.

**Theorem 3.** *Let $R$ be a PID and $M$ a finitely generated $R$-module. Then $M$ is free $\iff$ $M$ is torsion-free.*

*Proof.* $(\Rightarrow)$ : is because $R$ is a domain.
$(\Leftarrow)$ : Let $\{s_1, \dots, s_m\}$ be a generating set of $M$. Without loss of generality, assume $\{s_1, \dots, s_n\}$ is a maximal free subset $(n \leq m)$.

If $n = m$, we're done.
If $n > m$, $\{s_1, \dots, s_n, s_m\}$ is not free. We get

$$a_m s_m = \sum_{i=1}^{n} b_i s_i, \quad b_i \in R, a_m \neq 0$$

i.e. $a_m \cdot M \leq \sum_{i=1}^{m-1} R s_{i-1}$.
If $m - 1 > n$, we get $a_{m-1} \neq 0$ such that $a_{m-1} a_m M \leq \sum_{i=1}^{m-2} R s_i$. Continuing, we get $a \in R \setminus \{0\}$ so that $aM \leq \sum_{i=1}^{n} R s_i =: F$ where $F$ is free.

Consider $\varphi : M \to M, x \mapsto a \cdot x$. $M$ torsion-free $\implies$ $\varphi$ is injective.
$\varphi \in \mathrm{Hom}_R(M, M)$ and $\varphi(M) \leq F \leq M$ (so $\varphi(M)$ is free by Theorem 2).
Then $\varphi$ injective $\implies \varphi(M) \cong M \implies M$ is free. $\square$

**Note.** $(\Leftarrow)$ is false for modules that are not finitely generated. For example, $\mathbb{Q}$ is a torsion-free $\mathbb{Z}$-module but is not free because any 2 elements are linearly dependent, but $\mathbb{Q}$ also cannot be rank 1 because it has a divisibility property that e.g. $\mathbb{Z}$ does not.

**Corollary 4.** *Let $R$ be a PID and $M$ a finitely generated $R$-module. Then $M = M_{tor} \oplus F$ where $F \cong M/M_{tor}$ is free.*

### 1.2.2 Primary Modules

Let $R$ be a PID. Let $\mathbb{P}(R)$ be a system of representatives of prime elements in $R$.

**Definition.** Let $M$ be an $R$-module and $p \in \mathbb{P}(R)$.

$x \in M$ is called $p$-**primary** if $p^n x = 0$ for some $n \in \mathbb{N}$

The $p$-**primary part / $p$-torsion of** $M$ is $M_{(p)} := \{x \in M | x \text{ is } p\text{-primary}\}$

**Proposition 5.** *Let $R$ be a PID, $M$ be an $R$-module, $p \in \mathbb{P}(R)$. Then $M_{(p)} \leq M$.*

**Proposition 6.** *Let $R$ be a PID, $M$ be an $R$-module, $M = M_{tor}$. Then*

$$M = \bigoplus_{p \in \mathbb{P}(R)} M_{(p)}$$

*Proof.* Let $x \in M$. $\operatorname{Ann}(x) \trianglelefteq R$ and since $R$ is a PID, $\operatorname{Ann}(x) = (a)$ where $a \in R \setminus \{0\}$ since $M = M_{\text{tor}}$. Write

$$a \simeq \prod_{i=1}^{r} p_i^{e_i}$$

with $p_i$ distinct elements of $\mathbb{P}(R)$.
Write $a = a_i \cdot p_i^{e_i}$ where $a_i \in R$ (for each $i = 1, \ldots, r$). $\gcd(a_1, \ldots, a_r) = 1 \implies$ there exist $b_i \in R$ such that

$$1 = \sum_{i=1}^{r} a_i b_i$$

Let $x_i := a_i b_i x$. Then $p_i^{e_i} x_i = p_i^{e_i} a_i b_i = ax = 0$, so $x_i \in M_{(p_i)}$. Moreover,

$$\sum_{i=1}^{r} x_i = \left( \sum_{i=1}^{r} a_i b_i \right) x = x \implies M = \sum M_{(p)}.$$

Remains to show uniqueness (that this sum is indeed a direct sum):
Let $p_1, \ldots, p_n \in \mathbb{P}(R)$ be distinct. Suppose $\sum_i x_i = 0$ with $x_i \in M_{(p_i)}$. WTS: $x_i = 0$ for all $i$.

Let $a := \prod_{i=1}^{r} p_i^{e_i} \implies a = a_i p_i^{e_i}$ as before. Then there exist $b_i, c_i \in R$ so that $a_i b_i + c_i p_i^{e_i} = 1$. Hence $x_i = 1 \cdot x_i = (a_i b_i + c_i p_i^{e_i}) x_i = a_i b_i x_i$.

Fix $j$:

$$0 = \sum x_i = a_j b_j \sum x_i = \left( \sum_{i \neq j} a_j b_j x_i \right) + a_j b_j x_j = 0 + x_j = x_j \implies x_1 = \cdots = x_n = 0.$$

□

**Corollary 7.** *Let $M$ be finitely generated module over a PID $R$ and suppose $M = M_{tor}$. Then:*

*1.* $\mathrm{Ann}(M) = \cap_{x \in M} \mathrm{Ann}(x) = (a)$ *some* $a \neq 0$, *and* $\exists y \in M$ *so that* $\mathrm{Ann}(M) = \mathrm{Ann}(y)$.

*2.*

$$a \simeq \prod_{i=1}^{r} a_i p_i^{e_i}, \quad \textit{(the factorization into powers of distinct primes in } \mathbb{P}(R)\textit{)}$$
$$\implies M \simeq \bigoplus_{i=1}^{r} M_{(p_i)}, \quad \textit{where } \mathrm{Ann}(M_{(p_i)}) = (p_i^{e_i})$$

*Proof.* 1. Let $s_1, \ldots, s_r$ generate $M = M_{\mathrm{tor}}$. Easy to check:

$$\mathrm{Ann}(M) = \bigcap_{i=1}^{r} \mathrm{Ann}(s_i)$$

Write

$$\mathrm{Ann}(s_i) = (a_i), \quad a_i \in R \setminus \{0\}$$

Then $\mathrm{Ann}(M) = \bigcap_{i=1}^{r} (a_i) =: a$. Notice that $a_i | a$ for all $i = 1, \ldots, r$ and $a | a_1 \cdots a_r \implies a \neq 0$.

2. $a \simeq \prod_{i=1}^{r} p_i^{e_i} \implies M = \bigoplus_{i=1}^{r} M_{(p_i)}$ by the proof of Theorem 6. We know that $(a) = \mathrm{Ann}(M) = \bigcap \mathrm{Ann}(M_{(p_i)})$.
Fix $i$:
Take $y \in M_{(p_i)}$. We know $\mathrm{Ann}(y) = (p_i^{k_i})$ for some $k_i$. $ay = 0 \implies k_i \leq e_i$.

Suppose that for all $y \in M_{(p_i)}$, $k_i < e_i$. Then let $\tilde{a} := \prod_{i \neq j} p_j^{e_j} p_i^{e_i - 1}$. We still get $\tilde{a}M = 0 \rightarrow\leftarrow$ (contradiction since $\mathrm{Ann}(M) = (a)$, and not $\tilde{a}$).

Hence, for all $i \exists y_i \in M_{(p_i)}$ such that $\mathrm{Ann}(y_i) = (p_i^{e_i})$.

Now we just have to show the rest of (1):

Let $y = \sum_{i=1}^{r} y_i$. If $b \in \mathrm{Ann}(y), b \in \mathrm{Ann}(y_i) = (p_i^{e_i})$ for all $i$ because our sum is direct. Hence $p_i^{e_i} | b \implies a | b \implies \mathrm{Ann}(y) \subseteq (a) = \mathrm{Ann}(M) \implies \mathrm{Ann}(y) = \mathrm{Ann}(M)$. □

**Proposition 8.** *Let $R$ be a PID, $p \in \mathbb{P}(R)$, and $M$ a finitely generated $R$-module. Assume $p^k M = 0$ for some $k \in \mathbb{N}$ and let $x \in M$ such that $\mathrm{Ann}(x) = \mathrm{Ann}(M) = (p^k)$. Then $R \cdot x$ is a direct summand of $M$. i.e. $M = Rx \oplus N$ for some $N \leq M$.*

*Proof.* Let $\mathcal{U}$ be the set of submodules $U \subseteq M$ such that $U \cap Rx = 0$. Partially order by inclusion. $0 \in \mathcal{U}$ so $\mathcal{U} \neq \varnothing$. One can check: $\mathcal{U}$ satisfies the conditions for Zorn's Lemma. So $\mathcal{U}$ has a maximal element $N$.

Let $M' = Rx \oplus N \leq M$. WTS: $M' = M$.

Suppose not. Let $y \in M \setminus M'$. Note $y \notin N \implies N \subsetneq Ry + N$. Let $A := Ry \cap (Rx \oplus N)$. By maximality of $N$, $A \neq 0$.

So let $z \in A \setminus \{0\}$. Then $z = cx = ay + n$ for some $a, c \in R$, $n \in N$. Note: $cx \neq 0$ and $a \neq 0$ (since $a \notin N$). Also $a \notin R^\times$ (otherwise $y = a^{-1}(cx - n) \in M'$, but $y \notin M'$). So in summary:
For all $y \in M \setminus M', \exists a \in R \setminus \{0\}, a \notin R^\times, c \in R, cx \neq 0, n \in N$ such that $ay = cx + n$ (in particular, $ay \in M'$).

Decompose: $a \simeq p_1 \cdots p_r, p_i \in \mathbb{P}(R)$ (not necessarily distinct). Consider: $y \notin M', p_r y, p_{r-1}p_r y, \ldots ay \in M'$. We may assume $y \notin M', qy \in M'$ for some $q \in \mathbb{P}(R)$.

**We claim:** $qy \in N + qRx$.
Know: $qy \in N + Rx$.
**Case 1:** If $p \neq q$, then (since $q$ does not annihilate things) $rp^k + sq = 1$ some $r, s \in R \implies Rx = (p^k R + qR)x = qRx$ since $p^k x = 0 \implies qy \in M' = N + Rx = N + qRx$.

**Case 2:** If $p = q$, then $py \in M'$. So $py = n + dx$ some $n \in \mathbb{N}, d \in R$. We also know that $p^k M = 0$, so

$$0 = p^{k-1}py = p^{k-1}(n + dx) = p^{k-1}n + p^{k-1}dx \implies -p^{k-1}n = p^{k-1}dx$$

(noting that $-p^{k-1}n \in N$ and $p^{k-1}dx \in Rx$).
Hence, $p^{k-1}d \in \operatorname{Ann}(x) = (p^k) \implies p|d \implies py = n + dx \in N + pRx$.
This concludes our proof of the claim.

Now we write $qy = n + qz, n \in N, z \in Rx$. Then $q(y - z) = n \in N$, so $y - z \notin M'$. Then by our "summary" before, $\exists b \in R \setminus \{0\}, b \notin R^\times, x' \in Rx, x' \neq 0, n' \in N$ such that $b(y - z) = n' + x'$.

Suppose $q|b$. Then $b(y - z)$ is a multiple of $q(y - z) \in N$. $\rightarrow\leftarrow$
So we must have $q \nmid b$. Then we can write $r'q + s'b = 1$ for some $r', s' \in R$.
So $s'b(y - z) = s'n' + s'x'$ (because $b(y - z) = n' + x'$), and $r'q(y - z) = r'n$ (because $q(y - z) = n$).

Then $y - z = s'n' + r'n' + s'x' \in N \oplus Rx - M'$. $\rightarrow\leftarrow$ So $M' = M$. $\square$

**Theorem 9.** *Let $R$ be a PID, $p \in \mathbb{P}(R)$, and $M = M_{(p)}$ a finitely generated $R$-module. Then there exist $x_1, \ldots, x_s \in M$ such that*

$$M = \bigoplus_{i=1}^{s} Rx_i, \quad \text{and } \operatorname{Ann}(x_i) = (p_i^{e_i})$$

*for some $1 \leq e_1 \leq \cdots \leq e_s$. These $e_i$'s are uniquely determined.*

*Proof.* Say $M = \langle x_1, \ldots, x_n \rangle, \operatorname{Ann}(x_i) = (p_i^{e_i})$. Without loss of generality, assume $e_n = \max(e_1, \ldots, e_n)$.

By Proposition 8, $M = N \oplus Rx_n$ where $N \cong M/(Rx_n)$ is generated by $x_1 + Rx_n, \ldots, x_{n-1} +$

$Rx_n$. Then the first claim follows by induction.

Remains to show uniqueness of the $e_i$'s:

Let $d_i = \dim_K(M/p^{i-1}M)$ where $K := R/(p)$ is a field. Then $d_i = \#\{e_j | e_j \geq i\}$ and $d_i - d_{i-1} = \#\{e_j | e_j = i\}$ is uniquely determined. $\square$

**Corollary 10.**

$$M \cong \bigoplus_{i=1}^{s} R/(p_i^{e_i})$$

### 1.2.3 The Structure Theorem

**Theorem 11.** *Let $M$ be a finitely generated module over a PID. Then*

$$M \cong F \oplus \bigoplus_{i=1}^{r} \bigoplus_{j=1}^{t_i} Rx_{ij}$$

*where*

- $F \cong M/M_{tor}$ *is free*
- $\operatorname{Ann}(x_{ij}) = (p_i^{e_{ij}})$ *for $p_i \in \mathbb{P}(R)$ where $e_{i1} \leq \cdots \leq e_{it_i}$ for uniquely determined* ***structure constants*** $e_{ij}$.

**Definition.** Let $M$ be a finitely generated module over a PID. Then $\dim(M/M_{\text{tor}}) := \dim(M) := \operatorname{rk}(M)$ is the **dimension** or **rank** of $M$

**Example.** $R = \mathbb{Z}$, $M =$ finitely generated abelian group ($\mathbb{Z}$-module)

Theorem 11 gives the structure theorem for finitely generated abelian groups.

### 1.2.4 Applications to Linear Algebra

Let $K$ be a field, $V$ a finite-dimensional $K$-vector space, and $\varphi \in End(V)$ be fixed.

**Proposition 13.** *Let $R := K[x]$.*

1. *$V$ is an $R$-module via $R \times V \to V, (h, v) \mapsto h \cdot v := h(\varphi)(v)$. With this structure, $V$ is a torsion $R$-module.*
2. $\operatorname{Ann}(V) = (g_\varphi(x))$, *where $g_\varphi$ is the the minimal polynomial of $\varphi$ on $V$*

**Example.** $V \cong \mathbb{R}^2$, $\varphi$ given by $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ with respect to the standard basis,

$$h = x^2 + 1 \in \mathbb{R}[x] \implies h(\varphi) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^2 + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 0 & 2 \end{pmatrix}.$$

*Proof.* (Of Proposition 13)

1. Easy to check this defines an $R$-module structure (e.g. $1 \cdot bv = 1(\varphi)(v) - \mathrm{id}_V(v) = v$) $\dim V < \infty \implies g_\varphi$ exists and $g_\varphi(v) := g_\varphi(\varphi)(v) = 0$ for all $v \in V$ (by Cayley-Hamilton), so $V$ is a torsion module.

2. By definition of minimal polynomial.

$\square$

**Corollary 14.** *Let*

$$g_\varphi(x) = \prod_{i=1}^{r} p_i(x)^{e_i}$$

*be the factorization into powers of distinct monic prime polynomials. Then*

*1.*
$$V = \bigoplus_{i=1}^{r} \mathrm{Ker}(p_i(\varphi)^{e_i}) \quad \textit{and } \mathrm{Ann}(p_i(\varphi)^{e_i}) := \mathrm{Ann}(W_i) = (p_i^{e_i})$$

*2.*
$$W_i = \bigoplus_{j=1}^{t_i} K[x] \cdot s_{ij} := \bigoplus_{j=1}^{t_i} W_{ij}, \quad \mathrm{Ann}(s_{ij}) = (p_i^{e_{ij}})$$

**Note.**
1. $p_i^{e_i}$ is the minimal polynomial of $\varphi|_{W_i}$
2. $p_i^{e_{ij}}$ is the minimal polynomial of $\varphi|_{W_{ij}}$

**Definition.** Let $V$ be a $K$-vector space, $\varphi \in \mathrm{End}(V), W \leq V$. Then

- $W$ is $\varphi$**-stable** if $\varphi(W) \leq W$.
- $W$ is $\varphi$**- indecomposable** if it is $\varphi$-stable and there is no nontrivial decomposition into a direct sum of $\varphi$-stable subspaces.
- $W$ is $\varphi$**-primary** if it is $\varphi$-stable and $\mathrm{Ann}(W) = (p(x)^e)$ some $e \in \mathbb{N}, p(x) \in K[x]$ prime
- $W$ is $\varphi$**-cyclic** if $W = K[x] \cdot s$ for some $s \in W$.

**Note.** $\varphi$-cyclic $\Rightarrow$ $\varphi$-stable

**Theorem 15.** *$W$ is $\varphi$-indecomposable $\iff$ $W$ is $\varphi$-primary and $\varphi$-cyclic*

*Proof.* ($\Rightarrow$): $g_{(\varphi|_W)}(x) = \prod_{i=1}^r p_i^{e_i} \implies$ (by Corollary 14 part 1) $W = \bigoplus_{i=1}^r W_i$. So $r = 1 \implies W$ is $\varphi$-primary. By Corollary 14 part 2, $W$ is $\varphi$-cyclic.
($\Leftarrow$): $W = K[x] \cdot s$ is $\varphi$-cyclic. $\varphi$-primary $\implies$ $\mathrm{Ann}(W) = \mathrm{Ann}(S) = (p^e)$ for some $p \in K[x], e \in \mathbb{N}$.

Now suppose $W = \bigoplus_{i=1}^r W_i$ is a $\varphi$-stable decomposition, where $W_i \neq 0$ for all $i = 1, \ldots, r$ and each $W_i$ is indecomposable. Then suppose (for contradiction) that $r \geq 2$. By "$\Rightarrow$", we know that the $W_i$ is $\varphi$-primary and $\varphi$-cyclic for all $i$. That is, $\mathrm{Ann}(W_i) = (p^{e_i})$ since $\mathrm{Ann}(W_i) \supseteq \mathrm{Ann}(W)$. Now

$\dim W = e \cdot \deg p$ and $\dim W_i = e_i \cdot \deg p$. So $e_i < e$ for all $i = 1, \ldots, r$ (since $r > 1$). But then

$$\mathrm{Ann}(W) = \bigcap_{i=1}^r \mathrm{Ann}(W_i) = \bigcap_{i=1}^r (p^{e_i}) = (p^t)$$

some $t < e$. $\rightarrow\leftarrow$ $\square$

**Example.** Let $p = x^2 + 1 \in \mathbb{R}[x]$ be prime. Let $V$ be a $\mathbb{R}$-vector space and $\varphi \in \mathrm{End}(V)$. Say $W := \mathbb{R}[x] \cdot s$ is $\varphi$-cyclic. Assume $g_{(\varphi|_W)} = p(x)^2$. Then $(1, x, x^2, x^3)$ is a basis of $\mathbb{R}[x]/(p(x)^2)$. Then $S := (s, \varphi(s), \varphi^2(s), \varphi^3(s))$ is a basis for $W$ as a $\mathbb{R}$-vector space.

Goal: Find a new basis $T$ for $W$ such that $D_T(\varphi)$ is nice in some reasonable way.

$T = (s, \varphi(s), \ldots ?)$. Observe:

$$\varphi^2(s) = p(\varphi)(s) - s \text{ since } p(\varphi)(s) = (\varphi^2 + 1)(s) = \varphi^2(s) + s$$

Try: $T = (t_1, t_2, t_3, t_4) = (s, \varphi(s), p(\varphi)(s), \varphi(p(\varphi)(s)))$. Calculate:

$$\varphi(\varphi(p(\varphi))(s)) = \varphi^2(p(\varphi)(s)) = (p(\varphi) - \mathrm{id})p(\varphi)(s) = p^2(\varphi)(s) - (\varphi)(s) = -p(\varphi)(s)$$

Hence $D_T(\varphi|_W) =$

$$\begin{pmatrix} \varphi(t_1) & \varphi(t_2) & \varphi(t_3) & \varphi(t_4) \end{pmatrix} = \left(\begin{array}{cc|cc} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{array}\right) = \left(\begin{array}{c|c} C & 0 \\ \hline \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} & C \end{array}\right), \quad \text{where } C = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

We have two blocks because the minimal polynomial had degree 2. And the 1 in the bottom left section is because it (what?) is indecomposable.

**Proposition 16.** *Let $V$ be a finite-dimensional $K$-vector space, $\varphi \in \mathrm{End}(V)$. Let $W := K[x] \cdot s \leq V$ be $\varphi$-indecomposable with $g_{(\varphi|_W)}(x) = p(x)^e$ where $p(x) = \sum_{i=0}^d a_i x^i$ is monic. Then*

$$T = (s, \varphi(s), \ldots, \varphi^{d-1}(s), p(\varphi)(s), \varphi(p(\varphi)(s)), \ldots, \varphi^{d-1}(p(\varphi(s)), p^2(\varphi)(s), \varphi(p^2(\varphi)(s)), \ldots)$$

*is a basis of $W$. And*

$$D_T(\varphi_W) = \underbrace{\left(\begin{array}{c|c|c|c} C & & & \\ \hline E & C & & \\ \hline & E & C & \\ \hline & & \ddots & \ddots \end{array}\right)}_{e \cdot d} \qquad \textit{where } E := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \qquad \textit{and} \quad C = \underbrace{\begin{pmatrix} 0 & 0 & \cdots & -a_0 \\ 1 & 0 & \cdots & -a_1 \\ 0 & 1 & \ddots & \vdots \\ 0 & 0 & \cdots & -a_{d-1} \end{pmatrix}}_{d}$$

*$C$ is called the* ***companion matrix*** *of $\varphi$ on $W$.*

**Note.** If $p(x) = x - a$ for $a \in K$, $C = (a)$. Proposition 16 generalized Jordan Normal Form (for cyclic spaces).

**Theorem 17** (Jordan Normal Form)**.** *If $V$ is a finite-dimensional $K$-vector space, $\varphi \in \operatorname{End}(V), V = \bigoplus_{i,j} W_{ij}$ where $W_{ij} = K[x]s_{ij}$ is cyclic, $\operatorname{Ann}(s_{ij}) = (p_i^{e_{ij}}), d_i = \deg(p_i)$. Then $V$ has a basis $T = \sqcup T_{ij}$ (where $T_{ij}$ = basis of $W_{ij}$) of the form given in Proposition 16 and*

$$D_T(\varphi) = \begin{pmatrix} \square & & & \\ & \square & & \\ & & \ddots & \\ & & & \square \end{pmatrix} \text{ where each block corresponds to: } D_{T_{ij}}(\varphi|_{W_{ij}}) = \left(\begin{array}{c|c|c|c} C_i & & & \\ \hline E & C_i & & \\ \hline & E & \ddots & \\ \hline & & E & C_i \end{array}\right)$$

$$\text{such that } E := \begin{pmatrix} 0 & \cdots & 0 & 1 \\ 0 & \cdots & 0 & 0 \\ \vdots & & \vdots & \vdots \\ 0 & \cdots & 0 & 0 \end{pmatrix} \quad \text{and where } C_i \text{ is the companion matrix of } \varphi|_{W_{ij}}.$$

*Moreover,*

$$g_\varphi = \prod_{i=1}^{r} p_i^{k_i} \text{ where } k_i := \max_j \{e_{ij}\}.$$

**Corollary 18.** *Let $\varphi \in \operatorname{End}(V)$. Let $\chi_\varphi$ be the characteristic polynomial and $g_\varphi$ the minimal polynomial. Then if $n = \dim V < \infty$, $\chi_\varphi \mid g_\varphi^n$. In particular, the two have the same prime factors.*

**Example.** Let

$$A = \begin{pmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 2 & -1 & 1 \\ 2 & 0 & -1 & 0 \end{pmatrix} \in \operatorname{Mat}_4(\mathbb{R}).$$

Step 1: Compute $\chi_A(x) = (x^2+x+1)^2$. Then $p(x) = x^2+x+1$. Check: $p$ is irreducible in $\mathbb{R}[x]$.

Step 2: Find the minimal polynomial. $p(A) \neq 0$ so the minimal polynomial must equal the characteristic polynomial, hence $V$ is indecomposable, so cyclic. This mean we put a 1 in the top right corner of the bottom left block, to get:

$$D_T(\varphi) = \left(\begin{array}{cc|cc} 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ \hline 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{array}\right)$$

To find $T$:
$\ker(p(\varphi)) - \langle \beta_1 \rangle, \beta_1 = (e_1 + 2e_3, e_2)$

$\ker(p^2(\varphi)) = V = \beta_1 \sqcup \beta_2$
$\beta_2 = (e_1, e_4)$

Letting $s = e_1$ $T = (s, \varphi(s), p(\varphi)(s), \varphi(p(\varphi))(s))$

## 1.3 Homological Algebra

For this entire section, $R$ will be a commutative ring with 1.

### 1.3.1 Exact Sequences

**Definition.** If $M_i$ is an $R$-module for $i \in I \subseteq \mathbb{Z}$, $\varphi_i : M_i \to M_{i+1}$ are $R$-module homomorphisms, then:

$$\cdots \xrightarrow{\varphi_{i-2}} M_{i-1} \xrightarrow{\varphi_{i-1}} M_i \xrightarrow{\varphi_i} \cdots$$

is **exact at** $M_i$ if $\mathrm{Im}(\varphi_{i-1}) = \mathrm{Ker}(\varphi_i)$.

If the sequence is exact at $M_i$ for all $i$, it is an **exact sequence**.

**Note.** It is often enough to understand short exact sequences, because from longer exact sequences, we can build short exact sequences. For instance, from

$$\cdots \xrightarrow{\varphi_{i-2}} M_{i-1} \xrightarrow{\varphi_{i-1}} M_i \xrightarrow{\varphi_i} \cdots$$

we can build the short exact sequence

$$0 \longrightarrow \mathrm{Im}(\varphi_i) = \mathrm{Ker}(\varphi_{i+1}) \longrightarrow M_{i+1} \longrightarrow \mathrm{Im}(\varphi_{i+1}) \longrightarrow 0$$

**Theorem 1.** *1. If $M, N_i$ are $R$-modules and $0 \to N_1 \to N_2 \to N_3$ is exact, then $0 \to \mathrm{Hom}_R(M, N_1) \to \mathrm{Hom}_R(M, N_2) \to \mathrm{Hom}_R(M, N_3)$ is exact.*

*2. If $M_i, N$ are $R$-modules and $M_3 \to M_2 \to M_1 \to 0$ is exact, then $0 \to \mathrm{Hom}_R(M_1, N) \to \mathrm{Hom}_R(M_2, N) \to \mathrm{Hom}_R(M_3, N)$ is exact.*

*Proof.* We prove (2) ((1) is similar):
Say we have

$$M_3 \xrightarrow{\mu_2} M_2 \xrightarrow{\mu_1} M_1 \longrightarrow 0$$

Then we can construct a corresponding sequence

$$0 \longrightarrow \mathrm{Hom}_R(M_1, N) \xrightarrow{\psi_1} \mathrm{Hom}_R(M_2, N) \xrightarrow{\psi_2} \mathrm{Hom}_R(M_3, N)$$

where $\psi_1$ sends $\varphi_1$ to $\varphi_1 \circ \mu_1$ and $\psi_2$ sends $\varphi_2$ to $\varphi_2 \circ \mu_2$.

Want to show: $\ker \psi_1 = 0$. Now $\varphi_1 \in \ker \psi_1 \implies \varphi_1 \circ \mu_1 = 0 \implies \varphi_1 = 0$ since $\mu_1$ is surjective.

Want to show: $\mathrm{Im}(\psi_1) = \ker \psi_2$. For ($\subseteq$), if $\varphi = \varphi_1 \circ \mu_1$, then $\psi_2(\varphi) = \varphi_1 \circ \mu_1 \circ \mu_2 = \varphi_1 \circ 0 = 0 \implies \varphi \in \ker \psi_2$. for ($\supseteq$), if $\varphi_2 \in \ker \psi_2$, then $\varphi_2 \circ \mu_2 = 0$. But $\mathrm{Im}(\mu_2) = \ker \mu_1 \implies$ we can define $\varphi_1(x) := \varphi_2(y)$ where $y$ is such that $\mu_1(y) = x$. $\varphi_1$ is well-defined:
If $y_1 - y_2 \in \ker \mu_1 = \mathrm{Im}(\mu_2)$, then $\varphi_1(y_1 - y_2) = \varphi_2 \circ \mu_2(\cdots) = 0$.
By construction, $\varphi_1 \circ \mu_1 = \varphi_2$. So $\varphi_2 \in \mathrm{Im}(\psi_1)$. □

**Corollary 2.** *Let $M_i, N_i$ be R-modules. Let*

$$\begin{tikzcd} M_3 \arrow[r, "\mu_2"] \arrow[d, "\varphi_3"] & M_2 \arrow[r, "\mu_1"] \arrow[d, "\varphi_2"] & M_1 \arrow[r] & 0 \\ N_3 \arrow[r, "\gamma_2"] & N_2 \arrow[r, "\gamma_1"] & N_1 \arrow[r] & 0 \end{tikzcd}$$

*be a commutative diagram with exact rows. Then there exists $\varphi \in \operatorname{Hom}(M_1, N_1)$ such that $\varphi_1 \circ \mu_1 = \gamma_1 \circ \varphi_2$.*

$$\begin{tikzcd} M_3 \arrow[r, "\mu_2"] \arrow[d, "\varphi_3"] & M_2 \arrow[r, "\mu_1"] \arrow[d, "\varphi_2"] & M_1 \arrow[r] \arrow[d, dashed, "\varphi_1"] & 0 \\ N_3 \arrow[r, "\gamma_2"] & N_2 \arrow[r, "\gamma_1"] & N_1 \arrow[r] & 0 \end{tikzcd}$$

*Proof.* By Theorem 1 part b, taking $N = N_1$, we have that

$$0 \longrightarrow \operatorname{Hom}_R(M_1, N_1) \xrightarrow{\psi_1} \operatorname{Hom}_R(M_2, N_1) \xrightarrow{\psi_2} \operatorname{Hom}_R(M_3, N_1)$$

is exact. Now $\psi_2(\gamma_1\gamma_2) = \gamma_1\varphi_2 \circ \mu_2 = \gamma_1\gamma_2\varphi_3 = 0$ by the exactness of the second row. But then $\varphi_1 \circ \mu_1 = \psi_1(\varphi_1) = \gamma_1\varphi_2 \in \ker\psi_2 = \operatorname{Im}\psi_1$. $\square$

**Theorem 3** (Five Lemma)**.** *Given*

$$\begin{tikzcd} 0 \arrow[r] & M_1 \arrow[r, "\alpha_1"] \arrow[d, "\varphi_1"] & M_2 \arrow[r, "\alpha_2"] \arrow[d, "\varphi_2"] & M_3 \arrow[r] \arrow[d, "\varphi_3"] & 0 \\ 0 \arrow[r] & N_1 \arrow[r, "\beta_1"] & N_2 \arrow[r, "\beta_2"] & N_3 \arrow[r] & 0 \end{tikzcd}$$

*a commutative diagram with exact rows, if $\varphi_1, \varphi_2$ are isomorphisms, then $\varphi_2$ is also.*

*Proof.* $\varphi_2$ is injective:
Say $x \in M_2$ such that $\varphi_2(x) = 0$. Then $\beta_2(\varphi_2(x)) = 0$. So $\varphi_3(\alpha_2(x)) = 0$. Since $\varphi_3$ is an isomorphism, $\alpha_2(x) = 0 \implies x \in \ker\alpha_2 = \operatorname{Im}(\alpha_1)$. So $x = \alpha_1(y)$ for some $y \in M_1$. Now

$$0 = \varphi_2(x) = \varphi_2(\alpha_1(y)) = \beta_1(\varphi_1(y)) \implies \varphi_1(y) \in \ker\beta_1 = 0 \implies y = 0$$

$$\text{since } \varphi_1 \text{ is an isomorphism} \implies x = 0.$$

**Exercise: surjectivity**
If $y \in N_2$, then $\beta_2(y) \in N_3$. There exists $m \in M_3$ such that $\varphi_3(m) = \beta_2(y)$. Since $\alpha_2$ is surjective, there exists $l \in M_2$ so that $\alpha_2(l) = m$. Now $\beta_2(y) = \varphi_3(m) = \varphi_3(\alpha_2(l)) = \beta_2(\varphi_2(l)) \implies \varphi_2(l) = y$ since $\beta_2$ is injective. $\square$

**Theorem 4** (Snake Lemma)**.** *Suppose we have the following commutative diagram with exact rows:*

$$\begin{tikzcd} 0 \arrow[r] & M_1 \arrow[r] \arrow[d] & M_2 \arrow[r] \arrow[d] & M_3 \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & N_1 \arrow[r] & N_2 \arrow[r] & N_3 \arrow[r] & 0 \end{tikzcd}$$

*Then, in fact, we get the following commutative diagram with exact rows and columns:*

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \ker(M_1 \to N_1) \arrow[r] \arrow[d] & \ker(M_2 \to N_2) \arrow[r] \arrow[d] & \ker(M_3 \to N_3) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & M_1 \arrow[r] \arrow[d] & M_2 \arrow[r] \arrow[d] & M_3 \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & N_1 \arrow[r] \arrow[d] & N_2 \arrow[r] \arrow[d] & N_3 \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \operatorname{coker}(M_1 \to N_1) \arrow[r] \arrow[d] & \operatorname{coker}(M_2 \to N_2) \arrow[r] \arrow[d] & \operatorname{coker}(M_3 \to N_3) \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

*And then there exists* $\varphi \in \operatorname{Hom}(\ker(M_3 \to N_3), \operatorname{coker}(M_1 \to N_1))$ *so that*

$$\begin{tikzcd}
0 \to \ker(M_1 \to N_1) \arrow[r] & \ker(M_2 \to N_2) \arrow[r] & \ker(M_3 \to N_3) \arrow[dll, "\varphi"] \\
\operatorname{coker}(M_1 \to N_1) \arrow[r] & \operatorname{coker}(M_2 \to N_2) \arrow[r] & \operatorname{coker}(M_3 \to N_3) \to 0
\end{tikzcd}$$

*is exact.*

**Note.** $\operatorname{coker}(f : A \to B) = B/\operatorname{im}(f)$.

**Corollary 5.** *If $M$ is a finitely presented $R$-module and $S$ is a finite generating set for $M$, then the kernel of $F_S \to M$ is a finitely generated $R$-module.*

*Proof.* Let $F = F_S$. We get the short exact sequence $0 \to G \to F \to M \to 0$. We want to show $G$ is finitely generated.

By assumption, there exists some finitely generated free module $L$ and a finitely generated module $E$ so that we have the short exact sequence $0 \to E \to L \to M \to 0$. Then we have the diagram

$$\begin{tikzcd}
0 \arrow[r] & E \arrow[r] & L \arrow[r] & M \arrow[r] \arrow[d, "="] & 0 \\
0 \arrow[r] & G \arrow[r] & F \arrow[r] & M \arrow[r] & 0
\end{tikzcd}$$

By $F$ being free, we get a homomorphism $L \to F$ that fits into the diagram. And then by Corollary 2, we get a homomorphism $E \to G$ that fits into the diagram. So now we have:

$$\begin{tikzcd}
 & & & 0 \arrow[d] & \\
0 \arrow[r] & E \arrow[r] \arrow[d] & L \arrow[r] \arrow[d] & M \arrow[r] \arrow[d, "="] & 0 \\
0 \arrow[r] & G \arrow[r] \arrow[d] & F \arrow[r] \arrow[d] & M \arrow[r] \arrow[d] & 0 \\
 & \mathrm{coker}(E \to G) \arrow[r] & \mathrm{coker}(L \to F) \arrow[r] & 0 &
\end{tikzcd}$$

So the Snake Lemma applies and we get a short exact sequence $0 \to \mathrm{coker}(E \to G) \to \mathrm{coker}(L \to F) \to 0$, forcing $\mathrm{coker}(E \to G) \cong \mathrm{coker}(L \to F)$. We know $\mathrm{coker}(L \to F)$ is finitely generated, by being a quotient of $F$. $E$ is finitely generated and $E \to G \to \mathrm{coker}(E \to G)$ is exact $\implies$ $G$ is finitely generated. □

**Note.** 1. If $M$ is finitely generated, we can construct a "free resolution":

$$\begin{tikzcd}
G_1 \arrow[d] & & & \\
F_1 \arrow[d] \arrow[dr] & & & \\
G \arrow[r] \arrow[d] & F \arrow[r] & M \arrow[r] & 0 \\
0 & & &
\end{tikzcd}$$

This process keeps working its way up and ends at some point.

2. Over a noetherian ring, finitely generated $\implies$ finitely presented.

### 1.3.2 Tensor products and base changes

**Definition.** Let $M, N$ be $R$-modules. A **tensor product** of $M$ and $N$ is an $R$-module $T$ together with a bilinear map $t : M \times N \to T$ with the following UMP (where $\psi \in \mathrm{Hom}_R(T, P)$):

$$\begin{tikzcd}
M \times N \arrow[r] \arrow[d, "\text{bilinear } \varphi"'] & T \arrow[dl, dashed, "\exists! \psi"] \\
P &
\end{tikzcd}$$

Tensor products exist and are unique up to isomorphism.

**Example.**

1. Say $T := \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/3\mathbb{Z}$.

   Then $a \otimes b = 1 \cdot (a \otimes b) = (3-2)(a \otimes b) = 3(a \otimes b) - 2(a \otimes b) = a \otimes 3b - 2a \otimes b = 0$. So $T = 0$.

2. If $M, N$ are free of rank $m, n$ over $R$, then $M \otimes_R N \cong R^m \otimes_R (R \oplus \cdots \oplus R) \cong (R^m \otimes_R R) \oplus \cdots \oplus (R^m \otimes_R R) \cong R^{mn}$

3. In $\mathbb{Z} \otimes_{\mathbb{Z}} (\mathbb{Z}/2\mathbb{Z})$, $2 \otimes \overline{1} = 2(1 \otimes \overline{1}) = 1 \otimes \overline{2} = 0$.

   In $2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$,
   $\mathbb{Z} \cong 2\mathbb{Z}$ via $1 \mapsto 2$, so $2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$ via $2a \otimes b \mapsto \overline{ab}$ so here, $2 \otimes \overline{1} \neq 0$.

In particular, if $M' \leq M, N$ are $R$-modules, $M' \otimes N$ may not embed into $M \otimes N$.

**Note.** We can define the tensor product of homomorphisms the same way as for vector spaces, so this operation is a functor.

Applications: "extended scalars."

**Definition.** If $R$ is a ring, then an $R$**-algebra** is a ring $A$ which is also an $R$-module such that $A \times A \to A$ is $R$-bilinear: $r(a \cdot b) = (ra) \cdot b = a \cdot (rb)$ for all $a, b \in A, r \in R$.

A map $\varphi : A \to B$ between $R$-algebras is an $R$**-algebra homomorphism** if it is an $R$-linear ring homomorphism (in particular, $\varphi(1_A) = 1_B$).

**Note.**

1. This generalizes our past definition of a $K$-algebra
2. Equivalently, $(A, f)$ is a ring and $f : R \to A$ is a ring homomorphism such that $f(R) \subseteq Z(A)$

**Example.**

1. $A$ ring, $R \subseteq Z(A) \implies A$ is an $R$-algebra
2. Any ring is a $\mathbb{Z}$-algebra in a unique way ($\varphi : \mathbb{Z} \to R, 1 \mapsto R$)
3. $R[x_1, \ldots, x_n]$ is an $R$-algebra
4. $\mathrm{Mat}_n(R)$ is an $R$-algebra

**Note.** If $A, B$ are $R$-algebras, then the $R$-module $A \otimes_R B$ is an $R$-algebra via $(a_1 \otimes b_1)(a_2 \otimes b_2) = a_1a_2 \otimes b_1b_2$.

( Use $(A \times B) \times (A \times B) \to A \otimes B$ to get $(A \otimes B) \times (A \otimes B) \to A \otimes B$ via the UMP)

**Example.** $R[x,y] = R[x] \otimes_R R[y]$

**Definition.** Let $M$ be an $R$-module, $S$ an $R$-algebra. Then $M_S := M \otimes_R S$ is an $S$-module via $S \times M_S \to M_S, (a, m \otimes s) \mapsto m \otimes as$.

We say $M_S$ is obtained from $M$ by **extension of scalars** from $R$ to $S$.

**Lemma 6.** *If $M, N$ are $R$-modules and $S$ is a commutative $R$-algebra, then $(M \otimes_R N)_S \cong M_S \otimes_S N_S$.*

*Proof.* **Exercise** $\square$

**Proposition 7.** *If*

$$M_1 \xrightarrow{\varphi_1} M_2 \xrightarrow{\varphi_2} M_3 \longrightarrow 0$$

*is an exact sequence of $R$-modules, if $N$ is an $R$-modules, then*

$$M_1 \otimes N \xrightarrow{\varphi_1} M_2 \otimes N \xrightarrow{\varphi_2} M_3 \otimes N \longrightarrow 0$$

*is exact.*

*Proof.* For any $R$-module $P$,

$$0 \to \mathrm{Hom}_R(M_3, \mathrm{Hom}_R(N,P)) \to \mathrm{Hom}_R(M_2, \mathrm{Hom}_R(N,P)) \to \mathrm{Hom}_R(M_1, \mathrm{Hom}_R(N,P))$$

is exact by Theorem 1(b). Then

$$0 \to \mathrm{Hom}_R(M_3 \otimes N, P) \to \mathrm{Hom}_R(M_2 \otimes N, P) \to \mathrm{Hom}_R(M_1 \otimes N, P) \to 0$$

is exact and (by an exercise) we get

$$M_1 \otimes N \to M_2 \otimes N \to M_3 \otimes N \to 0$$

is exact. $\square$

### 1.3.3 Projective and flat modules

**Proposition 8.** *An $R$-module $P$ is projective $\iff$ $\mathrm{Hom}_R(P, -)$ is exact (i.e. preserves exact sequences),*

*Proof.* ($\Leftarrow$): We have

$$\begin{tikzcd} & & & P \arrow[d, "\psi"] & \\ 0 \arrow[r] & \ker\varphi \arrow[r] & N \arrow[r, "\varphi"] & M \arrow[r] & 0 \end{tikzcd}$$

We can then get

$$0 \to \mathrm{Hom}_R(P, \ker\varphi) \to \mathrm{Hom}_R(P, N) \to \mathrm{Hom}_R(P, M) \to 0$$

exact. Then $\psi \in \mathrm{Hom}_R(P, M) \implies$ we can pull back to some $\tilde{\psi} \in \mathrm{Hom}_R(P, N)$. So $\varphi\tilde{\psi} = \psi \implies P$ is projective.
($\Rightarrow$): If $P$ is projective, we have the short exact sequence $0 \to \ker\varphi \to N \to M \to 0$. We know $\mathrm{Hom}_R(P, N) \to \mathrm{Hom}_R(P, M)$ is surjective since $P$ is projective (+ Theorem 1(b)) $\implies$ claim. $\square$

**Theorem 9.** *If $P$ is projective, then $P \otimes_R -$ and $- \otimes_R P$ are exact.*

*Proof.* The proof for $(P \otimes_R -)$: We want to show that $P \otimes_R -$ preserves monomorphisms. So let $\varphi : M \to N$ be a monomorphism. If $P$ is free, $P \otimes_R M \cong M^I$ and $P \otimes_R N \cong N^I$ for some $I$. Easy to see: $1 \otimes \varphi$ is a monomorphism ($\varphi$ in each component).

If $P$ is not free, we can take $\phi : F \to P$ an epimorphism with $F$ free. $P$ projective $\implies$ we get $\psi : P \to F$ a splitting.

We see that $\varphi \otimes 1_M : F \otimes M \to P \otimes M$ and $\psi \otimes 1_M : P \otimes M \to F \otimes M \implies \psi \otimes 1_M$ is a monomorphism. Similarly for $psi \otimes 1_N$. Thus, we get:

$$\begin{tikzcd}
P \otimes M \arrow[r] \arrow[d, "\psi\otimes 1_M \text{ mono}"'] & P \otimes N \arrow[d, "\psi\otimes 1_N \text{ mono}"] \\
F \otimes M \arrow[r] & F \otimes N
\end{tikzcd}$$

So by the first part, $F \otimes M \to F \otimes N$ is a monomorphism. So $P \otimes M \to P \otimes N$ is a monomorphism. □

**Definition.** An $R$-module $M$ is **flat** if $M \otimes_R -$ preserves short exact sequences.

Thus Theorem 9 says projective $\implies$ flat.

**Proposition 10.** *If $M$ is flat over a domain, $M$ is torsion-free.*

*Proof.* Say $0 \neq x \in M$. Then $R \to R, r \mapsto x \cdot r$ is a monomorphism, so $M = R \otimes_R m \to M, a \otimes b \mapsto x \cdot (a \otimes b)$ is a monomorphism. Thus $M_{\text{tor}} = 0$. □

### 1.3.4 Injective modules

**Definition.** An $R$-module $M$ is **injective** if for all diagrams below with exact row

$$\begin{tikzcd}
0 \arrow[r] & N \arrow[r, "i"] \arrow[d, "\varphi"] & P \\
 & M &
\end{tikzcd}$$

there exists $\psi \in \mathrm{Hom}(P, M)$ so that $\psi i = \varphi$:

$$\begin{tikzcd}
0 \arrow[r] & N \arrow[r, "i"] \arrow[d, "\varphi"] & P \arrow[dl, dashed, "\exists\psi"] \\
 & M &
\end{tikzcd}$$

**Lemma 11.** *If $M$ is injective and we have the diagram*

$$\begin{tikzcd}
N_1 \arrow[r, "\alpha"] & N \arrow[r, "\beta"] \arrow[d, "\varphi"] & N_2 \\
 & M &
\end{tikzcd}$$

*with exact row, then if $\varphi\alpha = 0$, there exists $\psi \in \mathrm{Hom}(N_3, M)$ such that $\psi\beta = \varphi$ :*

$$\begin{tikzcd} N_1 \arrow[r, "\alpha"] & N \arrow[r, "\beta"] \arrow[d, "\varphi"] & N_2 \arrow[dl, dashed, "\exists\psi"] \\ & M & \end{tikzcd}$$

*Proof.* Let $X = \ker\beta = \mathrm{Im}\alpha$. Then if $\pi : N_2 \to N_2/X$ is the canonical homomorphism, then $\beta$ induces $\overline{\beta} : N_2/X \to N_3$.

$\varphi\alpha = 0 \implies X = \mathrm{Im}\alpha \subseteq \ker\varphi$. So $\overline{\varphi} : N_2/X \to M$ is well-defined.

$M$ injective $\implies \exists\psi : N_3 \to M$ such that $\overline{\varphi} = \psi\overline{\beta}$. But then $\varphi = \overline{\varphi}\pi = \psi\overline{\beta}\pi = \psi\beta$. $\square$

**Proposition 12.** *If $R$ is a ring and $M$ is an $R$-module, TFAE:*

1. *$M$ is injective*
2. *Every short exact sequence $0 \to M \to N \to P \to 0$ splits*
3. *$\mathrm{Hom}_R(-, M)$ preserves short exact sequences*

*Proof.* **Exercise.** $\square$

**Theorem 13** (Baer's Criterion)**.** *Let $R$ be a ring and $M$ be an $R$-module. Then TFAE:*

1. *$M$ is injective*
2. *For every ideal $I$ in $R$ and every $\varphi \in \mathrm{Hom}_R(I, M)$, there exists $\tilde{\varphi} \in \mathrm{Hom}_R(R, M)$ extending $\varphi$ ($\tilde{\varphi}|_I = \varphi$):*

$$\begin{tikzcd} 0 \arrow[r] & I \arrow[r] \arrow[d, "\varphi"] & R \arrow[dl, dashed, "\tilde{\varphi}"] \\ & M & \end{tikzcd}$$

*Proof.* ($\Rightarrow$): by definition.

($\Rightarrow$): Say we have the exact row $0 \to N \to N'$ and $\varphi : N \to M$. We want some $\psi \in \mathrm{Hom}_R(N', M)$ such that $\psi|_N = \varphi$ (we will view $N \leq N'$ as a submodule).

Let $S := \{(K, \varphi_K) \mid N \leq K \leq N', \varphi_K \in \mathrm{Hom}_R(K, M)$ such that $\varphi_K|_N = \varphi\}$. $S$ is the set of all partial extensions of $\varphi$.

Then $S$ is nonempty since $(N, \varphi) \in S$. And $S$ is partially ordered via $(K, \varphi_K) \leq (K', \varphi_{K'}) \iff K \supseteq K'$ and $\varphi_{K'}|_K = \varphi_K$. One can check: $S$ satisfies the hypotheses of Zorn's Lemma. Then $S$ has a maximal element, say $(K_0, \varphi_0)$.

We will show: $K_0 = N'$. Assume for contradiction that $m \in N' \setminus K_0$.

Define $I := \{r \in R \mid rm \in K_0\}$. Can check: this is an ideal in $R$.

Define $\zeta : I \to M$ so that $\zeta(r) = \varphi_0(rm)$. Can check: $\zeta$ is a homomorphism.

So $M$ injective $\implies \exists \chi : R \to M$ extending $\zeta$.

Consider $K' := K_0 + Rm \leq N'$. $K'$ properly contains $K_0$.

Define $\varphi' : K' = K_0 + Rm \to M$ by $\varphi(k + rm) := \varphi_0(K) + \chi(r)$.

Easy to check: $\varphi' \in \mathrm{Hom}_R(K', M)$ where $\varphi'|_N = \varphi$. $\to\leftarrow$ contradicts maximality of $K_0$. $\square$

**Corollary 14.** *If $R$ is an integral domain and $K = \mathrm{Frac}(R)$ is its fraction field, then $K$ is an injective $R$-module.*

*Proof.* Let $I \subseteq R$. Let $\varphi : I \to K$ be a homomorphism of $R$-modules. Then $r\varphi(s) = \varphi(rs) = s\varphi(r)$ i.e. in $K$, $\frac{\varphi(r)}{r} = \frac{\varphi(s)}{s} =: a$.

Define $\psi : R \to K, \psi(r) := ra$. Then $\psi \in \mathrm{Hom}_R(R, K)$ and $\psi|_I = \varphi$. By Baer, $K$ is injective. $\square$

**Note.** Injective is "dual" to projective. Every module is a quotient of a projective module. True: every module is a submodule of an injective module (hard to prove)

## 1.4 Localization

Let $R$ be a commutative ring with identity.

**Definition.** Take $S \subseteq R$ multiplicatively closed and $M$ an $R$-module. Define $(m, s) \sim (m', s')$ if there exists $u \in S$ such that $u(s'm - sm') = 0$.

The class of $(m, s)$ is written as $\frac{m}{s}$

The **localization of $M$ at $S$** is $S^{-1}M = \{\frac{m}{s} \mid m \in M, s \in S\}$ (this is an $S^{-1}R$-module)

If $S = \{a^n | n \in \mathbb{N}\}$, we write $M_a$ for $S^{-1}M$

If $S = R \setminus P$ for some prime ideal $P$ in $R$, we write $M_P$ for $S^{-1}M$

**Example.** If $S \subseteq R$ is multiplicatively closed and $I \trianglelefteq R$ then $S^{-1}I = I \cdot S^{-1}R$.

**Lemma 1** (Localization is extension of scalars)**.** *If $S \subseteq R$ is multiplicatively closed and $M$ is an $R$-module, $S^{-1}M \cong M \otimes_R S^{-1}R$.*

*Proof.* Define $\varphi : S^{-1}M \to M \otimes_R S^{-1}R, \quad \frac{m}{s} \mapsto m \otimes \frac{1}{s}$.

This map is well-defined: $\frac{m}{s} = \frac{m'}{s'} \iff u(s'm - sm') = 0$ for some $u \in S \iff m \otimes \frac{1}{s} - m' \otimes \frac{1}{s'} = ums' \otimes \frac{1}{uss'} - um's \otimes \frac{1}{uss'} = u(ms' - m's) \otimes \frac{1}{uss'} = 0 \iff \varphi(\frac{m}{s}) = \varphi(\frac{m'}{s'})$.

Now define a map $M \times S^{-1}R \to S^{-1}M$ ($R$-bilinear) by $(m, \frac{a}{s}) \mapsto \frac{am}{s}$.

This map is well-defined: $(m, \frac{a}{s}) = (m', \frac{a'}{s'}) \iff m = m'$ and $\exists u \in S$ such that $u(as' - a's) = 0 \iff u(ams' - a'm's) = um(as' - a's) = 0 \iff \frac{am}{s} = \frac{a'm'}{s}$.

UMP of tensor product gives us a homomorphism $\psi : M \otimes_R S^{-1}R \to S^{-1}M$ so that $\psi(m \otimes \frac{r}{s}) = \varphi(m, \frac{r}{s})$.

Observe: $\psi : M \otimes_R S^{-1}R \to S^{-1}M$ and $\psi(m \otimes \frac{a}{s}) = \frac{am}{s}$.

Check: $\psi$ is well-defined.

Then $\varphi\psi = \text{id}, \psi\varphi = \text{id}$ by construction. $\square$

**Note.** $\varphi : M \to N$ homomorphism of $R$-modules and $S \subseteq R$ multiplicatively closed $\implies \varphi \otimes \text{id} : M \otimes_R S^{-1}R \to N \otimes_R S^{-1}R, \quad (\varphi \otimes \text{id})(m \otimes \frac{a}{s}) = \varphi(m) \otimes \frac{a}{s}$.

Lemma 1 $\implies S^{-1}\varphi : S^{-1}M \to S^{-1}N \implies$ localization is a functor. (We also write $\varphi_a$ or $\varphi_P$ when $S^{-1}M = M_a$ or $M_P$)

**Proposition 2** (Exactness of Localization)**.** *Given a short exact sequence of $R$-modules*

$$0 \longrightarrow M_1 \xrightarrow{\varphi} M_2 \xrightarrow{\psi} M_3 \longrightarrow 0$$

*if $S \subseteq R$ is multiplicatively closed, $0 \to S^{-1}M_1 \to S^{-1}M_2 \to S^{-1}M_3 \to 0$ is also exact (i.e., $S^{-1}R$ is a flat $R$-module)*

*Proof.* By Lemma 1, $S^{-1}M_i = M_i \otimes_R S^{-1}R$. By Prop 3.7, we only need to show $S^{-1}\varphi$ is injective.

Let $\frac{m}{s} \in S^{-1}M$ with $0 = S^{-1}\varphi(\frac{m}{s}) = \frac{\varphi(m)}{s}$ i.e. $u\varphi(m) = \varphi(um) = 0$ for some $u \in S$. Then $\varphi$ injective $\implies um = 0 \implies \frac{m}{s} = \frac{1}{us}um = 0 \implies S^{-1}\varphi$ is injective. $\square$

**Example.** As a $\mathbb{Z}$-module, $\mathbb{Q}$ is:

- Not free
- Flat (Prop 2 + $\mathbb{Q} = \text{Frac}(\mathbb{Z})$)
- Injective (Corollary 14)
- Torsion-free

- Not projective (can prove this using the fact that the surjective map $\mathbb{Z}[x_1, \dots] \twoheadrightarrow \mathbb{Q}, \quad x_i \mapsto \frac{1}{i}$ has no lift $\mathbb{Q} \to \mathbb{Z}[x_1, \dots]$)

**Corollary 3.** *Let $S \subseteq R$ be multiplicatively closed. Then*

1. $\varphi \in \mathrm{Hom}_R(M, N) \implies \ker(S^{-1}\varphi) = S^{-1}\ker\varphi$ *and* $\mathrm{im}(S^{-1}\varphi) = S^{-1}\mathrm{im}\varphi$
2. $N \leq M \implies S^{-1}(M/N) \cong (S^{-1}M)/(S^{-1}N)$
3. $S^{-1}(M \oplus N) \cong S^{-1}M \oplus S^{-1}M$

### 1.4.1 Nakayama's Lemma

**Proposition 4** (Generalized Cayley-Hamilton)**.** *Let $M$ be a f.g. $R$-module, $I \trianglelefteq R$, and $\varphi : M \to M$ an $R$-module homomorphism such that $\varphi(M) \subseteq IM$. Then there is a monic polynomial $\chi = x^n + a_{n-1}x^{n-1} + \dots + a_0 \in R[x]$ with $a_i \in I$ such that $\chi(\varphi) = 0$ in $End_R(M)$.*

**Note.** If $R$ is a field and $M$ is an $R$-vector space and $I = R$ then unless $\varphi = 0$ this proof will give us the characteristic polynomial.

*Proof.* Let $x_1, \dots, x_n$ generate $M$. Then $\varphi(x_i) \in IM \implies \varphi(x_i) = \sum_{j=1}^n a_{ij}x_j$ for $a_{ij} \in I$.

Consider $M$ as an $R[x]$ module via $X \cdot m := \varphi(m)$. Then

$$\sum_{j=1}^{n}(\delta_{ij}x - a_{ij})x_j = \sum_{j=1}^{n}\delta_{ij}\varphi(x_j) - a_{ij})x_j = \varphi(x_i) - \sum a_{ij}x_j = 0$$

for $1 \leq i \leq n$.
We can think of this as the linear system:

$$(x \cdot I_n - A)\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = 0, \quad A = (a_{ij})$$

Multiply on the left with the cofactor matrix to get $I_n \cdot \det(x \cdot I_n - A)$. Then we get $\det((\delta_{i_k}x - a_{i_k})_{i_k})x_j = 0$ for all $j$. $\chi := \det((\delta_{i_k}x - a_{i_k})_{i_k})$.

Then $\chi(\varphi)x_j = 0$ for all $j \implies \chi(\varphi) = 0$ in $\mathrm{End}_R(M)$, because $x_1, \dots, x_n$ generate $M$. $\square$

**Corollary 5** (Nakayama's Lemma)**.** *If $M$ is f.g. and $I \trianglelefteq R$ so that $IM = M$, then there exists $i \in I$ such that $im = m$ for all $m \in M$.*

*Proof.* Say $IM = M$. Apply Cayley-Hamilton to $\varphi = \mathrm{id} \implies \exists a_i \in I$ so that $\mathrm{id}^n + a_{n-1}\mathrm{id}^{n-1} + \dots + a_0 = 0$ in $\mathrm{End}_R(M)$ i.e. $(1 + a_{n-1} + \dots + a_0)\mathrm{id} = 0$.

Let $i := -(a_{n-1} + \dots + a_0)$. Then $(1 - i)m = 0$, i.e. $im = m$. $\square$

**Corollary 6** (Nakayama, Local Version)**.** *If $R$ is local with unique maximal ideal $P$ and $M$ is a f.g. $R$-module:*

1. *If $I \lneq R$ such that $IM = M$, then $M = 0$.*

2. *If $m_1, \ldots, m_k$ are preimages of a basis for $M/PM$ as an $R/P$-vector space, then $m_1, \ldots, m_k$ generate $M$.*

*Proof.* 1. Corollary 5 $\implies \exists x \in R$ such that $x \equiv 1 \pmod{P}$, $\quad xM = 0$. But $x$ is a unit since $R$ is local $\implies M = x^{-1}(xM) = 0$.

2. $N := \langle m_1, \ldots, m_k \rangle \leq M$. Then we have the surjection $N \to M \to M/PM \implies N + PM = M$. So $P(M/N) = (PM + N)/N \cong M/N \implies$ (by 1) $M/N = 0 \implies N = M$.

□

**Corollary 7.** *$M$ f.g. projective module over a local ring $R \implies M$ is free.*

*Proof.* Let $R$ be a local ring with maximal ideal $P$. Let $M$ be a f.g. projective $R$-module. Take a basis $\overline{x_1}, \ldots, \overline{x_n}$ of $M/PM$. Then

$$F := \bigoplus_{i=1}^{n} R \implies F/PF \cong M/PM$$

We get the following diagram with exact rows:

$$\begin{tikzcd}
0 \arrow[r] & PF \arrow[r] & F \arrow[r, "\pi_F"] \arrow[dr, "\alpha"] & F/PF \arrow[r] \arrow[d, "\psi"] & 0 \\
0 \arrow[r] & PM \arrow[r] & M \arrow[r, "\pi_M"] & M/PM \arrow[r] & 0
\end{tikzcd}$$

Then using freeness of $F$, $\exists \varphi : F \to M$ so that $\alpha = \pi_M \varphi$. We claim: $\varphi$ is an isomorphism.

If $f \in F, \psi\pi_F(f) = \psi(f + PF) = \varphi(f) + PM$. $\psi$ surjective $\implies$ $\operatorname{im}\varphi + PM = M$. Nakayama $\implies$ $\operatorname{im}\varphi = M \implies \varphi$ surjective.

$M$ projective $\implies F \cong \ker\varphi \oplus X$ since $X \cong M \implies PF \cong P\ker\varphi \oplus PX \implies \ker\varphi = P\ker\varphi + (\ker\varphi \cap PX) \subseteq \ker\varphi \cap X = 0$.

By Nakayama, $\ker\varphi = P\ker\varphi \implies \ker\varphi = 0 \implies \varphi$ is an isomorphism. □

### 1.4.2 Local properties

**Definition.** We say that a property (*) of modules over some ring $R$ is **localizable** if for all $R$-modules $M$:
(*) holds for $M \implies$ (*) holds for $M_P$ for all primes $P \trianglelefteq R$

Conversely, we say that (*) is **local-to-global** if for all $R$-modules $M$,
(*) holds for $M_P$ for all $P \trianglelefteq R$ prime $\implies$ (*) holds for $M$

We say (*) is **local** if it is localizable and local-to-global

A similar definition can be made for homomorphisms, sequences, etc.

**Proposition 8.** *1. $M$ $R$-module wth $M_P = 0$ for all maximal $P \trianglelefteq R \implies M = 0$*
*Hence, being trivial is a local property*

*2. Given a sequence of $R$-modules*

$$0 \longrightarrow M_1 \xrightarrow{\varphi_1} M_2 \xrightarrow{\varphi_2} M_3 \longrightarrow 0$$

*so that*

$$0 \longrightarrow (M_1)_P \xrightarrow{(\varphi_1)_P} (M_2)_P \xrightarrow{(\varphi_2)_P} (M_3)_P \longrightarrow 0$$

*is exact for all maximal ideals $P \trianglelefteq R$, then*

$$0 \to M_1 \to M_2 \to M_3 \to 0$$

*is exact.*
*Hence (by Prop 2) exactness is a local property.*

*Proof.* 1. Say $M \neq 0$. Let $m \in M \setminus \{0\}$. Then $1 \notin \operatorname{Ann}(m) \trianglelefteq R \implies \exists P \trianglelefteq R$ maximal so that $\operatorname{Ann}(m) \subseteq P$.

$u \in R \setminus P \implies u \notin \operatorname{Ann}(m) \implies um \neq 0 \implies \frac{m}{1} \neq 0$ in $M_P \implies M_p \neq 0$.

2. Suffices to show: if for

$$(*) \qquad\qquad L \xrightarrow{\varphi} M \xrightarrow{\psi} N$$

the localization at each maximal ideal is exact, then the original sequence (*) is exact.

Say $S = R \setminus P, P \trianglelefteq R$ maximal. Then

$\psi(\varphi(L))_P = \{\frac{\psi\varphi(m)}{s} \mid m \in L, s \in S\} = \{\psi_P\varphi_P(\frac{m}{s}) \mid m \in L, s \in S\} = \psi_P\varphi_P(L_P) = 0$

by assumption.

By (1), $\psi(\varphi(L)) = 0$, i.e. $\operatorname{im}\varphi \subseteq \ker\psi$. So (*) is a complex.

Now consider $\ker\psi/\operatorname{im}\varphi$. Then by Corollary 3,

$$(\ker\psi/\operatorname{im}\varphi)_P = (\ker\psi)_P/(\operatorname{im}\varphi)_P = \ker(\psi_P)/\operatorname{im}(\varphi_P) = 0$$

for all maximal ideals $P \trianglelefteq R$. So then by (1), $\ker\psi/\operatorname{im}\varphi = 0 \implies \ker\psi = \operatorname{im}\varphi$.

□

**Corollary 9.** *Let $M$ be an $R$-module. Then $M$ is flat over $R \iff M_P$ is flat over $R_P$ for all maximal ideals $P$ in $R$.*
*Hence flatness is a local property.*

*Proof.* $(\Leftarrow)$ : Say $M_P$ is flat over $R_P$ for all maximal $P \trianglelefteq R$. Then for all short exact sequences

$$0 \longrightarrow A \xrightarrow{\varphi} B \xrightarrow{\psi} C \longrightarrow 0$$

we want to show:

$$0 \longrightarrow M \otimes_R A \longrightarrow M \otimes_R B \longrightarrow M \otimes_R C \longrightarrow 0$$

is exact.

The functor $M \otimes_R -$ is always right exact, so we just have to show that for every injective map of $R$-modules $f : N \to N'$, the induced map $\mathrm{id}_M \otimes f : M \otimes_R N \to M \otimes_R N'$ is also injective.

Since $M_P$ is flat and $f$ is injective, $(\mathrm{id}_M)_P \otimes f_P : M_P \otimes_{R_P} N_P \to M_P \otimes_{R_P} (N')_P$ is also injective.

Then $(\ker(\mathrm{id}_M \otimes f))_P = \ker((\mathrm{id}_M \otimes f)_P) = \ker((\mathrm{id}_M)_P \otimes f_P) = 0$, and since being trivial is local, we must have $\ker(\mathrm{id}_M \otimes f) = 0$ i.e. $\mathrm{id}_M \otimes f$ is injective.

$(\Rightarrow)$ : Exercise. $\square$

### 1.4.3 Z-local properties

**Definition.** A family $\{f_i \mid i \in I\}$ of elements in $R$ is a **Z-family** if $(f_i \mid i \in I) = R$.

i.e. $\bigcup D(f_i) = \operatorname{Spec} R$

**Note.** If $\{f_i \mid i \in I\}$ is a Z-family, then there exists a finite $J \subseteq I$ so that $\{f_i \mid i \in J\}$ is still a Z-family.

**Definition.** A property (*) of $R$-modules is **Z-local** if:
(*) holds over $R \iff \exists$ a Z-family $\{f_i \mid i \in I\}$ so that (*) holds over all $R_{f_i}$

**Proposition 10.** *Say $M, N$ are $R$-modules, $\varphi \in \operatorname{Hom}_R(M, N)$, $P \trianglelefteq R$ prime.*

1. *If $N$ is finitely generated and $\varphi_P$ is surjective, then there exists $f \in R \setminus P$ such that $\varphi_f$ is surjective.*
2. *Surjectivity of $\varphi$ is Z-local*
3. *If $M$ is finitely generated and $N$ finitely presented and $\varphi_P$ an isomorphism, then there exists $f \in R \setminus P$ so that $\varphi_f$ is an isomorphism*

*4. If $M$ is finitely generated and $N$ is finitely presented, then bijectivity of $\varphi$ is Z-local*

**Lemma.** *Say $0 \to M_1 \to M_2 \to M_3 \to 0$ is a short exact sequence with $M_2$ finitely generated and $M_3$ finitely presented. Then $M_1$ is finitely generated.*

*Proof.* Exercise. $\square$

*Proof.* (Of Proposition 10)

1. Consider the exact sequence

$$0 \longrightarrow \ker\varphi \longrightarrow M \xrightarrow{\varphi} N \longrightarrow N/\mathrm{im}\varphi = \mathrm{coker}\varphi \longrightarrow 0$$

   Then by Corollary 3, $(\mathrm{coker})_P = \mathrm{coker}_P) = 0$ since $\varphi_P$ is surjective. So for all $x \in \mathrm{coker}\varphi, \exists f_x \in R \setminus P$ such that $f_x \cdot x = 0$.

   $N$ finitely generated $\implies$ $\mathrm{coker}\varphi$ is finitely generated, say by $x_1, \dots, x_n$. Define $f := f_{x_1} \dots f_{x_n}$. Then $f \in R \setminus P$ and $f \cdot \mathrm{coker}\varphi = 0 \implies \varphi_f$ is surjective.

2. $\varphi$ surjective $\implies$ $\varphi_f$ is surjective for all $f$, since localization = extension of scalars, which preserves exact sequences.

   Conversely, let $\{f_i \mid i \in I\}$ be a Z-family so that $\varphi_{f_i}$ is surjective for all $i \in I$. $P \trianglelefteq R$ primes $\implies \exists i \in I$ so that $f_i \in R \setminus P \implies \varphi_P$ is a localization of $\varphi_{f_i} \implies \varphi_P$ is surjective. So $\varphi$ is surjective by Prop 8 part 2.

3. Let $M$ be finitely generated, $N$ finitely presented, and $\varphi_P$ an isomorphism. Then there exists $f_1 \in R \setminus P$ so that $(\mathrm{coker}\varphi)_{f_1} = 0$ by (1), i.e. $0 \to (\ker\varphi)_{f_1} \to M_{f_1} \to N_{f_1} \to 0$ is exact.

   $N$ finitely generated $\implies$ $N_{f_1}$ is finitely generated over $R_{f_1}$ $\implies$ by the lemma, $(\ker\varphi)_{f_1}$ is finitely generated. As in the proof of (1), we get $f_2 \in R \setminus P$ such that $f := f_1 f_2 \ker\varphi = 0 \implies \varphi_f : M_f \to N_f$ is an isomorphism.

4. Exercise.

$\square$

**Corollary 11.** *If $M$ is finitely presented, TFAE:*

1. *There exists a Z-family $\{f_i \mid i \in I\}$ such that $M_{f_i}$ is a free $R_{f_i}$-module for all $i$*
2. *For all prime ideals $P$ in $R$, $M_P$ is a free $R_P$-module*
3. *For all maximal ideals $P$ in $R$, $M_P$ is a free $R_P$-module*

*Proof.* $1 \implies 2$: $P \trianglelefteq R$ prime $\implies \exists i \in I$ so that $f_i \in R \setminus P \implies M_p \cong M_{f_i} \otimes_{R_{f_i}} R_P$ is free since $M_{f_i}$ is.
$2 \implies 3$: ✓
$3 \implies 1$: Suffices to show:
For all $P \trianglelefteq R$ maximal, there exists $f_p \in R \setminus P$ such that $M_{f_p}$ is free. Then tak e$\{f_p \mid P \trianglelefteq R \text{ maximal}\}$. This is a Z-family because $(f_p \mid P \trianglelefteq R)$ cannot be contained in any maximal ideal.

Now fix $P \trianglelefteq R$ maximal. Let $x_1, \ldots, x_n \in M$ be such that the images form an $R_P$-basis of $M_P$.

Define $\varphi : R^n \cong \bigoplus_{i=1}^n Re_i \to M, \quad e_i \mapsto x_i$. Then $\varphi_P$ is an isomorphism by construction. So by Prop 10 part 3, there exists $f_p \in R \setminus P$ so that $\varphi_{f_p}$ is an isomorphism. Thus $M_{f_p}$ is free. □

**Note.** By Corollary 7, finitely generated projective modules are locally free.

**Theorem 12.** *Let $R$ be a ring and $M$ an $R$-module. Then TFAE:*

1. *$M$ is f.g. projective*
2. *$M$ is f.p. and locally free*
3. *$M$ is f.p. and $M_P$ is free over $R_P$ for all $P \trianglelefteq R$ maximal*
4. *For all $P \trianglelefteq R$ maximal, $\exists\, f \in R \setminus P$ so that $M_f$ is a free $R_f$-module of finite rank*
5. *$\exists$ a finite $Z$-family $\{f_i\}$ so that $M_{f_i}$ is a free $R_{f_i}$-module of finite rank for all $i \in I$*

**Definition.** An $R$-module $N$ is **faithfully flat** if for every sequence of $R$-modules

$$(*) \quad 0 \to M_1 \to M_2 \to M_3 \to 0,$$

the sequence

$$0 \to M_1 \otimes N \to M_2 \otimes N \to M_3 \otimes N \to 0 \text{ is exact}$$
$$\iff (*) \text{ is.}$$

**Lemma 13.** *$N$ is faithfully flat $\iff$ $N$ is flat and $PN \neq N$ for every prime ideal $P \trianglelefteq R$.*

*Proof.* Exercise. □

**Lemma 14.** *$R$ ring, $\{f_i \mid i \in I\}$ finite Z-family*

$$\implies R' := \prod_{i \in I} R_{f_i} \text{ is a faithfully flat } R\text{-module.}$$

*Proof.* Each $R_{f_i}$ is flat as an $R$-module, since flatness is localizable and $R$ is a flat $R$-module. So $R'$ is flat (tensor products distribute over finite direct sums).

Let $P \trianglelefteq R$ be prime. Then there exists $f_i \notin P$. $P_{f_i} = PR_{f_i} \subsetneq R_{f_i} \implies PR' \subseteq PR_{f_i} \times \prod_{i \neq j} PR_{f_j} \neq R' \implies$ (by Lemma 13) $R'$ is faithfully flat over $R$. □

**Lemma 15.** *$R$ ring, $\{f_i \mid i \in I\}$ finite $Z$-family, $M$ $R$-module. If $M_{f_i}$ is finitely presented for all $i \in I$, then $M$ is finitely presented.*

*Proof.* Let $M' = \prod_{i \in I} M_{f_i}$. $M'$ is an $R'$-module ($R' = \prod_{i \in I} R_{f_i}$ as in Lemma 14). Take $m, k$ so that there are exact sequences

$$R_{f_i}^m \to R_{f_i}^k \to M_{f_i} \to 0 \quad \text{for all } i.$$

Then $(R')^m \to (R')^k \to M' \to 0$ is exact. Thus $M'(= M \otimes_R R')$ is finitely presented $\implies R' \otimes_R R^m \to R' \otimes_R R^k \to R' \otimes_R M \to 0$ is exact. So since $R'$ is faithfully flat, $R^m \to R^k \to M \to 0$ is exact. So $M$ is finitely presented. $\square$

**Note.** The proof also shows that $M$ is finitely generated if all $M_{f_i}$ are.

**Lemma 16.** *Let $M$ be a finitely presented $R$-module, $N$ an $R$-module, $S \subseteq R$ multiplicatively closed $\implies S^{-1}(\mathrm{Hom}_R(M, N)) \cong \mathrm{Hom}_{S^{-1}R}(S^{-1}M, S^{-1}N)$.*

*Proof.* The map $\mathrm{Hom}_R(M, N) \to \mathrm{Hom}_{S^{-1}R}(S^{-1}M, S^{-1}N)$, $\varphi \mapsto S^{-1}\varphi$ induces a map $\alpha : \mathrm{Hom}_R(M, N)_{S^{-1}R} \to \mathrm{Hom}_{S^{-1}R}(S^{-1}M, S^{-1}N)$ via $\frac{\varphi}{t} \mapsto \frac{1}{t} S^{-1}\varphi$.

$M$ finitely presented $\implies \exists K \to F \to M \to 0$ exact with $K, F$ free of finite rank $\implies S^{-1}K \to S^{-1}F \to S^{-1}M \to 0$ is exact by Prop 2. Moreover, $0 \to \mathrm{Hom}_R(M, N) \to \mathrm{Hom}_R(F, N) \to \mathrm{Hom}_R(K, N)$ is exact. So we get commutative diagram with exact rows:

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Hom}_R(M,N)_{S^{-1}R} \arrow[r] \arrow[d, "\alpha"] & \mathrm{Hom}_R(F,N)_{S^{-1}R} \arrow[r] \arrow[d, "\beta"] & \mathrm{Hom}_R(K,N)_{S^{-1}R} \arrow[d, "\gamma"] \\
0 \arrow[r] & \mathrm{Hom}_{S^{-1}R}(S^{-1}M, S^{-1}N) \arrow[r] & \mathrm{Hom}_{S^{-1}R}(S^{-1}F, S^{-1}N) \arrow[r] & \mathrm{Hom}_{S^{-1}R}(S^{-1}K, S^{-1}N)
\end{tikzcd}$$

If $F = R$, $\mathrm{Hom}_R(R, N)_{S^{-1}R} \cong S^{-1}N \cong \mathrm{Hom}_{S^{-1}R}(S^{-1}R, S^{-1}N)$, so $\beta$ and $\gamma$ are isomorphisms (by $K, F$ being f.g.). By a diagram chase, $\alpha$ is an isomorphism. $\square$

**Theorem 12.**
$R$ ring, $M$ $R$-module. TFAE:

1. $M$ f.g. projective.
2. $M$ f.p. and locally free.
3. $M$ f.g. and $M_P$ is a free $R_P$-module for all $P \trianglelefteq R$ maximal.
4. For all $P \trianglelefteq R$ maximal, $\exists f \in R \setminus P$ such that $M_f$ is a free $R_f$-module of finite rank.
5. $\exists$ a finite $Z$-family such that $M_{f_i}$ is free $R_{f_i}$-module of finite rank for all $i \in I$.

*Proof.* ($1 \Rightarrow 2$) $M$ f.g. projective $\implies$ by Corollary 7, $M$ is locally free.
We get $F \to M \to 0$ exact, $F$ free finite rank.
Since $M$ is projective, the sequence splits $\Rightarrow M$ is f.p. (kernel is quotient of $F$) .

($2 \Rightarrow 3$) Clear.

($3 \Rightarrow 4$) Corollary 11.

($4 \Rightarrow 5$) $P \trianglelefteq R$ maximal ideal $\implies$ choose $f_P \in R \setminus P$ such that $M_{f_P}$ is a free $R_{f_P}$-module of finite rank.
$\{f_P \mid P \text{ maximal}\}$ is a $Z$-family, has a finite subfamily.

($5 \Rightarrow 1$) Lemma 15 $\implies$ $M$ f.p. and Corollary 11 $\Rightarrow M$ is locally free.
WTS: if $N' \to N \to 0$ is exact then $\mathrm{Hom}_R(M, N') \to \mathrm{Hom}_R(M, N) \to 0$ is exact (Prop 3.8 says this is equivalent to projective). So by Corollary 3, it suffices to show that $\mathrm{Hom}_R(M, N')_P \to \mathrm{Hom}_R(M, N)_P \to 0$ is exact for all $P \trianglelefteq R$ maximal.

Lemma 16 $\implies$ $(\mathrm{Hom}_R(M, N'))_P \cong \mathrm{Hom}_{R_P}(M_P, N'_P)$ and $(\mathrm{Hom}_R(M, N))_P \cong \mathrm{Hom}_{R_P}(M_P, N_P)$

$\mathrm{Hom}_{R_P}(M_P, N'_P) \to \mathrm{Hom}_{R_P}(M_P, N_P) \to 0$ is exact since $M_P$ is free for all $P$ (hence projective). So the sequence $\mathrm{Hom}_R(M, N')_P \to \mathrm{Hom}_R(M, N)_P \to 0$ is exact. □

**Corollary 17.** *Every f.p. flat module is projective.*

*Proof.* Flatness is local but locally flat is locally free for f.p. modules (by Corollary 7 or the proof of Theorem 12). □

**Note.** Not every finitely generated flat module is projective.

## 1.5 Chain Conditions

Let $R$ be commutative with 1.

### 1.5.1 Noetherian modules

**Definition.** An $R$-module is **Noetherian** if every submodule is finitely generated.

**Note.**

1. $R$ is a noetherian ring $\iff$ $R$ is noetherian as an $R$-module
2. Noetherian modules arise naturally and behave well under constructions
3. By definition, noetherian modules are finitely generated

**Theorem 1.** *TFAE ($R$ ring, $M$ $R$-module):*

1. *$M$ Noetherian.*

2. *Every ascending chain $N_1 \leq N_2 \leq \ldots$ of $R$-submodules of $M$ is eventually constant (ACC).*

3. *Every nonempty collection of submodules has a maximal element.*

**Theorem 2.** *Let $R$ be a ring, $M$ an $R$-module, $N \subseteq M$ a submodule. Then $M$ is Noetherian $\iff$ $N$ and $M/N$ are Noetherian.*

*Proof.* ($\Rightarrow$) $M$ Noetherian $\implies$ $N$ Noetherian (by def).
$\overline{L} \leq M/N$, $\overline{L} = L/N$ for some $L \leq M$.
$L$ is f.g. so $\overline{L}$ is f.g.
($\Leftarrow$) Suppose $M/N, N$ are Noetherian. Let $L \leq M$. Then $L \cap N$ is f.g. and $L/N$ is f.g. $\implies$ $L$ is finitely generated (by generators of $L \cap N$ and preimage of generators of $L/N$). ☐

**Theorem 3.** *$M, N$ noetherian $\implies$ $M \oplus N$ is noetherian.*

*Proof.* $(M \oplus N)/(M \oplus 0) \cong N$ and use Theorem 2. ☐

**Theorem 4.** *$R$ Noetherian ring, $M$ $R$-module:*
*$M$ is Noetherian $\iff$ $M$ is finitely generated.*

*Proof.* ($\Rightarrow$) by def. ($\Leftarrow$): $M$ f.g. $\implies$ $R^k \to M \to 0$ for some $k$.

$R$ Noetherian $\implies$ $R^k$ Noetherian by Theorem 3 $\implies$ $M$ Noetherian by Theorem 2. ☐

**Theorem 5.** *$R$ Noetherian, $M, N$ finitely generated $\implies$ $Hom_R(M, N)$ is finitely generated.*

*Proof.* $M$ finitely generated $\implies$ $M \cong R^k/L$ for some $k, L \leq R^k$.
$\mathrm{Hom}_R(M/N) \cong \mathrm{Hom}_R(R^k/L, N)$.

We have the short exact sequence $0 \to L \to R^k \to M \to 0$ so we get the exact sequence

$$0 \to \mathrm{Hom}_R(M, N) \to \mathrm{Hom}_R(R^k, N) \cong N^k$$

$N^k$ is Noetherian by Theorem 3 and 4. So $\mathrm{Hom}_R(M, N)$ is noetherian, by Theorem 2. ☐

**Example.** $R$ ring, $I \trianglelefteq R$ ideal. $\mathrm{Hom}_R(R/I, R) \cong \{r \in R \mid Ir = 0\}$

e.g. $K$ field, $R = K[x_1, x_2, \ldots]/(\ldots, x_i x_j, \ldots)$

$I \trianglelefteq R$ coset of polynomials with no constant term

$If = 0$ $\iff$ $f$ has no constant term $\iff$ $f \in I$. Then $\mathrm{Hom}(R/I, R) \cong I$ is not finitely generated.

**Corollary 6.** 1. *$R$ noetherian, $M, N$ noetherian $\implies$ $Hom_R(M, N)$ is noetherian*

2. *$R$ noetherian, $M$ finitely generated $\implies$ $M^* = Hom_R(M, R)$ the set of linear forms is finitely generated*

**Lemma 7.** *If $R$ is noetherian, $M$ is finitely generated $\iff$ $M$ is finitely presented.*

*Proof.* $(\Leftarrow)$ : clear.

$(\Rightarrow)$ : $M$ finitely generated $\implies$ we get $R^k \xrightarrow{\varphi} M \to 0$ exact.

$\ker\varphi \leq R^k$ (and $R^k$ is noetherian by Theorem 3) $\implies$ $\ker\varphi$ is finitely generated $\implies$ $M$ is finitely presented. $\square$

**Corollary 8.** *$M$ is finitely generated over $R$ noetherian. Then TFAE:*

1. *$M$ is projective*
2. *$M$ is locally free*
3. *$M$ is flat*

*Proof.* (1) $\Rightarrow$ (2): by Theorem 1.2

(2) $\Rightarrow$ (3): $M$ is locally free $\implies$ $M$ is locally flat (free modules are flat) $\implies$ $M$ is flat (flatness is local).

(3) $\Rightarrow$ (1): Lemma 7 $\implies$ $M$ is finitely presented. Finitely presented and flat $\implies$ projective (Corollary 7). $\square$

### 1.5.2 Artinian modules

**Definition.** An $R$-module $M$ is **Artinian** if it satisfies the descending chain condition (DCC):
For every chain of $R$-submodules, $N_1 \geq N_2 \geq \dots$ is eventually constant.

**Example.**

1. $R$ field $\implies$ every finite-dimensional vector space over $R$ is an artinian $R$-module (also noetherian)
2. $R = K[x]$ is not artinian as an $R$-module: $(x) \supsetneq (x^2) \supsetneq \cdots$

**Proposition 9.** *If $0 \to M' \to M \to M'' \to 0$ is a short exact sequence of $R$-modules, then: $M$ is artinian $\iff$ $M', M''$ are artinian.*

**Proposition 10.** *If $R$ artinian ring (i.e. $R$ is artinian as an $R$-module), then if $M$ is finitely generated, $M$ is an artinian $R$-module.*

*Proof.* **Exercise** $\square$

**Theorem 11.** *$R$ ring, $M$ $R$-module:*
*$M$ is Artinian $\iff$ every nonempty collection of $R$-submodules of $M$ has a minimal element.*

*Proof.* $(\Leftarrow)$ : Given a chain $N_1 \geq N_2 \geq \cdots$,
$\{N_k\}$ has a minimal element. So the chain stabilizes at that minimal element.

$(\Rightarrow)$ : Given a nonempty collection of submodules $\{N_k\}$, if there is no minimal element, then for eah $N_k$ there exists some $N_{k+1}$ with $N_k \geq N_{k+1}$. Continuing, we'd get an infinitely ascending chain $\rightarrow\leftarrow$ . $\square$

**Example.**

1. An integral domain is artinian $\iff$ it is a field
2. $\mathbb{Z}/n\mathbb{Z}$ is artinian
3. $K[x]/(x^n)$ is artinian (as a $K[x]$-module)

### 1.5.3 Normal series

**Definition.** A **normal series** of an $R$-module $M$ is a chain $M = M_0 \geq M_1 \geq \cdots \geq M_k = 0$.

The **length** of the series is the number of strict inclusions (i.e. the number of nontrivial quotients)

A **refinement** of a normal series is obtained by inserting a finite number of additional submodules (proper if length increases)

Two series are equivalent if there is a 1-1 correspondence between the nontrivial factors so that the corresponding factors are isomorphic as modules

A **composition series** is a normal series where each $M_i/M_{i+1}$ is nonzero and simple.

**Definition.** $\ell(M) :=$ smallest length of a composition series (or $\infty$) $=$ **length** of $M$.

**Proposition 12.** *$N \leq M \Rightarrow \ell(N) \leq \ell(M)$.*
*If $\ell(M) < \infty$, then equality holds $\iff$ $N = M$.*

*Proof.* WLOG $\ell(M) < \infty$. Let $(M_i)$ be a composition series of length $\ell(M)$.

Define $N_i := N \cap M_i$. Then $N_i/N_{i+1} \leq M_i/M_{i+1}$ is simple $\implies$ $N_i/N_{i+1} = \begin{cases} M_i/M_{i+1} \text{ simple} \\ 0 \end{cases}$
$\implies$ $(N_i)$ is a composition series of $N$ (after possibly removing repeated terms)
$\implies$ $\ell(N) \leq \ell(M)$.

If $\ell(M) = \ell(N)$, $N_i/N_{i+1} \cong M_i/M_{i+1}$ for all $i$. So $M_{n-1} = N_{n-1} \implies$ (inductively) $M_i = N_i$ for all $i \implies M = M_0 = N_0 = N$. $\square$

**Corollary 13.** *Any normal series of $M$ has length at most $\ell(M)$ (if $\ell(M) < \infty$).*

*Proof.* $M = M_0 \geq M_1 \geq \cdots \geq M_k. = 0$ normal series.

Without loss of generality assume this series is length $k$. By Prop 12, $\ell(M) = \ell(M_0) > \ell(M_1) > \cdots > \ell(M_n) = 0 \implies \ell(M) \geq k$. □

**Corollary 14.** *If $\ell(M) < \infty$, all composition series have the same length, and every normal series can be refined to a composition series.*

*Proof.* If $(M_i)$ is a composition series of $M$ of length $k$, then $k \leq \ell(M)$ by Corollary 13, so $k = \ell(M)$ by definition. And the last assertion is clear. □

**Theorem 15.** *$M$ has a composition series $\iff$ $M$ is both artinian and noetherian.*

*Proof.* $(\Rightarrow)$: Corollary 13.
$(\Leftarrow)$: $M = M_0$ has a maximal submodule $M_1 \subsetneq M_0$ since $M$ is noetherian. Inductively, we get $M = M_0 \supsetneq M_1 \supsetneq \cdots$ which has to be finite since $M$ is artinian, so we get a composition series. □

**Proposition 16.** *$\ell(M)$ is additive on exact sequences. In particular, for the short exact sequence $0 \to M' \to M \to M'' \to 0$, $\ell(M) = \ell(M') + \ell(M'')$ when $\ell(M) < \infty$.*

*Proof.* **Exercise**. □

**Proposition 17.** *For a $K$-vector space $V$, TFAE:*

1. *$V$ has finite dimension*
2. *$V$ has finite length*
3. *$V$ is Noetherian*
4. *$V$ is Artinian*

*Proof.* $(1) \Rightarrow (2)$: $\ell(V) \leq \dim V$.

$(2) \Rightarrow (3), (4)$: by Theorem 15.

$(3), (4) \Rightarrow (1)$:
Suppose $\dim V = \infty$. Let $(x_n)_{n \geq 0}$ be an infinite sequence of elements that are linearly independent.
Let $U_n := \langle x_1, \ldots, x_n \rangle$.
$V_n := \langle x_{n+1}, \ldots \rangle$.
Then $U_n$ is strictly increasing and $V_n$ is strictly decreasing. So $V$ is neither noetherian nor artinian. □

**Corollary 18.** *If $R$ is a ring such that $0 = m_1 \cdots m_k$ for some maximal ideals $m_1, \ldots, m_k$ then $R$ is noetherian $\iff$ $R$ is artinian.*

*Proof.* **Exercise** □

### 1.5.4 Krull's Intersection Theorem

**Theorem 19.** *Let $R$ be a Noetherian ring and $I \trianglelefteq R$ an ideal. Let $M$ be a finitely generated $R$-module. Let $L = \bigcap_{n=1}^{\infty} I^n M$. Then $IL = L$.*

*Proof.* Since $M$ is finitely generated and $R$ is noetherian, by Theorem 4, $M$ is noetherian.

Let $S := \{N \leq M \mid N \cap L = IL\} \neq \varnothing$ because $IL \in S$. So Theorem 1 $\implies \exists N_0 \in S$ maximal.

Claim: $I^m N \subseteq N_0$ for sufficiently large $m$.
Proof of claim:
Fix $a \in I$. Define: $N_i := \{x\ inM \mid a^i x \in N_0\}$.
$M$ noetherian $\implies N_0 \geq N_1 \geq \cdots$ stabilizes, i.e. there exists $k$ such that $N_k = N_{k+1} = \cdots$

Let $z \in (a^k M + N_0) \cap L$. Write $z = a^k x + y, x \in M, y \in N_0$. Then $az \in IL \leq N_0$.
$az = a^{k+1}x + ay \implies$ since $ax\mathbb{Z} \in N_0$ and $ay \in N_0$, $a^{k+1}x \in N_0$. So $z \in N_0 \cap L = IL$.

Thus, $(a^k M + N_0) \cap L = IL$.
By maximality of $N_0$, $a^k M \leq N_0$. But $a \in I$ arbitrary and $I$ finitely generated $\implies I^m M \leq N_0$ for $m$ large. Hence

$$L = \bigcap_{n \geq 1} I^n M \subseteq I^m M \leq N_0$$

for large enough $m$. Hence $L = N_0 \cap L = IL$. $\square$

**Example.** Let $R = \mathbb{Q}[x, z, y_1, y_2, \ldots]/(x - zy_1, z - z^2 y_2, x - z^3 y_3, \ldots)$.
Let $M = R$ and $I = (\overline{z})$.

Then $\bigcap_{n \geq 1} I^n M = (\overline{x})$ and $\overline{z} \cdot (\overline{x}) \neq \overline{x}$. So the example violates the theorem.

**Corollary 20.** *If $R$ is Noetherian and $M$ is a finitely generated $R$-module,*

$$J \subseteq \mathrm{Jac} R = \bigcap_{P \trianglelefteq R, P \text{ maximal}} P \implies \bigcap_{n \geq 1} J^n M = 0.$$

*In particular, if $R$ is a Noetherian local ring with maximal ideal $\mathfrak{m}$, then $\bigcap_{n=1}^{\infty} \mathfrak{m}^n = 0$.*

*Proof.* Let $L = \bigcap_{n \geq 0} J^n M$. By Theorem 19, $JL = L$. So by Nayakama, there exists $a \in J$ so that $(1 - a)x = 0$ for all $x \in L$.

But $a \in \mathrm{Jac} R \implies 1 - a \in R^{\times} \implies L = 0$. $\square$

## 1.6 Homological Algebra

### 1.6.1 Exact functors and derived functors

**Definition.** A functor between abelian categories is **additive** if it induces a homomorphism on the Hom groups.

An additive functor $F : \mathcal{C} \to \mathcal{D}$ between abelian categories is **left exact** if for any exact sequence $0 \to X \to Y \to Z$ the sequence $0 \to F(X) \to F(Y) \to F(Z)$ is also exact and **right exact** if for any exact sequence $X \to Y \to Z \to 0$, the sequence $F(X) \to F(Y) \to F(Z) \to 0$ is also exact.

$F$ is **exact** if $F$ is both left and right exact.

**Example.** Let $R$ be commutative with 1 and $M$ an $R$-module.

1. $M \otimes_R -$ is right exact
2. $M \otimes_R -$ is exact if $M$ is flat
3. $\mathrm{Hom}_R(M, -)$ is left exact

**Definition.** A sequence $\{X_i\}_{i \in \mathbb{N}}$ of $R$-modules with homomorphisms $X_{n+1} \to X_n$ for all $n$ together with an **augmentation map** $\varepsilon_0 : X_0 \to M$ ($\varepsilon_0 \in \mathrm{Hom}_R(X_0, M)$) is a **left resolution** of $M$ if

$$\cdots \to X_{n+1} \to X_n \to \cdots \to X_0 \xrightarrow{\varepsilon_0} M \to 0$$

is exact.

Notation: $X_\bullet \to M \to 0$.

A **right resolution** of $M$ is a sequence of $R$-modules $\{Y^i\}_{i \in \mathbb{N}}$ with homomorphisms $Y^n \to Y^{n+1}$ for all $n$ together with $\varepsilon_0 \in \mathrm{Hom}_R(M, Y^0)$ so that

$$0 \to M \to Y^0 \to Y^1 \to \cdots$$

is exact. Notation: $0 \to M \to Y^\bullet$

A **projective resolution** is a left resolution with $X_i$ projective for all $i$.

A **injective resolution** is a right resolution with $Y^i$ injective for all $i$.

**Theorem 1.** *Let $M$ be an $R$-module.*

1. *$M$ admits a projective resolution*
2. *$M$ admits an injective resolution*

Now let $X_\bullet \to M \to 0$ be a projective resolution of $M$. Let $F$ be an additive functor. Take $X_\bullet \to 0$ which is exact except at 0.

$$H_0(X_\bullet) = X_0/\mathrm{Im}(X_1 \to X_0) = X_0/\ker(X_0 \to M) \cong M \quad \text{and } H_n(X_\bullet) = 0 \text{ for all } n \geq 1.$$

We get a new complex $FX_\bullet$ by applying $F$.

Define $(L_n F)(M) := H_n(FX_\bullet)$.

**Proposition 2.** *$(L_0F)(M) = F(M)$ if $F$ is right exact.*

*Proof.* Given
$$\cdots \to X_1 \to X_0 \to M \to 0 \text{ exact}$$
we get
$$FX_1 \to FX_0 \to FM \to o \text{ exact}$$
So $\mathrm{Im}(FX_1 \to FX_0) = \ker(FX_0 \to FM) \implies$

$$H_0(FX_\bullet) = \ker(FX_0 \to 0)/\mathrm{Im}(FX_1 \to FX_0) \cong \ker(FX_0 \to 0)/\ker(FX_0 \to FM) \cong F(M).$$

□

**Definition.** Let $C_\bullet, D_\bullet$ be two (chain) complexes of modules. A homomorphism $f : C_\bullet \to D_\bullet$ is a collection of homomorphisms $f_n : C_n \to D_n$ such that the following diagram commutes:

$$\begin{tikzcd}
\dots \arrow[r] & C_n \arrow[r, "d_n"] \arrow[d, "f_n"] & C_{n-1} \arrow[r] \arrow[d, "f_{n-1}"] & \dots \\
\dots \arrow[r] & D_n \arrow[r, "d_n"] & D_{n-1} \arrow[r] & \dots
\end{tikzcd}$$

**Definition.** Two homomorphisms $f, g : C_\bullet \to D_\bullet$ are **chain homotopic** if there exist $s_n \in \mathrm{Hom}(C_n, D_{n+1})$ for all $n$ such that $f_n - g_n = d_{n+1}s_n + s_{n-1}d_n$.

$$\begin{tikzcd}
\dots \arrow[r] & C_{n+1} \arrow[r, "d_{n+1}"] \arrow[d, "(f-g)_{n+1}"'] & C_n \arrow[r, "d_n"] \arrow[d, "(f-g)_n"'] \arrow[dl, "s_n"'] & C_{n-1} \arrow[r] \arrow[d, "(f-g)_{n-1}"] \arrow[dl, "s_{n-1}"'] & \dots \\
\dots \arrow[r] & D_{n+1} \arrow[r, "d_{n+1}"] & D_n \arrow[r, "d_n"] & D_{n-1} \arrow[r] & \dots
\end{tikzcd}$$

$s = \{s_n\}$ is the **chain homotopy** from $f$ to $g$.

**Note.** Chain homotopy defines an equivalence relation.

**Definition.** $f : C_\bullet \to D_\bullet$ is a **chain homotopy equivalence** if there exists $g : D_\bullet \to C_\bullet$ such that $fg$ and $gf$ are both homotopic to the identity.

**Proposition 3.** *If $f, g$ are chain homotopic, then $H_n(f) = H_n(g)$ for all $n$.*

**Theorem 4** (Comparison Theorem for Resolutions)**.** *If $X_\bullet \to M \to 0$ is a projective resolution, $N$ is an $R$-module, $f_{-1} \in \mathrm{Hom}_R(M, N)$ and $Y_\bullet \to N \to 0$ is a left resolution, then there exists $\eta : (X_\bullet \to M \to 0) \to (Y_\bullet \to N \to 0)$ extending $f_{-1}$ and $\eta$ is unique up to chain homotopy.*

$$\begin{tikzcd}
\cdots \arrow[r] & X_1 \arrow[r] \arrow[d, dashed] & X_0 \arrow[r] \arrow[d, dashed] & M \arrow[r] \arrow[d, "f_{-1}"] & 0 \\
\cdots \arrow[r] & Y_1 \arrow[r] & Y_0 \arrow[r] & N \arrow[r] & 0
\end{tikzcd}$$

Now let $X_\bullet \to M \to 0$ and $X'_\bullet \to M \to 0$ be projective resolutions. Using the theorem, we get

$$\begin{tikzcd}
\cdots \arrow[r] & X_1 \arrow[r] \arrow[d, "\eta_1"] & X_0 \arrow[r] \arrow[d, "\eta_2"] & M \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \\
\cdots \arrow[r] & X'_1 \arrow[r] & X'_0 \arrow[r] & M \arrow[r] & 0
\end{tikzcd}$$

But we can also use it the other way around and get

$$\begin{tikzcd}
\cdots \arrow[r] & X_1 \arrow[r] & X_0 \arrow[r] & M \arrow[r] & 0 \\
\cdots \arrow[r] & X'_1 \arrow[r] \arrow[u, "\eta'_1"] & X'_0 \arrow[r] \arrow[u, "\eta'_2"] & M \arrow[r] \arrow[u, "\mathrm{id}"] & 0
\end{tikzcd}$$

So we get $\eta : (X_\bullet \to M \to 0) \to (X'_\bullet \to M \to 0)$ and $\eta' : (X'_\bullet \to M \to 0) \to (X_\bullet \to M \to 0)$.

$\eta \circ \eta'$ and $\eta' \circ \eta$ are chain homotopic to the identity by the comparison theorem. So $\eta$ is a chain homotopy equivalence. Exercise: $f(\eta)$ is also a chain homotopy equivalence. So by Proposition 3, $H_n(f(\eta)) : H_n(FX_\bullet) = (L_nF)(M) \to H_n(FX'_\bullet)$ is an isomorphism. From this we conclude that different projective resolutions give isomorphic modules $(L_nF)(M)$. So $L_nF$ is well-defined, called the **n-th left derived functor of** $F$.

**Example.** Let $M$ be projective and $F$ right exact.

$\cdots \to M \xrightarrow{\mathrm{id}} M \xrightarrow{0} M \xrightarrow{\mathrm{id}} M \to 0$ is a projective resoluton.

We get $FM \xrightarrow{\mathrm{id}} FM \xrightarrow{0} FM \to 0$ so

$$L_n(F)(M) = H_n(FX_\bullet) = \begin{cases} FM, & n = 0 \\ 0, & n > 0 \end{cases}$$

**Theorem 5.** *Let $0 \to M_1 \to M_2 \to M_3 \to 0$ be a short exact sequence of $R$-modules. Let $F$ be a right exact functor of $R$-modules. Then there is a long exact sequence*

$$\cdots \to (L_2F)(M_3) \xrightarrow{\partial} (L_1F)(M_1) \to (L_1F)(M_2) \to (L_1F)(M_3) \xrightarrow{\partial} FM_1 \to FM_2 \to FM_3 \to 0$$

*This is functorial in the following sense:*
*If $0 \to N_1 \to N_2 \to N_3 \to 0$ is a short exact sequence with homomorphisms $N_i \to M_i$ such*

*that*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M_1 & \longrightarrow & M_2 & \longrightarrow & M_3 & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & N_1 & \longrightarrow & N_2 & \longrightarrow & N_3 & \longrightarrow & 0 \end{array}$$

*commutes, then there is an induced commutative diagram*

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & (L_2F)(M_3) & \xrightarrow{\partial} & (L_1F)(M_1) & \longrightarrow & (L_1F)(M_2) & \to & \cdots \\ & & \downarrow & & \downarrow & & \downarrow & & \\ \cdots & \longrightarrow & (L_2F)(N_3) & \xrightarrow{\partial} & (L_1F)(N_1) & \longrightarrow & (L_1F)(N_2) & \to & \cdots \end{array}$$

**Note.**

1. The sequence is called the long exact homology sequence associated to this short exact sequence (and $F$)

2. $\partial$ are called connecting homomorphisms (from the snake lemma)

Dually: if $F$ is a left exact functor on $R$-modules, define right derived functor $R^nF$ as follows:
Choose $0 \to M \to Y^\bullet$ injective resolution.
Take $(R^nF)(M) = H^n(FY^\bullet)$ where for

$$0 \to Y^0 \xrightarrow{d^0} Y^1 \xrightarrow{d^1} \cdots, \quad H^i(Y^\bullet) := \ker(d^i)/\mathrm{Im}(d^{i-1})$$

And one can show:

- $(R^0F)(M) = F(M)$
- $R^nF$ is well-defined
- The analog of Theorem 5 holds

### 1.6.2 Tor and Ext

Let $M$ be an $R$-module, $\mathrm{Mod}_R$ = the category of $R$-modules, $F : \mathrm{Mod}_R \to \mathrm{Mod}_R, N \mapsto M \otimes_R N$.

We saw (3.7) that $F$ is right exact so has left derived functors $L_nF$.

**Definition.** $\mathrm{Tor}_n(M, N) := (L_nF)(N)$ is called the Tor functor

**Theorem 6** (Balancing Tor)**.** *Say $M, N$ are $R$-modules and $n \geq 0$. Then* $\mathrm{Tor}_n(M, N)$ *is naturally isomorphic to* $\mathrm{Tor}_n(N, M)$.

*Proof.* Theorem 2.7.2 in Weibel. □

**Example.** $\mathrm{Tor}_i(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}) =?$ (as $\mathbb{Z}$-modules)

Projective resolution of $\mathbb{Z}/n\mathbb{Z}$ as a $\mathbb{Z}$-module:

$$0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \xrightarrow{\text{mod } n} \mathbb{Z}/n\mathbb{Z} \to 0$$

Tensor with $\mathbb{Z}/m\mathbb{Z}$ and remove the last term:

$$0 \to \mathbb{Z} \otimes \mathbb{Z}/m\mathbb{Z} \to \mathbb{Z} \otimes \mathbb{Z}/m\mathbb{Z} \to 0$$

Recall that $\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/m\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z}$.

So $\mathrm{Tor}_0(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}) = \mathbb{Z}/m\mathbb{Z} \otimes \mathbb{Z}/n\mathbb{Z} = \mathbb{Z}/d\mathbb{Z}$ where $d = \gcd(m, n)$.

$\mathrm{Tor}_1(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}) = \ker(\cdot n) = \{i \in \mathbb{Z}/m\mathbb{Z} \mid ni = 0\}$.
If $m = dm', ni = 0 \iff m \mid ni \iff m'd \mid ni \iff m' \mid i$. So $\ker(\cdot n) \cong m'\mathbb{Z}/m\mathbb{Z} \cong \mathbb{Z}/d\mathbb{Z}$.

$\mathrm{Tor}_i(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}) = 0$ for $i > 1$.

**Definition.** Let $M$ be an $R$-module, $F : \mathrm{Mod}_R \to \mathrm{Mod}_R, N \mapsto \mathrm{Hom}_R(M, N)$.
This is left exact (Theorem 3.1).
Define $\mathrm{Ext}^n(M, N) = (R^n F)(M)$.
Note we can define this alternatively but it is not symmetric.

**Theorem 7.** *Let $P$ be an $R$-module. Then*

1. *$P$ is projective $\iff$ $\mathrm{Ext}^1_R(P, M) = 0$ for all $R$-modules $M$*
2. *$P$ is injective $\iff$ $\mathrm{Ext}^1_R(M, P) = 0$ for all $R$-modules $M$*

*Proof.* We'll prove (2).
($\Rightarrow$): $P$ injective $\iff$ $\mathrm{Hom}_R(-, P)$ is exact $\implies$ $\mathrm{Ext}^1_R(M, P) = 0$ for all $M$.
($\Leftarrow$): $0 \to M_1 \to M_2 \to M_3 \to 0$ short exact sequence $\implies$ we get the long exact sequence
$\cdots \to 0 \to \mathrm{Hom}(M_3, P) \to \mathrm{Hom}(M_2, P) \to \mathrm{Hom}(M_1, P) \xrightarrow{\partial} \mathrm{Ext}^1_R(M_3, P) \to \cdots$
and $\mathrm{Ext}^1_R(M_3, P) = 0$ by assumption, so we get a short exact sequence, hence $P$ is injective.

(1) is similar. $\square$

### 1.6.3 Ext and extensions

**Definition.** Let $M, N$ $R$-modules. An **extension of $M$ by $N$** is a short exact sequence of $R$-modules

$$0 \to N \to E \to M \to 0$$

A **map of extensions** is a commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & N \arrow[r] \arrow[d, "\varphi_1"] & E \arrow[r] \arrow[d, "\varphi_2"] & M \arrow[r] \arrow[d, "\varphi_3"] & 0 \\ 0 \arrow[r] & N' \arrow[r] & E' \arrow[r] & M' \arrow[r] & 0 \end{tikzcd}$$

where $\varphi_i$ is a commutative diagram.

A map of extensions is an **equivalence** if $\varphi_1$ and $\varphi_3$ are the identity ($N = N', M = M'$)

$\operatorname{Ext}(M, N)$ is a set of extensions up to equivalence

**Facts:**

1. $\operatorname{Ext}(M, -)$ is a covariant functor and $\operatorname{Ext}(-, N)$ is a contravariant functor
2. $\operatorname{Ext}(M, N)$ is a group via **Baer sum**

$$0 \to N \to E \to M \to 0$$

$$0 \to N \to E' \to M \to 0$$

$$\begin{tikzcd}
0 \arrow[r] & N \oplus N \arrow[r] \arrow[d, "+"] & E \oplus E' \arrow[r] \arrow[d] & M \oplus M \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & N \arrow[r] \arrow[d, equal] & \tilde{E} \arrow[r] & M \oplus M \arrow[r] & 0 \\
0 \arrow[r] & N \arrow[r] & E + E' \arrow[r] & M \arrow[r] \arrow[u, "\triangle"] & 0
\end{tikzcd}$$

3. $\operatorname{Ext}(M, N)$ is naturally isomorphic to $\operatorname{Ext}^1(M, N)$.

### 1.6.4 Applications of Tor

**Theorem 8.** *If $M$ is an $R$-module, $M$ is flat $\iff$ for all ideals $I \trianglelefteq R$, $I \otimes_R M \to IM, i \otimes m \mapsto im$ is an isomorphism.*

*Proof.* Note: $I \otimes_R M \to IM$ is always surjective.
$(\Rightarrow)$ : is by definition. $0 \to I \to R$, tensor with $M$.

$(\Leftarrow)$ : $0 \to N' \to N$. WTS: $- \otimes_R M$ preserves this. Note: suffices to show this for $N, N'$ finitely generated (every element is contained in a finitely generated submodule)

Consider a sequence
$$N' = N_0 \leq N_1 \leq \cdots \leq N_k = N$$
where each $N_{i+1} = N_i + R\gamma_i$ (add one generator each time). Then

$$N_{i+1}/N_i \cong R/I_i$$

for some ideal $I_i \trianglelefteq R$. Suffices to show: $N_i \otimes M \to N_{i+1} \to M$ is injective for all $i$.
Theorem 5 $\implies \exists$ exact sequence associated to $0 \to N_i \to N_{i+1} \to N_{i+1}/N_i \to 0$:

$$\operatorname{Tor}_1(M, N_{i+1}/N_i) \to N_i \otimes M \to N_{i+1} \otimes M$$

But $\mathrm{Tor}_1(M, N_{i+1}/N_i) \cong \mathrm{Tor}_1(M, R/I_i) = 0$ by assumption and we also have the exact sequence

$$\mathrm{Tor}_1(M, R/I_i) \to M \otimes I_i \to M \otimes R \cong M$$

and the short exact sequence $0 \to I_i \to R \to R/I_i \to 0$ which proves the claim. □

**Corollary 9.** *$R$ PID, $M$ $R$-module. Then $M$ is flat $\iff$ $M$ is torsion-free.*

*Proof.* $M$ torsion free $\iff$ $M \xrightarrow{\cdot a} M$ is injective for all $a \in R$.

The sequence $0 \to (a) \to R \to R/(a) \to 0$ would then imply $\mathrm{Tor}_1(M, R/(a)) = 0$ for all $a \in R$. Since $R$ is a PID, $\mathrm{Tor}_1(M, R/I) = 0$ for all $I \trianglelefteq R$ and then by Theorem 8, we get that $M$ is flat.

We saw $\Rightarrow$ earlier. □

**Theorem 10** (Homological Criterion for Flatness)**.** *Let $M$ be an $R$-module. TFAE:*

1. *$M$ flat*
2. *$\mathrm{Tor}_i(M, N) = 0$ for all $R$-modules $N$ and all $i > 0$*
3. *$\mathrm{Tor}_1(M, N) = 0$ for all $R$-modules $N$*
4. *$\mathrm{Tor}_1(M, R/I) = 0$ for all $I \trianglelefteq R$*

*Proof.* (2) $\implies$ (3) $\implies$ (4) is clear. And (4) $\implies$ (1) is Theorem 8.
Suffices to show (1) $\implies$ (2).
Take $X_\bullet \to N \to 0$ projective resolution of $N$. Then $\cdots \to X_n \otimes M \to \cdots \to X_0 \otimes M \to 0$ is exact since $M$ is flat. So $\mathrm{Tor}_i(M, N) = 0$ for all $i > 0$. □

### 1.6.5 Limits

**Definition.** A partially ordered set $A$ is **directed** if for all $\alpha, \beta \in A$ there is $\gamma \in A$ so that $\alpha \leq \gamma, \beta \leq \gamma$.

If $\mathcal{C}$ is a category, then a **directed system** in $\mathcal{C}$ consists of a directed partially ordered set $A$ and a collection of objects $\{X_\alpha\}_{\alpha \in A}$ in $\mathcal{C}$ and morhphism $i_{\alpha\beta} : X_\alpha \to X_\beta$ for any $\alpha \neq \beta$ so that

1. $i_{\alpha\alpha} = \mathrm{id}_{X_\alpha}$ for all $\alpha \in A$
2. $i_{\beta\gamma} \circ i_{\alpha\beta} = i_{\alpha\gamma}$ when $\alpha \leq \beta \leq \gamma$

**Definition.** If $\mathcal{C}$ is a category and $(A, \{X_\alpha\}_{\alpha \in A}., \{i_{\alpha\beta}\})$ is a directed system in $\mathcal{C}$, then a **direct limit** if this system is an object $X$ in $\mathcal{C}$ with morphisms $i_\alpha : X_\alpha \to X$ for all $\alpha \in A$ so that the following holds:

1. If $\alpha \leq \beta$ then
$$\begin{tikzcd} X_\alpha \arrow[r, "i_\alpha"] \arrow[d, "i_{\alpha\beta}"] & X \\ X_\beta \arrow[ur, "i_\beta"] & \end{tikzcd}$$
commute
2. UMP: Given $Y \in \mathrm{Obj}\mathcal{C}$ with $\varphi_\alpha : X_\alpha \to Y$ so that
$$\begin{tikzcd} X_\alpha \arrow[r, "\varphi_\alpha"] \arrow[d, "i_{\alpha\beta}"] & X \\ X_\beta \arrow[ur, "\varphi_\beta"] & \end{tikzcd}$$
commutes for all $\alpha \leq \beta$ in $A$, there exists a unique morphism $\varphi : X \to Y$ so that
$$\begin{tikzcd} X_\alpha \arrow[r, "i_\alpha"] \arrow[d, "\varphi_\alpha"] & Y_\alpha \arrow[dl, "\varphi"] \\ Y & \end{tikzcd}$$
commutes for all $\alpha \in A$.

**Note.** If a direct limit exists, it is unique up to isomorphism.
Notation: $\lim_{\xrightarrow{\alpha}} X_\alpha$

**Example.**

1. $\mathcal{C}$ = sets, $\{X_\alpha\}$ chain of subsets of a set $Y$, partially ordered by inclusion
Then $\lim_{\xrightarrow{\alpha}} X_\alpha = \bigcup X_\alpha$.
2. $\mathcal{C}$ = sets, (A, $\{X_\alpha\}_{\alpha\in A}, \{i_{\alpha\beta}\}$) arbitrary direct system,

   Define $\sim$ on $\bigcup X_\alpha$ as follows:
   $a \in X_\alpha, y \in X_\beta$: $x \sim y \iff \exists \gamma \in A$ so that $i_{\alpha\gamma}(x) = i_{\beta\gamma}(y)$.

   This defines an equivalence relation and $\lim_{\xrightarrow{\alpha}} X_\alpha = (\bigcup X_\alpha)/\sim$ .
3. $\mathcal{C}$ = linear algebraic groups over $k$, $X_n = \mathrm{Gl}_n(K)$,
$$\mathrm{Gl}_n(K) \hookrightarrow \mathrm{Gl}_{n+1}(K), A \mapsto \begin{pmatrix} 1 & \\ & A \end{pmatrix}$$
Then $\lim_{\xrightarrow{\alpha}} \mathrm{Gl}_n(K) =: \mathrm{Gl}(K)$.
4. Every $R$-module is the direct limit of its finitely generated submodules

**Definition.** Let $\mathcal{C}$ be a category. An **inverse system** in $\mathcal{C}$ consists of a directed set $A$, a collection of objects $\{X_\alpha\}_{\alpha \in A}$ in $\mathcal{C}$, and morphisms $\pi_{\beta\alpha} : X_\beta \to X_\alpha$ for all $\alpha \leq \beta$ so that

1. $\pi_{\alpha\alpha} = \mathrm{id}_{X_\alpha}$ for all $\alpha \in A$
2. $\pi_{\beta\alpha} \circ \pi_{\gamma\beta} = \pi_{\gamma\alpha}$ for $\alpha \leq \beta \leq \gamma$

An **inverse limit** of an inverse system $(A, \{X_\alpha\}, \{\pi_{\beta\alpha}\})$ is an object $X$ in $\mathcal{C}$ together with morphisms $\pi_\alpha : X \to X_\alpha$ for $a \in A$ such that

1. For $\alpha \leq \beta$,

$$\begin{tikzcd} X \arrow[r, "\pi_\alpha"] \arrow[d, "\pi_\beta"'] & X_\alpha \\ X_\beta \arrow[ur, "\pi_{\beta\alpha}"] & \end{tikzcd}$$

   commutes

2. UMP: Given $Y \in \mathrm{Obj}\mathcal{C}$ together with $\varphi_\alpha : Y \to X_\alpha$ so that

$$\begin{tikzcd} Y \arrow[r, "\varphi_\alpha"] \arrow[d, "\varphi_\beta"'] & X_\alpha \\ X_\beta \arrow[ur, "\pi_{\beta\alpha}"] & \end{tikzcd}$$

   commutes, there exists a unique morphism $\varphi : Y \to X$ so that

$$\begin{tikzcd} Y \arrow[r, "\varphi"] \arrow[d, "\varphi_\alpha"'] & X \arrow[dl, "\pi_\alpha"] \\ X_\alpha & \end{tikzcd}$$

   commutes for all $\alpha \in A$.

**Note.**

1. If it exists, it is unique up to isomorphism. Notation: $\lim_{\overleftarrow{\alpha}} X_\alpha$.
2. Inverse limits exist in many categories (sets, groups, rings, etc.)
   $\lim_{\leftarrow} X_\alpha = \{(X_\alpha) \in \prod X_\alpha \mid \pi_{\beta\alpha}(X_\beta) = X_\alpha \text{ for all } \alpha \leq \beta\}$.

**Example.**

1. $X_n = \mathbb{Z}/p^n\mathbb{Z}, n \geq 0, p$ prime. Then $\{X_n\}$ defines an inverse system over $\mathbb{N}$ with $\pi_{mn} : X_m \to X_n$ the natural epimorphism

   $\lim_{\leftarrow} \mathbb{Z}/p^n\mathbb{Z} = \mathbb{Z}_p$ $p$-adic integers
   $a \in \mathbb{Z}_p \implies a = (a_1, \dots) = b_0 + b_1 p + b_2 p^2 + \cdots,\ b_0 = a_1,\ b_n = \frac{a_{n+1} - a_n}{p^n}$

2. $R$ ring, $I$ ideal
   $R_n := R/I^n, n \geq 0$
   $\{R_n\}_{\mathbb{N}\in\mathbb{N}}$ inverse system over $\mathbb{N}$ with $\pi_{mn} : R_m \to R_n$ natural epimorphism
   $\lim_{\leftarrow} R_n := \hat{R}_I$ is the **$I$-adic completion**

3. Index set $A = \mathbb{N}$, $X_n = \mathbb{N}$ all $n$
   $\pi_{mn} : X_m \to X_n, x \mapsto x + (m - n)$
   Check: $\lim_{\leftarrow} X_n = \varnothing$

4. $G$ group, $\mathcal{N}$ = set of normal subgroups of finite index, ordered by reverse inclusion
   $(\mathcal{N}, \{G/N\}_{N\in\mathcal{N}}, \{\pi_{K,N}\})$ inverse system
   $\pi_{K,N} : G/K \to G/N$ epimorphism for $N \subseteq K$

   $\hat{G} := \lim_{\underleftarrow{\mathcal{N}}} G/N$ is the **profinite completion of $G$**

Some applications:
Can show that $\mathrm{Tor}_1(-, N)$ commutes with direct limits
$\mathrm{Tor}_1^R(\lim_{\rightarrow} M_i, N) = \lim_{\rightarrow} \mathrm{Tor}_1^R(M_i, N) \quad (*)$

**Theorem 11.** *$M = \lim_{\rightarrow} M_i$, $M_i$ flat $R$-modules $\implies$ $M$ is flat.*

*Proof.* (*) $\mathrm{Tor}_1^R(M, N) = \lim_{\rightarrow} \mathrm{Tor}_1^R(M_i, N) = 0$ by homological criterion for flatness
So by the homological criterion again, $M$ is flat. □

**Fact**: $M$ an $R$-module:
$M$ is flat $\iff$ $M$ is the direct limit of finitely generated free submodules.

# 2 Commutative Rings

All rings are commutative with 1.

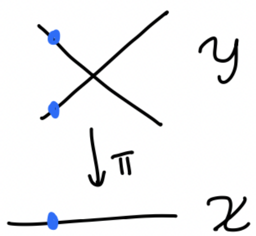
## 2.1 Integral extensions revisited

### 2.1.1 Integrality and finiteness

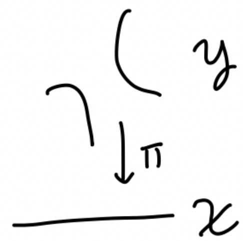
**Definition.** $A \subseteq B$ rings. We say that $B$ is **finite** over $A$ if $B$ is a finitely generated $A$-module.

**Geometric examples.** $A = \mathbb{C}[x]$,
$B = \mathbb{C}[x][y]/(f) = A[y]/(f)$, $f \in A[y] \setminus A$ polynomial
$\mathcal{X} = \mathbb{A}^1$ affine line, $\mathcal{Y}$ = curve associated with $B$
$A \hookrightarrow B$ corresponds to $\pi : \mathcal{Y} \to \mathcal{X}$

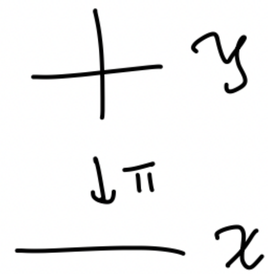
In (a), we have a monic relation so $\overline{y}$ is integral over $A$ (we will soon show that $B$ is integral over $A$)




Case (a): $f = y^2 - x^2$




Case (b): $f = xy - 1$




Case (c): $f = xy$

Geometrically any value of $x$ gives a quadratic equation so we get 2 points in the preimage (when counted with multiplicity)

In (b), the equation is not monic and cannot be rewritten as monic. So $\overline{y}$ is not integral over $A \implies B$ is not integral over $A$
Geometrically, $x \neq 0$ works fine, the relation is linear. But $x = 0$ is constant.
$|\pi^{-1}(x)| = \begin{cases} 1, & x \neq 0 \\ 0, & x = 0 \end{cases}$
(c) is not integral, not linear when $x = 0$, but $|\pi^{-1}(0)| = \infty$

What we should expect: integral extension $\to$ morphism which is surjective with finite fibers

**Proposition 1.** *$B/A$ ring extension:*
*$B$ is finite over $A \iff B = A[b_1, \ldots, b_n]$ for some $b_i \in B$ which are integral over $A$*

*Proof.* $(\Rightarrow) : B = \sum_{i=1}^n Ab_i$ finitely generated $A$-module $\implies B = A[b_1, \ldots, b_n]$.
Let $b \in B$. Consider $\varphi : B \to B, x \mapsto xb$. By Cayley-Hamilton, there exists $\varphi^{+} c_{k-1}\varphi^{k-1} + \cdots + c_0 \cdot \mathrm{id} = 0 \in \mathrm{Hom}_A(B, B), c_i \in A$.

$b = \varphi(1) \implies$ plug in 1 : get $\varphi(1)^k + c_{k-1}\varphi(1)^{k-1} + \cdots + c_0 = 0 \implies b^k + c_{k-1}b^{k-1} + \cdots + c_0 = 0 \implies b$ is integral over $A$.

$(\Leftarrow)$: $B = A[b_1, \ldots, b_n]$ with $b_i$ integral over $A \implies$ for $b \in B$, $b = \sum c_I b^I$, $I = 9i_1, \ldots, i_n), C_I \in A, b^I = b^{i_1} \cdots b^{i_n}$.

The $b_i$ satisfy monic relations of degree $r_i \implies I = (i_1, \ldots, i_n) \leq (r_1, \ldots, r_n)$ for all $I \implies B$ is a finitely-generated $A$-module. $\square$

**Proposition 2.** *$A \leq B \leq C$ rings,*

1. *$A \leq B, B \leq C$ finite $\implies A \leq C$ is finite*
2. *$A \leq B, B \leq C$ integral $\implies A \leq C$ is integral*

*Proof.* 1. clear

2. $c \in C \implies c^n + b_{n-1}c^{n-1} + \cdots + b_0 = 0$ some $b_i \in B \implies c$ is integral over $A[b_0, \dots, b_{n-1}] =: \tilde{A} \implies$ by (1), $\tilde{A}$ over $A$ is finite $\implies \tilde{A}[c]$ over $A$ is finite $\implies$ by Prop 1, $c$ is integral over $A$.

$\square$

**Proposition 3.** *$A \leq B$ integral*

1. *$I \trianglelefteq B \implies B/I$ is integral over $A/A \cap I$*

2. *If $S \subseteq A$ is multiplicatively closed, $S^{-1}B$ over $S^{-1}A$ is integral*

3. *$A[x] \leq B[x]$ is integral*

*Proof.* exercise $\square$

**Example.** $A = k[\mathcal{X}]$ coordinate ring of some irreducible affine algebraic set $\mathcal{X}$
$k(\mathcal{X})$ fraction field of $A$

What does it mean for $A$ to be integrally closed?
$f \in k[\mathcal{X}] \implies f = \frac{f_1}{f_2}, f_i \in A$ well-defined except at $\{x \in \mathcal{X} \mid f_2(x) = 0\}$
$k[\mathcal{X}]$ integrally closed $\iff$ every rational function that satisfies a monic relation over $k[\mathcal{X}]$ is well-defined on all of $\mathcal{X}$

1. $A = \mathbb{C}[x]$, $\mathcal{X} = \mathbb{A}^1_{\mathbb{C}}$
   $A$ is a UFD, so is integrally closed
   Geometrically, if $f \in K(\mathcal{X})$ is not well-defined on all of $\mathcal{X}$ it has a pole, so cannot be integral over $k[\mathcal{X}]$

2. $\mathcal{X} = V(y^2 - x^2 - x^3) \subseteq \mathbb{A}^2_{\mathbb{R}}$
   $A = k[\mathcal{X}] = \mathbb{R}[x, y]/(y^2 - x^2 - x^3)$

   $f = \frac{y}{x} \in k(\mathcal{X})$, $(\frac{y}{x})^2 - x - 1 = 0$ is a monic relation over $A$
   So $A$ is not integrally closed.

   Think of not being integrally closed as related to singularities

3. $A$ as in (2)
   $L = \mathrm{Quot}(A)$
   $t = \frac{y}{x} \in L$. One may show:

$$A^c = A[t] = \left(\mathbb{R}[x, y]/(y^2 - x^2 - x^3)\right)\left[\frac{y}{x}\right] \cong \mathbb{R}[t]$$

   $t^2 = (\frac{y}{x})^2 = 1 + x, \quad (t^2 - 1)t = y$
   So $A = \mathbb{R}[t^2 - 1, t^3 - t]$
   $A \hookrightarrow A^c$ corresponds to $a \mapsto (a^2 - 1, a^3 - a), \{\pm 1\} \mapsto (0, 0)$ (think of wrapping the interval around in a loop)

Goal: Study prime ideals in integral extensions

### 2.1.2 Lying over and incomparability

Recall: $A \leq B$ rings, $P \trianglelefteq A \implies P^e = PB \trianglelefteq B$
$Q \trianglelefteq B \implies Q^c = Q \cap A \trianglelefteq A$
$Q$ prime $\implies Q \cap A$ prime

**Proposition 4** (Lying Over). *$A \leq B, P \trianglelefteq A$ prime*

1. *$P = Q \cap A$ for some $Q \trianglelefteq B$ prime $\iff PB \cap A \leq P$*
   *In this case, we say $Q$ "lies over" $P$*

2. *If $A \leq B$ is integral, this is always the case ($Q$ always lies over $P$)*

*Proof.*

1. Proved last semester

2. $f \in PB \cap A \implies f = p_1 b_1 + \cdots + p_k b_k$, $b_i \in B$, $p_i \in P \implies f \in PB' \cap A$ where $B' = A[b_1, \ldots, b_k]$, which is finite over $A$ by Proposition 1.

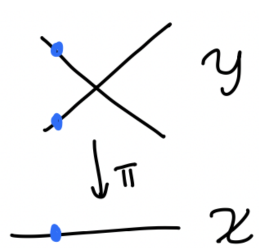
   So $B'$ is a finitely generated $A$-module. Consider: $\varphi : B' \to B', b \mapsto fb \implies \mathrm{Im}(\varphi) \subseteq PB'$

   (Strong version of) Cayley-Hamilton $\implies \varphi^n + a_{n-1}\varphi^{n-1} + \cdots + a_o \cdot \mathrm{id} = 0 \in \mathrm{Hom}_A(B', B')$, coefficients $a_i \in P$
   $\implies \varphi(1)^n + a_{n-1}\varphi(1)^{n-1} + \cdots + a_0 = 0.$

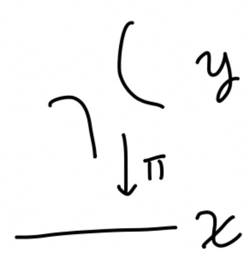
   $f^n + a_{n-1} f^{n-1} + \cdots + a_0 = 0 \implies f^n \in P \implies f \in P$, since $P$ is prime.

$\square$

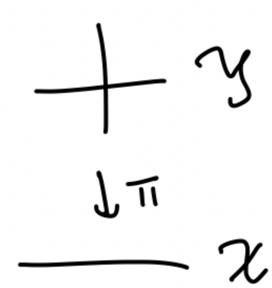
**Geometric examples.** $A = \mathbb{C}[x]$,
$B = \mathbb{C}[x][y]/(f) = A[y]/(f)$, $f \in A[y] \setminus A$ polynomial
$\mathcal{X} = \mathbb{A}^1$ affine line, $\mathcal{Y} =$ curve associated with $B$
$A \hookrightarrow B$ corresponds to $\pi : \mathcal{Y} \to \mathcal{X}$




Case (a): $f = y^2 - x^2$




Case (b): $f = xy - 1$




Case (c): $f = xy$

In (a), we have lying over (every $P$ has two $Q$'s)

In (b), at 0 we have a $P$ with no $Q$, so we do not have lying over.

In (c), at 0 we have a $P$ with infinitely many $Q$'s. But we still do have lying over.

**Proposition 5** (Incomparability)**.** *$A \leq B$ integral. If $Q, Q'$ are distinct prime ideals in $B$ and $Q \cap A = Q' \cap A$ then $Q' \not\subseteq Q$ and $Q \not\subseteq Q'$.*

*Proof.* Assume $Q \subseteq Q'$. Suppose for contradiction that there exists $f \in Q' \setminus Q$.
Then $B/Q$ is integral over $A/Q \cap A = A/P$. (Prop 3).
Then we have $\bar{f}^n + \bar{a}_{n-1}\bar{f}^{n-1} + \cdots + \bar{f}_0 = 0$ in $B/Q$ with $\bar{a}_i \in A/P, a_i \in A$
Without loss of generality, assume $n$ is minimal. $f \in Q' \implies \bar{a_0} \in Q^c \cap A/Q \cap A = 0$.
This contradicts the minimality of $n$ ($B/Q$ is a domain since $Q$ is prime). $\square$

**Proposition 6** (Going up)**.** *$A \subseteq B$ integral extension, $P \subseteq P' \trianglelefteq A$ prime, $Q \trianglelefteq B$ prime so that $Q \cap A = P$. Then there exists $Q'$ prime in $B$ so that $Q \subseteq Q', Q' \cap A = P$.*

*Proof.* $B/A$ integral $\implies$ $B/Q$ is integral over $A/p$ by Prop 3.
Applying lying over to $P'/P$ to get $Q'/P$ and hence $Q'$ (check details). $\square$

For $L/K$ field extension, $\alpha \in L$ algebraic over $K$, we get $m_\alpha$ minimal polynomial of $\alpha$ over $K$.

**Lemma 7.** *Let $A \leq B$ be an integral extension of integral domains so that $A$ is integrally closed in Quot$(A) =: K$.*

1. *$b \in B \implies m_b \in A[x]$ i.e. the minimal polynomial has coefficients in $A$*
2. *If $b \in PB$ for some $P \trianglelefteq A$ then the coefficients of $m_b$ except for the leading one are in $P$*

*Proof.* Exercise $\square$

**Proposition 8** (Going down)**.** *$A \leq B$ integral extension of integral domains, $A$ integrally closed*
*$P \subseteq P'$ prime in $A$, $Q' \trianglelefteq B$ prime lying over $P'$ $\implies \exists Q \trianglelefteq B$ prime so that $Q \subseteq Q'$ and $Q$ lies over $P$.*

*Proof.* $B \mapsto B_{Q'}$ is injective since $B$ is a domain. So view $A, B$ as subrings of $B_{Q'}$.
$Q'B_{Q'}$ is the maximal ideal in $Q'B_0$.

Prop 4.1 $\implies$ it suffices to show that $PB_{Q'} \cap A \subseteq P$ (then the ideal lying over $P$ in $B_{Q'}$ will contact to the desired ideal in $B$)

Let $0 \neq f \in PB_{Q'} \cap A, \quad f = \frac{p}{s}, \quad p \in PB, \quad s \in B \setminus Q'$.

By Lemma 7, the minimal polynomial $m_p$ of $p$ over Quot$(A)$ looks like

$$m_p(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_1X + a_0$$

for $a_i \in P$ (and irreducible).
Let
$$\tilde{m} = X^n + \frac{a_{n-1}}{f}X^{n-1} + \cdots + \frac{a_1}{f^{n-1}}X + \frac{a_0}{f^n} = \frac{1}{f^n}m_p(f - X)$$
Note that $\tilde{m}$ is also irreducible and monic, but $\tilde{m}(s) = m_s$ is the minimal polynomial of $s$ over $\mathrm{Quot}(A) \implies$ by Lemma, 7 $m_s$ has coefficients in $A$ since $s \in B$ is integral over $A$.

Assume for contradiction that $f \notin P$. $f^i \cdot \frac{a_{n-i}}{f^i} = a_{n-i} \in P$. So $\frac{a_{n-i}}{f^i} \in P$ for all $i$. Hence $s^n \in PB \subseteq P'B = Q' \implies s \in Q' \rightarrow\leftarrow$ . So $f \in P$. $\square$

## 2.2 Dimension

### 2.2.1 Definition and basic properties

**Definition.** The **Krull dimension** of a ring $R$ is the maximal $n \in \mathbb{N}$ such that there is a chain of prime ideals
$$P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_n$$
of length $n$ in $R$ (or $\infty$ if no such $n$ exists)

The **codimension / height** of a prime ideal $P$ is the maximal $n \in \mathbb{N}$ so that there is a chain as above with $P_n \subsetneq P$.

Notation: $\dim R, \mathrm{codim}_R P = \mathrm{codim} P$.

"Maximal chain of prime ideals" means it cannot be extended

**Lemma 1.** *$R$ ring, $P \trianglelefteq R$ prime.*

1. *codim*$P = \dim R_P$
2. $\dim R \geq \dim(R/P) + \dim R_P$
3. $\dim R = \sup\{\dim R_P \mid P \trianglelefteq R \text{ maximal}\} = \sup\{\text{codim}P \mid P \trianglelefteq R \text{ maximal}\}$

*Proof.* (1), (2) we proved last semester
(3): $P_0 \subsetneq \cdots \subsetneq P_n$ chain of maximal ideals containing $P_n \implies$ localized chain in $R_P$ has same length and conversely. $\square$

**Lemma 2.** *If $R$ is a ring of finite dimension in which all maximal chains of prime ideals have the same length and $P \trianglelefteq R$ is prime:*

1. *$R/P$ is also a ring of finite dimension in which all maximal chains of prime ideals have the same length*
2. $\dim R = \dim R/P + \dim R_P$
3. $\dim R_P = \dim R$ *if $P$ is maximal*

*Proof.* A chain of prime ideals in $R/P$ corresponds to a chain $Q_0 \subsetneq \cdots \subsetneq Q_r$ of prime ideals in $R$ containing $P \implies \dim R/P \leq \dim R < \infty$
If the chain in $R/P$ is maximal, $Q_0 = P$, hence it can be extended to a maximal chain $P_0 \subsetneq \cdots \subsetneq P_m = P = Q_0 \subsetneq \cdots \subsetneq Q_r$ of prime ideals in $R$. So $\text{codim} P \geq m, \quad \dim R/P \geq r \implies \dim R = m + p$ by assumption.
By Lemma 1, $\text{codim} P = m$ and $\dim R/P = r$. This shows (1) and (2).

For (3), $P$ maximal $\implies \dim R/P = 0 \implies \dim R_P = \dim R - 0 = \dim R$. ☐

**Lemma 3.** *$R \leq S$ integral extension of rings $\implies \dim R = \dim S$.*

*Proof.* $\dim R \leq \dim S$:
Take a chain of ideals $P_0 \subsetneq \cdots \subsetneq P_n$ in $R$.
Lying over for $P_0$ and successively going up gives $Q_0 \subsetneq \cdots \subsetneq Q_n$ such that $Q_i$ lies over $P_i$.
The inclusions must all be strict $\implies \dim R \leq \dim S$.

For $\geq$: Let $Q_0 \subsetneq \cdots \subsetneq Q_n$ be a chain of prime inclusions in $S$. Let $P_i := Q_i \cap R$. This gives a chain of prime ideals in $R$. The inclusions must be strict in our new chain by incomparability. So $\dim S \leq \dim R$. ☐

**Proposition 4.** *Let $K$ be a field.*

1. $\dim K[x_1, \ldots, x_n] = n$
2. *All maximal chains in $K[x_1, \ldots, x_n]$ have length $n$*

*Proof.* Induction on $n$:
$n = 0$ ✓
$n \geq 1$: Let $P_0 \subsetneq \cdots \subsetneq P_m$ be a chain of prime ideals in $K[x_1, \ldots, x_n]$. We want to show that $m \geq n$ and $m = n$ if the chain is maximal.
Without loss of generality, assume $P_0 = 0, P_1$ is a minimal prime ideal, and $P_m$ is a maximal prime (extend chain if necessary).

$f \in P_1$ irreducible $\implies f$ is prime, since we are in a UFD $\implies (f)$ is a nonzero prime ideal contained in $P_1 \implies (f) = P_1$ by minimality.

Assume that $f$ is monic in $X_n$ (do a variable change as in the proof of Noether normalization).

$K[x_1, \ldots, x_n]/P_1 = K[x_1, \ldots, x_n]/(f)$ is integral over $K[x_1, \ldots, x_n]$ by Prop 1.1.

Then given $P_1 \subsetneq \cdots \subsetneq P_m$, mod out by $P_1$ to get $P_1/P_1 \subsetneq \cdots \subsetneq P_m/P_1$ in $B = K[x_1, \ldots, x_n]/(f)$. Letting $Q_i := P_i/P_1 \cap K[x_1, \ldots, x_n]$, this chain corresponds to a chain $Q_1 \subsetneq \cdots \subsetneq Q_m$ in $A = K[x_1, \ldots, x_{n-1}]$.

The extension to the quotient $K[x_1, \ldots, x_n] \to B$ preserves prime ideals, strict inclusions, and maximal chains (we proved this last semester).

The contraction $B \to A$ preserves prime ideals and strict inclusions (by incomparability).

We just need to show that the contraction $B \to A$ preserves maximal chains. There are 3 possible ways to extend a chain:

1. Left: but $Q_1 = 0$ so we cannot extend on the left
2. Right: $P_m/P_1 = Q'_m$ is maximal in $B \implies B/Q'_m$ is a field. $B/Q'_m$ is integral over $A/Q_m$ (integrality is preserved by quotients) so $A/Q_m$ is a field $\implies Q_m$ is maximal. So we cannot extend on the right.
3. Middle: If $Q_0 \subsetneq \cdots \subsetneq Q_m$ were extendable in the middle, once can show that $P_1/P_1 \subsetneq \cdots P_m/P_1$ would also be extendable. We will leave this as an exercise.

Thus $Q_0 \subsetneq \cdots \subsetneq Q_m$ is maximal. By induction, $m-1 \le n-1$ with equality if the chain is maximal. □

**Corollary 5.** *$A$ finitely generated $K$-algebra, $K$ field $\implies$ $A$ has finite dimension.*
*If $A$ is also a domain, every maximal chain of prime ideals has length* $\dim A$.

**Note.** There exist Noetherian rings of infinite Krull dimension, and there exist non-Noetherian rings of finite Krull dimension (e.g. $K[x_1, \dots]/(x_1^2, x_2^2, \dots)$ has dimension 0).

### 2.2.2 Artinian rings

**Proposition 6.** *$R$ artinian ring.*

1. *There are maximal ideals $P_1, \dots, P_m \trianglelefteq R$ such that $P_1 \cdots P_m = 0$ (not necessarily distinct)*
2. *$R$ has only finitely many prime ideals. All of them are maximal and occur in (1)*

*Proof.* 1. Let $S = \{P_1 \cdots P_n \mid P_i \text{ maximal}, n \in \mathbb{N}\}$.
$R$ has maximal ideals so $S \neq \varnothing$.
Let $I = P_1 \cdots P_m$ be a minimal element of $S$ (which exists since $R$ is artinian).
$I^2 \in S$ and $I^2 \subseteq I \implies I^2 = 0$ (by minimality)
Let $P \trianglelefteq R$ be any maximal ideal. Then $PI \in A$ and $PI \subseteq I \implies PI = I$ (again by minimality)
Thus $I$ is contained in every maximal ideal: $I = PI \subseteq P \cap I \subseteq P$.

Now suppose for contradiction that $I \neq 0$. $R$ artinian $\implies \exists J \trianglelefteq R$ minimal such that $IJ \neq 0$. $J$ is generated by any $b \in J$ with $I \cdot (b) \neq 0$ (by minimality).
So $IJ \cdot I = I^2 \cdot J = I \cdot J \neq 0 \implies IJ = J$ by minimality.

Then by Nakayama's Lemma, we get $a \in I$ such that $(1-a)J = 0$.
Note: $J \neq 0 \implies 1 - a \in R^\times \implies (1-a) \neq R \implies (1-a) \subseteq P \implies 1 \in P \to\leftarrow$ .

2. Let $P \trianglelefteq R$ be any prime ideal. Suppose $P_1, \ldots, P_m$ are as in (1) and assume (for contradiction) that $P_i \nsubseteq P$ for all $i$. Then there exist $a_1, \ldots, a_m$ with $a_i \in P_i \setminus P$ for all $i$. Then $a_1 \cdots a_m \in P_1 \cdots P_m = 0 \subseteq P \rightarrow\leftarrow$ ($P$ prime).
   So $P_i \subseteq P$ for some $i \implies P_i = P$ by maximality.

□

**Corollary 7.** *$R$ artinian $\implies \dim R = 0$.*

**Theorem 8** (Hopkins)**.** *$R$ Artinian $\iff$ $R$ noetherian of dimension 0.*
*In particular, every Artinian ring is Noetherian.*

**Note.** This is not true for $R$-modules.

*Proof.* $(\Rightarrow)$ : $P_1 \cdots P_m = 0$ as in Proposition 6 $\implies Q_i := P_{i+1} \cdots P_m, \quad i = 0, \ldots, m \implies$ $0 = Q_0 \subsetneq \cdots \subsetneq Q_m = R$.

$Q_i/Q_{i-1} = Q_i/P_iQ_i$ is Artinian as a module, since $Q_i$ is and $Q_i \leq R$. But $Q_i/Q_{i-1}$ is also an $R/P_i$-vector space (since $P_i$ is maximal). So it is also Noetherian as an $R/P_i$-vector space and then as an $R/P_i$-module.
Inductively, this shows $Q_i$ is a Noetherian $R$-module for all $i$. Hence $Q_m = R$ is Noetherian. Moreover, $R$ has dimension 0 by Corollary 7.

$(\Leftarrow)$ : Assume $R$ is Noetherian but not Artinian. We will show: there exists $P \trianglelefteq R$ prime which is not maximal ( $\implies \dim R \geq 1$).

Let $S = \{I \trianglelefteq R \mid R/I \text{ is not Artinian}\}$. Note $0 \in S$ so $S \neq \varnothing$.

$R$ Noetherian $\implies$ $S$ has maximal element $P$.

Note: $P$ is not a maximal ideal, since $R/P$ is not Artinian. So it suffices to show that $P$ is prime i.e. $\overline{R} := R/P$ is a domain.
Let $a \in R$. Consider the following short exact sequence induced by $\cdot a$:

$$0 \to \overline{R}/\operatorname{Ann}(\overline{a}) \to \overline{R} \to \overline{R}/(\overline{a}) \to 0$$

$\overline{R}$ is not Artinian, so either $\overline{R}/\operatorname{Ann}(\overline{a})$ is not Artinian or $\overline{R}/(\overline{a})$ is not Artinian.
But these are both quotients of $R/P$, and proper quotients of $R/P$ are Artinian.
Hence $\operatorname{Ann}(\overline{a}) = 0$ or $(\overline{a}) = 0 \implies R/P$ is a domain $\implies P$ is prime. □

**Example.** Let $I = (a)$ be a nonzero prime ideal in a PID $R$.

So $a = p_1^{a_1} \cdots p_m^{a_n}$ where $a_i \in \mathbb{N}_{>0}$ and $p_i$ are distinct prime elements

Let $S = R/I$. We recall that ideals in $S$
$\leftrightarrow$ ideals in $R$ containing $I$
$\leftrightarrow$ $b \in R$ so that $b \mid a$

$\leftrightarrow b = p_1^{b_1} \cdots p_n^{b_n}$ so that $b_i \leq a_i$ for all $i$
So $S$ has finitely many ideals, hence is Noetherian and Artinian.

$R$ PID $\implies$ every nonzero prime ideal is maximal $\implies \dim S = 0$.
Maximal ideals in $S$ are of the form $(\overline{p_i})$ for $i = 1, \ldots, n$ and $(\overline{p_1})^{a_1} \cdots (\overline{p_n})^{a_n} = 0$.

This checks all statements of Prop 6 and Theorem 8.

**Theorem 9** (Structure Theorem for Artinian Rings)**.** *Every Artinian ring is a finite product of Artinian local rings. More precisely, let $P_1, \ldots, P_n$ be the distinct maximal ideals of $R$ as in Prop 6. Then $R_{P_i}$ is Artinian for all $i$ and $R \cong R_{P_1} \times \cdots \times R_{P_n}$.*

*Proof.* By Prop 6, $P_1^k \cdots P_n^k = 0$ for some $k \in \mathbb{N}$. Since $P_i$ are pairwise coprime, $P_i^k$ are pairwise coprime ($\sqrt{P_i^k + P_j^k} \supseteq P_i + P_j = 1$)

By Prop II.1.7.1 from last semester, $R \cong R/P_1^k \times \cdots \times R/P_n^k$.
$R/P_i^k$ is Artinian, so suffices to show that $R/P_i^k \cong R_{P_i}$ for all $i = 1, \ldots, n$. Without loss of generality, consider $i = 1$.
Consider $R \to R_{P_1}, a \mapsto \frac{a}{1}$. This has kernel containing $P_1^k$:
$a \in P_1^k \implies$ choose $a_j \in P_j \setminus P_1, j = 2, \ldots, n \implies u := a_2^k \cdots a_n^k \notin P_1$ but $a \cdot u \in P_1^k \cdots P_n^k = 0 \implies \frac{a}{1} = 0$ in $R_{P_1}$.

So $\varphi : R/P_1^k \to R_{P_1}, \overline{a} \mapsto \frac{a}{1}$ is well-defined.

We may construct an inverse as follows: $R \xrightarrow{\pi} R/P_1^k, a \mapsto \overline{a}$ maps $u \in R \setminus P_1$ to a unit (otherwise $u \in P$ for some $P/P_1^k \trianglelefteq R/P_1^k$ maximal $\implies P \supseteq P_1 \implies P = P_1$ since $P$ is maximal $\implies u \in P_1 \rightarrow\leftarrow$).

Hence $\pi$ induces a homomorphism $\psi : R_{P_1} \to R/P_1^k, \frac{a}{1} \mapsto \overline{a}$ which is an inverse to $\varphi$. □

**Note.** Now it is easy to prove Corollary I.5.18: If $R$ is a ring with $m_1 \cdots m_k = 0$ for maximal $m_i \trianglelefteq R$ then $R$ is Artinian $\iff$ $R$ is Noetherian.

### 2.2.3 Krull's principal ideal theorem

**Definition.** $P \trianglelefteq R$ prime ideal: $P^{(k)} := \{a \in R \mid ab \in P^k \text{ some } b \in R \setminus P\}$ is called the **k-th symbolic power of $P$**

**Lemma 10.** *$P \trianglelefteq R$ prime, $k \in \mathbb{N}$*

1. $P^k \subseteq P^{(k)} \subseteq P$
2. $P^{(k)}$ *is $P$-primary*
3. $P^{(k)} R_P = P^k R_P$

**Note.** $P^k$ need not be primary.
One can show: $P^{(k)}$ is the smallest primary ideal containing $P^k$. (HW 10 last semester)

*Proof.* 1. $P^k \subseteq P^{(k)}$ by definition
$a \in P^{(k)} \implies ab \in P^k$ some $b \notin P \implies a \in P$ since $P$ is prime

2. $P = \sqrt{P} = \sqrt{P^k} \subseteq \sqrt{P^{(k)}} \subseteq \sqrt{P} = P \implies \sqrt{P^{(k)}} = P$.
$ab \in P^{(k)} \implies abc \in P^k$ some $c \notin P$
If $b \notin \sqrt{P^{(k)}} = P$ then $bc \notin P \implies a \in P^{(k)}$ by definition $\implies$ (2).

3. $\frac{b}{s} \in P^{(k)}R_P \implies bc \in P^k$ some $c \notin P \implies \frac{b}{s} = \frac{bc}{sc} \in P^k R_P$.
The reverse inclusion is clear.

□

**Theorem 11** (Krull's principal ideal theorem)**.** *$R$ Noetherian ring, $a \in R$.*
*$P \supseteq (a)$ minimal prime ideal $\implies$ codim$P \leq 1$.*

*Proof.* $Q' \subseteq Q \subsetneq P$ chain of prime ideals. WTS: $Q' = Q$.
First step: reduce to nice $R$.
Given $Q' \subseteq Q \subsetneq P$ in $R$, reduce mod $Q'$ to get
$0 \subseteq Q/Q' \subsetneq P/Q'$ in $R/Q'$. Now localize at $P$ to get
$0 \subseteq (Q/Q')_P \subsetneq (P/Q')_P$ in $(R/Q')_P$.

$R/Q'$ is Noetherian as an $R$-module and hence as an $R/Q'$-module. So $(R/Q')_P$ is Noetherian (by HW problem).
So we may replace $R$ with $(R/Q')_P$ and assume that $0 \subseteq Q \subsetneq P$, $R$ is a domain, local with maximal ideal $P$. WTS: $Q = 0$.

Consider the symbolic powers $Q^{(k)}$ of $Q$. It is clear that $Q^{(n+1)} \subseteq Q^{(n)}$ for all $n$.

Second step: show $Q^{(n)} = Q^{(n+1)}$ for large $n$.
$R/(a)$ has dimension 0 (because $P$ is both a minimal prime and maximal ideal) and is noetherian, so by Theorem 8, $R/(a)$ is Artinian.

Thus
$$(Q^{(0)} + (a))/(a) \supseteq (Q^{(1)} + (a))/(a) \supseteq \cdots$$
stabilizes, meaning
$$(Q^{(n)} + (a))/(a) = (Q^{(n+1)} + (a))/(a)$$
for large $n$. So $Q^{(n)} + (a) \subseteq Q^{(n+1)} + (a)$. (*)

We claim: $Q^{(n)} = Q^{(n+1)} + PQ^{(n)}$.
$\supseteq$ is clear by definition.
For $\subseteq$, let $b \in Q^{(n)}$. Then $(*) \implies b = c + ar$ where $c \in Q^{(n+1)}$ and $r \in R$. So $ar = b - c \in Q^{(n)}$. But $a \notin Q$ ($P$ was minimal over $(a)$ and $Q \subsetneq P$). So $r \in Q^{(n)}$ by Lemma 10 ($Q^{(n)}$ is $Q$-primary), meaning $b \in Q^{(n+1)} + PQ^{(n)}$. This proves the claim.

So $Q^{(n)} = Q^{(n+1)} + PQ^{(n)}$ for large $n$. Then
$$Q^{(n)}/Q^{(n+1)} \cong PQ^{(n)}/Q^{(n+1)}.$$

So by Local Nakayama, $Q^{(n)}/Q^{(n+1)} = 0$, meaning $Q^{(n)} = Q^{(n+1)}$.

Final step: We know that $Q^{(n)} = Q^{(n+1)}$. So by Lemma 10, $Q^{(n)}R_Q = Q^{(n+1)}R_Q = Q(Q^nR_Q)$. Thus by Local Nakayama, $Q^nR_Q = 0$. Since $R$ is a domain, $R_Q$ is a domain. So we must have $Q = 0$. $\square$

**Corollary 12.** *Let $R$ be Noetherian, $a \in R$ not a zero-divisor, $P \supseteq (a)$ minimal prime. Then codim$P = 1$.*

*Proof.* Let $P_1, \ldots, P_n$ be the minimal primes over $(0)$ (these are the isolated ideals in $\mathrm{Ass}(I)$).

Write $P_i = \sqrt{(0 : b_i)}$ for some (nonzero) $b_i \in R$. Suppose $a \in P_i$ for some $i$. Then $a^rb_i = 0$ for some $r$. Without loss of generality choose $r$ minimally. Then $a(a^{r-1}b_i) = 0 = 0 \rightarrow\leftarrow$ . So $a \notin P_i$ for all $i$.

Since $a \in P$, $P$ is not minimal over $(0)$. Then codim$P \geq 1$. This along with Theorem 11 $\implies$ codim$P = 1$. $\square$

**Corollary 13.** *$R$ Noetherian, $a_1, \ldots, a_n \in R, P \supseteq (a_1, \ldots, a_n)$ minimal prime $\implies$ codim$P \leq n$.*

*Proof.* Homework. $\square$

**Example.** $\mathcal{X} = V(x^2 + y^2 - 1) \subseteq \mathbb{A}^2_{\mathbb{R}}$.

$\dim \mathcal{X} \equiv \dim k[\mathcal{X}] = \dim \mathbb{R}[x, y]/(x^2 + y^2 - 1)$.

Let $a = x^2 + y^2 - 1$. Then $(x^2 + y^2 - 1)$ is a minimal prime in $\mathbb{R}[x, y]$ containing $(a)$ and $a$ is not a zero-divisor. So by Corollary 12, $\mathrm{codim}(x^2 + y^2 - 1) = 1$. Then

$$\dim \big(\mathbb{R}[x, y]/(x^2 + y^2 - 1)\big) = \dim \mathbb{R}[x, y] - \mathrm{codim}(x^2 + y^2 - 1) = 2 - 1 = 1.$$

### 2.2.4 Dimension and transcendence degree

Recall:

**Definition.** $K \subseteq L$ field extension, $B \subseteq L$.
Let $K(B)$ denote the smallest subfield of $L$ containing $B$ and $K$.

$B$ is **algebraically dependent** over $K$ is there exists a nonzero polynomial $f \in K[x_1, \ldots, x_n]$ for some $n$ with $f(b_1, \ldots, b_n) = 0$ for distinct $b_i \in B$.

Otherwise $B$ is **algebraically independent**.

If $B$ is algebraically independent and every element of $L$ is algebraic over $K(B)$ then $B$ is a **transcendence basis**.

**Proposition 14.** *Let $K \subseteq L$ be a field extension.*
*Let $A \subseteq L$ be algebraically independent over $K$.*
*Let $C \subseteq L$ so that $L$ is algebraic over $K(C)$.*
*Then there exists a transcendence basis $B$ of $L$ over $K$ such that $A \subseteq B$ and $B \setminus A \subseteq C$.*
*In particular, every field extension has a transcendence basis.*
*If $L$ has a finite transcendence basis, then all are finite of the same length.*

*Proof.* Let $\mathcal{M} := \{B \mid B \text{ is algebraically independent, } A \subseteq B \subseteq A \cup C\}$.
$A \in \mathcal{M}$ so $\mathcal{M} \neq \varnothing$.
Apply Zorn's Lemma. $\mathcal{M}$ has a maximal element $B_0$. By construction, $B_0$ is algebraically independent and by maximality, $K(C)/K(B_0)$ is algebraic.

We know that $L/K(C)$ is algebraic, so $L/K(B_0)$ is algebraic.

Now let $B$ be a finite transcendence basis of $L$: $B = \{b_1, \dots, n\}, |B| = n$.

Let $B'$ be another transcendence basis. Suffices to show: $|B'| \leq n$. Proof by induction on $n$:
If $n = 0$, ✓.
If $n > 0$: If $B' \subseteq B$ there is nothing to show. So let $b \in B' \setminus B$. Note: $b$ is not algebraic over $K$.
Take $A = \{b\}, C = B$. There exists a transcendence basis $B'' = \{b, b''_1, \dots, b''_k\}$ where $b''_i \in B \setminus \{b\}$. Note that $k \leq n-1$ since $\{b, b_1, \dots, b_n\}$ is algebraically independent.

So $B' \setminus \{b\}$, and $B'' \setminus \{b\}$ are transcendence bases of $L$ over $K(b)$. Thus

$$|B' \setminus \{b\}| \leq |B'' \setminus \{b\}| \leq n-1 \implies |B'| \leq n$$

(where the first inequality is by induction and the second is because $k \leq n-1$). □

**Definition.** The **transcendence degree of $L$ over $L$** is $\operatorname{trdeg}(L/K) := \operatorname{trdeg}_K(L) := |B|$ where $B$ is a transcendence basis of $L/K$ ($= \infty$ if no finite $B$ exists).

**Example.**

1. $\operatorname{trdeg}_K(L) = 0 \iff L/K$ is algebraic
2. $\operatorname{trdeg}_{\mathbb{Q}}(\mathbb{C}) = \infty$ ($\pi \in \mathbb{C}$)
3. $K(x_1, \dots, x_n) = \operatorname{Quot}(K[x_1, \dots, x_n])$ has transcendence degree $n$ over $K$

**Proposition 15.** *$R$ finitely generated $K$-algebra which is a domain $\implies \dim R = \operatorname{trdeg}_K(\operatorname{Quot}(R))$.*

*Proof.* Let $R \supseteq K[x_1, \dots, x_n]$ be integral using Noether Normalization. Then $\dim R = n$.

$0 \neq \frac{a}{b} \in \operatorname{Quot}(R) \implies a, b$ are integral over $K[x_1, \dots, x_n] \implies a, b$ are algebraic over $K(x_1, \dots, x_n) \implies \frac{a}{b}$ is algebraic over $K(x_1, \dots, x_n)$. □

### 2.2.5 Regular local rings

Motivation: $\mathcal{X} \subseteq \mathbb{A}^n(K)$ affine algebraic set with coordinate ring $k[\mathcal{X}] = K[x_1, \ldots, x_n]/I(\mathcal{X})$.
$a = (a_1, \ldots, a_n) \in \mathcal{X} \implies$ set $y_i := x_i - a_i$. Consider

$$d_a : K[x_1, \ldots, x_n] \to K[y_1, \ldots, y_n], f \mapsto \sum \frac{df}{dx_i}(a) y_i = \text{ linear term in Taylor series expansion of } f \text{ at } a$$

Define the **tangent space of $\mathcal{X}$ at $a$**

$$T_a\mathcal{X} := V(\{d_a f \mid f \in I(\mathcal{X})\}) \subseteq K^n$$

Easy to check: $I(\mathcal{X}) = (f_1, \ldots, f_m) \implies T_a\mathcal{X} = V(\{d_a f_1, \ldots, d_a f_m\})$.

Algebraic definition:
$P = I(a) = (\overline{y_1}, \ldots, \overline{y_m}) \trianglelefteq k[\mathcal{X}]$
$\overline{y_i} = y_i + I(\mathcal{X})$
Obtain

$$\varphi : P \to \mathrm{Hom}_K(T_a\mathcal{X}, K) = (T_a\mathcal{X})^*, \quad \overline{f} \mapsto d_a f|_{T_a\mathcal{X}}$$

Observe:

1. $\varphi$ is surjective since any linear map on $T_a\mathcal{X}$ must be a linear combination of the coordinate functions
2. $L : -\{d_a g \mid g \in I(\mathcal{X})\} \leq \langle y_1, \ldots, y_n \rangle$
   $K := \dim L \implies \dim T_a\mathcal{X} = n - k \implies \dim T_a\mathcal{X}^\perp = k \implies L = T_a\mathcal{X}^\perp$ since $L \subseteq T_a\mathcal{X}$

Suppose that $\varphi(\overline{f}) = 0$. Then $d_a f \in L \implies d_a f = d_a g$ for some $g \in I(\mathcal{X}) \implies d_a(f - g) = 0 \implies \overline{f} = \overline{f} - \overline{g} \in P^2$.
On the other hand, $P^2$ is generated as a $K$-vector space by $\overline{f_1 f_2}$ for $\overline{f_i} \in P$ and $\varphi(\overline{f_1 f_2}) = f_1(a)\varphi(\overline{f_2}) + f_2(a)\varphi(\overline{f_1}) = 0 \implies \ker\varphi = P^2$.

So $P/P^2 \cong (T_a\mathcal{X})^*$, i.e. $(P/P^2)^* \cong T_a\mathcal{X}$.

Note: If $P \trianglelefteq R$ maximal, $S := R \setminus P \implies P/P^2$ is a vector space over $K := R/P$.
Elements of $S$ map to units in $K$ so $P/P^2 \cong S^{-1}(P/P^2) \cong (S^{-1}P)/(S^{-1}P^2)$. So we can instead work in the local ring at $a$.

**Proposition 16.** *$R$ Noetherian local with maximal ideal $P$, $K = R/P$.*

1. *$\dim_K(P/P^2)$ is the minimum number of generators for $P$ as an ideal*
2. *$\dim R \leq \dim_K(P/P^2) < \infty$*

*Proof.* 1. Nakayama

2. $\dim R = \dim R_P = \mathrm{codim} P \leq n$ by Corollary 13 (where $n =$ minimum number of generators of $P$)

□

**Example.**

1. $\mathcal{X}_1 \subseteq \mathbb{A}^2(\mathbb{R})$
   $I(\mathcal{X}_1) = (x_2 - x_1^2)$
   $T_0(\mathcal{X}_1) = V(x_2) = x_1$-axis

2. $\mathcal{X}_2 = V(x_2^2 - x_1^2 - x_1^3) \subseteq \mathbb{A}^2(\mathbb{R})$
   $T_0(\mathcal{X}_2) = \mathbb{R}^2$ (Note $\dim \mathcal{X}_2 = 1$ but $\dim T_0(\mathcal{X}_2) = 2$)

**Definition.** Let $R$ be a local ring with maximal ideal $P$. Let $K = R/P$.
$R$ is **regular** if it is Noetherian and $\dim R = \dim_K(P/P^2)$.

If $\mathcal{X}$ is an affine algebraic set over $K$, $a \in \mathcal{X}$ is a **regular point** or a **smooth point** if $k[\mathcal{X}]_{I(a)}$ is a regular local ring. Otherwise, $a$ is a **singular point**.

$\mathcal{X}$ is **smooth** if all its points are regular and **singular** otherwise

**Example.**

1. Any field is a regular local ring (both dimensions are 0)

2. $\mathcal{X} \subseteq \mathbb{A}^n(K)$ affine algebraic, $I(\mathcal{X}) = (f_1, \dots, f_r) \trianglelefteq K[x_1, \dots, x_n]$
   $T_a\mathcal{X}$ is the kernel of $Jf(a) := \left(\frac{\partial f_i}{\partial x_j}\right)_{1 \le i \le r, 1 \le j \le n}$ Jacobian matrix
   Hence $a$ is regular $\iff \dim \mathcal{X} = \dim \ker(Jf(a))$
   In particular, this happens when $Jf(a)$ has maximal rank $r$ ($\dim \mathcal{X} \le \dim T_a\mathcal{X} = \dim \ker(Jf(a)) \le n - r \le \dim \mathcal{X}$)

   In the examples from before, for $V(x_2 - x_1^2)$ we get the Jacobian $(-2x_1, 1)$, which is always nonzero so has full rank. But for $V(x_2^3 - x_1^2 - x_1^3)$ we get the Jacobian $(-2x_1 - 3x_1^2, 2x_2)$, which has nonzero kernel.

**Proposition 17.** *A regular local ring is an integral domain.*

*Proof.* $R$ regular local $\implies R$ has unique maximal ideal $P$. Let $K = R/P$.
Let $n = \dim R (= \dim_K(P/P^2))$.
Induction on $n$:
When $n = 0$, $0 = \dim R = \dim_K(P/P^2) \implies P^2 = P \implies$ by Nakayama, $P = 0$. Then $R = K$ so we have a field.
When $n > 0$, let $P_1, \dots, P_r$ be the minimal primes over $(0)$.

Last semester we saw: for an ideal $I$ in $R$, the isolated primes of $I$ = the minimal primes containing $I$.

Suppose (for contradiction) that $P \subseteq P^2 \cup P_1 \cup \dots \cup P_r$.
$P \not\subseteq P^2 \implies P \subseteq P_i$ for some $i$ (using prime avoidance + an exercise to show a little more)
Thus $P = P_i$ for some $i$, since $P$ is maximal ($P_i \subseteq P$ for all $i$) $\implies r = 1$.

So $P$ is the only prime above $0 \implies n = 0$. $\to\leftarrow$

Hence there exists $a \in P$ so that $a \notin P^2$ and $a \notin P_i$ for all $i = 1, \ldots, r$.

$R/(a)$ is local with maximal ideal $P/(a)$ and Noetherian since $R$ is.

Note:
$$(P/(a))/(P/(a))^2 \cong P/(P^2 + (a))$$
Claim: $R/(a)$ is regular of dimension $n-1$.

1. $a \notin P^2$ so $\overline{a} \in P/P^2$ is nonzero. $P/P^2$ is a $K$-vector space $\implies$ $\overline{a}$ extends to a basis of $P/P^2$: $(\overline{a}, \overline{a_2}, \ldots, \overline{a_n})$.

   So $\overline{a_2}, \ldots, \overline{a_n}$ generate $P/(P^2 + (a)) \implies$
   $$\dim(P/(a))/(P/(a))^2 \leq n - 1.$$

2. $Q_0 \subsetneq Q_1 \subsetneq \cdots \subsetneq Q_n = P$ maximal chain in $R$.
   $a \in P \implies$ wlog assume $a \in Q_1$. (homework)

   So $Q_1/(a) \subsetneq \cdots \subsetneq Q_n/(a0 = P/(a)$ is a chain in $R/(a)$ of length $n-1 \implies \dim R/(a) \geq n - 1$.

So $R/(a)$ is regular local of dimension $n-1$. Then by induction, $R/(a)$ is an integral domain.

$(a) \trianglelefteq R$ prime ideal $\implies P_i \subseteq (a)$ for some $i$. Let $b \in P_i$. $b = ca$ for some $c \in R$. But $a \notin P_i \implies c \in P_i$.

So $P = PP_i \implies$ by Nakayama, $P_i = 0$.
So $(0)$ is a prime ideal, meaning $R$ is a domain. $\square$

## 2.3 Rings of dimension 1

### 2.3.1 Valuation rings

**Lemma 1.** *$R$ regular local of dimension 1, $P \trianglelefteq R$ maximal. Then*

1. *$P$ is nonzero principal*
2. *$a \in R \setminus \{0\} \implies \exists! n \in \mathbb{N}_0$ so that $(a) = P^n$*

*Proof.* 1. $\dim R = \dim P/P^2 = 1$
Proposition 2.15 $\implies$ $P$ is principal and nonzero
Write $P = (\pi)$

2. Prop 2.16 $\implies$ $R$ integral domain $\implies$ $(0)$ is the unique minimal prime. The only other prime ideal is $P$ since $\dim R = 1$.

   Let $a \in R \setminus \{0\}$. By Krull intersection theorem, $\bigcap_{n=0}^{\infty} P^n = (0)$.
   So there is $n$ maximal with $a \in P^n$ ($n = 0$ if $a \in R^\times$). Then $a = \pi^n u$ and $u \in R^\times$ by maximality of $n$ $\implies$ $(a) = (\pi^n) = P^n$.

$\square$

**Note.** (Geometric interpretation)
$\mathcal{X}$ affine curve (i.e. affine algebraic set of dimension 1)
$x \in \mathcal{X}$ smooth

$R = k[\mathcal{X}]_{I(x)} \implies R$ is a regular local ring of dimension 1

$P \trianglelefteq R$ maximal $\implies$ $P$ consists of functions vanishing at $x$

$P = (\pi)$ as in Lemma 1 $\implies$ $\pi$ vanishes to order 1 at $x$

If $a \in R$, $(a) = (\pi^n) \iff a$ vanishes to order $n$ at $x$

**Definition.** An **ordered group** is an abelian group $G$ with a total order " $\leq$ " such that $m \leq n \implies m + k \leq n + k$ for all $m, n, k \in G$.

Let $K$ be a field. A **valuation** on $K$ is a map $v : K^\times \to G$ where $G$ is an ordered group such that for all $a, b \in K^\times$,

1. $v(ab) = v(a) + v(b)$ i.e. $v : K^\times \to G$ is a group homomorphism
2. $v(a + b) \geq \min\{v(a), v(b)\}$ if $a + b \neq 0$

We extend $v$ to $K$ by letting $v(0) = \infty$ (then (1),(2) holds for all $a, b \in K$)

If $v : K^\times \to G$ is a valuation, then
$v(K^\times) \leq G$ is the **value group**
$O_v := \{a \in K \mid v(a) \geq 0\}$ is the **valuation ring of** $v$
$m_v := \{a \in K \mid v(a) > 0\}$ is the maximal ideal of $v$
$O_v/m_v$ is called the **residue field** of $v$ (See Lemma 2)

A valuation $v : K^\times \twoheadrightarrow \mathbb{Z}$ is called a **discrete valuation** and $(K, v)$ is then called a **discretely valued field** and $(O_v, v)$ a **discrete valuation ring**

$\pi \in O_v$ with $v(\pi) = 1$ is called a **uniformizer**

**Example.**

1. $K$ any field
   $v : K^\times \to \{0\}$ **trivial valuation**

Value group $\{0\}$
Valuation ring $K$

2. $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ are ordered groups

3. $R$ UFD, $p \in R$ prime, $K = \mathrm{Quot}(R)$
$a \in R \implies a = up^n, n \in \mathbb{N}, n = \frac{u_1}{u_2}, p \nmid u_1 u_2$
We get $v : K^\times \to \mathbb{Z}, a \mapsto n$

Claim: this is a discrete valuation

$a = up^n, b = wp^m$. Without loss of generality, $n \geq m$.

$v(a+b) = v(up^n + wp^m) = v(p^m(up^{n-m} + w)) = m + v(up^{n-m} + w) \geq m$

$O_v = R_{(p)}$ localization of $R$ at $p$
$m_v = pR_{(p)}$

$R = \mathbb{Z}, p$ prime $\to$ get $p$-adic valuation

4. $L$ field,

$$K := L((t)) = \mathrm{Quot}(L[[t]]) = \left\{ \sum_{n=m}^{\infty} a_n t^n \mid a_i \in L, m \in \mathbb{Z} \right\} = \text{ Laurent series field}$$

$v : K^\times \to \mathbb{Z}, \quad \sum_{n=m}^{\infty} a_n t^n \mapsto m$ gives a discrete valuation

This is also a special case of (3)

5. $L$ field, define the field of formal Puiseux series:

$$K = \left\{ \sum_{q \in \mathbb{Q}} a_q t^q \mid \{q \in \mathbb{Q} \mid a_q \neq 0\} \text{ is bounded below and has bounded denominators} \right\}$$

Let $v : K^\times \to \mathbb{Q}, \sum a_q t^q \mapsto \min\{q \in \mathbb{Q} \mid a_q \neq 0\}$.

This is a valuation, not discrete.

**Lemma 2** (Properties of valuation rings)**.** *If $R = O_v$ for some valuation $v : K^\times \to G$ for some field $K$, ordered group $G$, then*

1. *$a \in K^\times \implies a \in R$ or $a^{-1} \in R$*
2. *For $a, b \in R$, $v(a) \leq v(b) \iff b \in (a)$*
3. *$R^\times = \{a \in R \mid v(a) = 0\}$*
4. *$R$ is a local ring with maximal ideal $m_v$*

5. *$R$ is a normal domain (integrally closed in its field of fractions)*

*Proof.* 1. $v(a) + v(a^{-1}) = v(1) = 0 \implies v(a) \geq 0$ or $v(a^{-1}) \geq 0 \implies a \in R$ or $a^{-1} \in R$ because $R = O_v = \{a \in K \mid v(a) \geq 0\}$

2. Without loss of generality, assume $a \neq 0$. If $v(a) \leq v(b), v(\frac{a}{b}) = v(b) - v(a) \geq 0 \implies \frac{b}{a} \in R \implies b = \frac{b}{a} \cdot a \in (a)$.
The converse is clear.

3. $a \in R, a \neq 0 \implies v(a) \geq 0$.
$a \in R^\times \iff a^{-1} \in R \iff -v(a) = v(a^{-1}) \geq 0 \iff v(a) = 0$

4. $m_v$ is an ideal:$m_v = \{a \mid v(a) > 0\}$
If $a, b \in m_v, r \in R$, then $v(a+b) \geq \min\{v(a), v(b)\} > 0$ and $v(ra) = v(r) + v(a) > 0$

If $a \notin m_v$ then $v(a) = 0 \implies$ by (3), $a \in R^\times \implies m_v$ is maximal in $R$ and $R$ is local

5. $a \in K^\times$ integral over $R \implies a^n + c_{n-1}a^{n-1} + \cdots + c_0 = 0 \quad (c_i \in R, n \in \mathbb{N})$.

If $a \notin R$ then by (1), $a^{-1} \in R \implies a = -c_{n-1} - \cdots - c_0 a^{-n+1} \in R \rightarrow\leftarrow$
So $a \in R$.

□

**Note.** If $O_v = R$ is the valuation ring of $v : K^\times \to G$, then $v$ can be "recovered up to isomorphism" from $R$:

$K = \mathrm{Quot}(R)$ by Lemma 2.1
$v : K^\times \to G$ has kernel $R^\times$ by Lemma 2.3
$v(K^\times) \cong K^\times / R^\times$ so $v$ is the quotient map
For $a, b \in K^\times$, $v(a) \leq v(b) \iff \frac{b}{a} \in R$, so $R$ determines the order on $v(K^\times)$

**Proposition 3.** *If $R$ is a ring, TFAE:*

1. *$R$ is the valuation ring of a valuation $v : K^\times \to G$*
2. *$R$ is an integral domain and for all $a \in QuotR \setminus \{0\}$, either $a \in R$ or $a^{-1} \in R$*

*Proof.* (1) ⇒ (2) by Lemma 2.1
For (2) ⇒ (1), let $K := \mathrm{Quot}(R)$ and $G := K^\times / R^\times$ (written additively)

Define $\overline{a} \leq \overline{b} \iff \frac{b}{a} \in R$
This gives a well-defined order on $G$.

Let $v : K^\times \to G$ be the quotient map. Check: this is a valuation.

1. $v(ab) = v(a) + v(b)$ because $v$ is a homomorphism

2. Let $a, b \in K^\times$. Wlog $\frac{a}{b} \in R$ (otherwise $\frac{b}{a} \in R$)
   Then $\frac{a+b}{b} = \frac{a}{b} + 1 \in R \implies v(a+b) \geq v(b) = \min\{v(a), v(b)\}$.

Clearly $O_v = R$ since $v(a) \geq 0 \iff \frac{a}{1} \in R$. □

**Lemma 4.** *Let $R \leq R' \leq Quot(R) = K$. Then if $R$ is a valuation ring, so is $R'$.*

*Proof.* $a \in K^\times \implies a \in R$ or $a^{-1} \in R \implies a \in R'$ or $a^{-1} \in R' \implies R'$ is a valuation ring by Proposition 3. □

**Proposition 5.** *If $R$ is an integral domain then if $R^c$ denotes the integral closure of $R$,*

$$R^c = \bigcap_{\substack{R \subseteq R' \subseteq \mathrm{Quot}(R) \\ R' \textit{ valuation ring}}} R'$$

*Proof.* ($\subseteq$): $x \in R^c, R' \supseteq R$ valuation ring $\implies x$ is integral over $R$ so also over $R'$. Thus $x \in R'$ by Lemma 2.
($\supseteq$): Let $x \notin R^c$. We construct a valuation ring $R'$ such that $R \subseteq R' \subseteq \mathrm{Quot}(R)$ and $x \notin R$.

Note: $x \notin R[\frac{1}{x}]$, since otherwise:

$$x = a_0 + a_1 \frac{1}{x} + \cdots + a_k \frac{1}{x^k} \text{ some } k \text{ and } a_i \in R \implies x^{k+1} - a_0 x^k - \cdots = 0$$

gives an integral equation for $x$ over $R \rightarrow\leftarrow$ .

Define $\mathcal{M} := \{R' \mid R; \text{ ring}, R[\frac{1}{x}] \subseteq R' \subseteq \mathrm{Quot}(R), x \notin R'\}$.
$M \neq \varnothing$ since $R[\frac{1}{x}] \in \mathcal{M}$.
Apply Zorn's Lemma to take a maximal element $R_0$.

Suffices to show: $R_0$ is a valuation ring.
Assume $a \notin R_0$ and $a^{-1} \notin R_0$.
Then $R_0 \subsetneq R_0[a]$ and $R_0 \subseteq R_0[a^{-1}] \implies x \in R_0[a] \cap R_0[a^{-1}]$ by maximality of $R_0$.
Write

$$x = \sum_{i=0}^{r} r_i a^i = \sum_{j=0}^{m} s_j a^{-j}$$

for $m, n$ minimal and $r_i, s_j \in R_0$. Without loss of generality assume $m \leq n$.

$$x = s_0 + (x - s_0)\left(x^{-1} \sum_{i=0}^{n} r_i a^i\right) = s_0 + (1 - \frac{s_0}{x}) \sum_{i=0}^{n-1} r_i a^i + \sum_{j=1}^{m} s_j a^{-j} x^{-1} r_n a^n$$

This is a polynomial of degree $< n$ in $a \rightarrow\leftarrow$.
So $R_0$ is a valuation ring. □

### 2.3.2 Discrete valuation rings

Recall:
A discrete valuation ring (DVR) is a valuation ring of a discrete valuation
A discretely valued field is a fraction field or a DVR

**Proposition 6.** *$R$ valuation ring, $P \trianglelefteq R$ maximal. TFAE:*

1. *$R$ is Noetherian but not a field*
2. *$R$ is a PID but not a field*
3. *$R$ is a DVR*

*Proof.* (1) $\Rightarrow$ (2): Let $I \trianglelefteq R$. Since $R$ is Noetherian, $I = (a_1, \ldots, a_n)$.

Let $i$ be such that $v(a_i)$ is minimal. Then by Lemma 2, $a_j \in (a_i)$ for all $j \implies I = (a_i)$.
(2) $\Rightarrow$ (3): $P \neq 0$ since $R$ is a field. $P = (\pi)$ for some $\pi \in R$ prime.

$R = R_{(\pi)}$ is the localization of a UFD at $(\pi) \implies R$ is a DVR as in our examples.
(3) $\Rightarrow$ (1): If $R$ a field then $R^\times = K^\times$ so $v(R^\times) = v(K^\times) = \{1\}$, hence a DVR is not a field.

So say $0 \neq I \trianglelefteq R$. Let $a \in I$ be such that $v(K^\times) \subseteq \mathbb{N}$. So $I = (a)$ by Lemma 2. $\square$

**Theorem 7.** *Let $R$ be a Noetherian local ring. Then TFAE:*

1. *$R$ is a DVR*
2. *$R$ is a PID but not a field*
3. *$R$ is a UFD of dimension 1*
4. *$R$ is a normal domain of dimension 1*
5. *$R$ is a 1-dimensional regular (local) ring*

*Proof.* (1) $\Rightarrow$ (2): by Proposition 6

(2) $\Rightarrow$ (3): local PID but not field $\implies$ dim $= 1$.

(3) $\Rightarrow$ (4): UFDs are integrally closed.

(4) $\Rightarrow$ (5): $R$ local domain of dimension 1, normal
$P$ maximal ideal $\implies P, (0)$ are the only prime ideals.

WTS: $P$ is principal ($P$ is principal $\iff 1 = \dim_K(P/P^2) =$ minimal number of generators).

$0 \neq a \in P \implies \sqrt{(a)} =: P$

$\mathcal{M} = \{(a : y) \mid y \in R \setminus (A)\}$
Recall: $(a : y) = \{x \in R \mid xy \in (a)\}$

$R$ Noetherian $\implies \mathcal{M}$ has a maximal element $Q$
$Q = (a : b)$ for some $b \in R \setminus (a)$

We claim: $Q$ is prime.
$b \notin (a) \implies Q \neq R$.
If $x, y \notin Q = (a : b)$, $xb \notin (a)$ and $(a : b) \leq (a : xb) \in \mathcal{M} \implies (a : xb) = Q$ by maximality.

So $y \notin (a : xb) \implies xy \notin (a : b) = Q$ so $Q$ is prime.

But $(a) \subseteq Q \implies Q = P$ maximal ideal of $R \implies (a : b) = P$.

Let $I := \frac{b}{a} \cdot P \subseteq \mathrm{Quot}(R)$ $R$-submodule
$I \subseteq R$ since $P = (a : b) \implies I \trianglelefteq R$.

Assume $I \subseteq P$ (i.e. $I \neq R$).

Then $P$ is a module over $R[\frac{b}{a}]$.
Consider $\varphi : P \to P, \quad x \mapsto \frac{b}{a} \cdot x$.
As usual, Cayley-Hamilton $\implies \frac{b}{a}$ is integral over $R \implies \frac{b}{a} \in R$, since $R$ is normal $\implies b \in (a) \implies I = R \rightarrow\leftarrow$ So we must have $I = R$.

But $I = R \implies P = \frac{a}{b} \cdot R \implies P$ is principal.

(5) $\Rightarrow$ (1): Exercise. Use Lemma 1. $\square$

**Corollary 8.** *$R$ DVR with maximal ideal $P \implies$ the nonzero ideals in $R$ are exactly the powers of $P$. They form a strictly decreasing chain $R = P^0 \supsetneq P \supsetneq P^2 \supsetneq \cdots$.*

*Proof.* Take an ideal $I \trianglelefteq R$. By Proposition 6, $I$ is principal. Say $I = (a)$ for some $a$. Then $I = P^n$ where $n = v(a)$.

If $P^{n+1} = P^n$ for some $n$, then $P \cdot P^n = P^n \implies P^n = 0$ by Nakayama. So $P = 0$. $\square$

### 2.3.3 Dedekind domains

**Definition.** An integral domain $R$ is a **Dedekind domain** if $R$ is Noetherian of dimension 1 and for all maximal ideals $P \trianglelefteq R$, $R_P$ is a regular local ring.

**Lemma 9.** *A 1-dimensional Noetherian domain is a Dedekind domain $\iff$ it is normal.*

*Proof.* Exercise. $\square$

**Example.**

1. $\mathcal{X}$ irreducible curve over an algebraically closed field $K$, smooth
   Then $k[\mathcal{X}]$ is a domain (because $\mathcal{X}$ is irreducible)
   Is 1-dimensional
   And all localizations at maximal ideals are regular (by $\mathcal{X}$ being smooth + Nullstellensatz)
   So $k[\mathcal{X}]$ is a Dedekind domain.

2. $\mathbb{Z}$ (or any PID that is not a field)

**Proposition 10.** *If $K$ is a number field (i.e. $K \geq \mathbb{Q}$ finite extension) and $R \subseteq K$ is the integral closure of $\mathbb{Z}$, then $R$ is a Dedekind domain.*

*Proof.* $R \leq K$ subring so $R$ is an integral domain.
$\dim R = \dim \mathbb{Z} = 1$ since $R/\mathbb{Z}$ is integral.
$R$ is normal because integrality is transitive.
By Lemma 9, just left to show $R$ is Noetherian.

Step 1: $|R/pR| < \infty$ for all $p \in \mathbb{Z}$ prime:
$R/pR$ is an $\mathbb{F}_p$-vector space so suffices to show that $\dim_{\mathbb{F}_p}(R/pR) < \infty$.
We'll show $\dim_{\mathbb{F}_p}(R/pR) \leq [K : \mathbb{Q}]$.
Let $\overline{a_1}, \ldots, \overline{a_n} \in R/pR$ be $\mathbb{F}_p$-linearly independent. Take preimages $a_i \in R$.
Suppose for contradiction that $\lambda_1 a_1 + \cdots + \lambda_n a_n = 0$ for $\lambda_i \in \mathbb{Q}$ not all 0.
Without loss of generality assume $\lambda_i \in \mathbb{Z}$ and there exists $j$ such that $p \nmid \lambda_j$.
Reduce mod $p$: $= \overline{\lambda_1}\overline{a_1} + \cdots + \overline{\lambda_n}\overline{a_n}$ is a nontrivial $\mathbb{F}_p$ relation on $\overline{a_1}, \ldots, \overline{a_n}$ $\rightarrow\leftarrow$

Step 2: $|R/mR| < \infty$ for all $m \in \mathbb{Z} \setminus \{0\}$:
We can assume $m$ is not prime, so say $m = m_1 m_2$. We get a short exact sequence

$$0 \to R/m_1R \to R/m_1m_2R \to R/m_2R \to 0$$

and

$$|R/mR| = |R/m_1R| \cdot |R/m_2R|$$

so the claim follows by induction on Step 1.

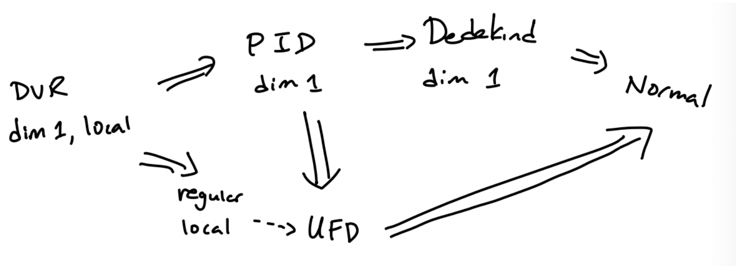
Step 3: We'll show any $0 \neq I \trianglelefteq R$ must contain some $m \in \mathbb{Z}, m \neq 0$.

Suppose not. Then $\dim(R/I) = \dim(\mathbb{Z}/\mathbb{Z} \cap I) = \dim \mathbb{Z} = 1$, but $I$ is nonzero, so $\dim(R/I) < \dim R = 1$ $\rightarrow\leftarrow$

Now pick $m \in \mathbb{Z} \cap I, m \neq 0$. Then $mR \subseteq I$.
$|I/mR| \leq |R/mR| < \infty$ by Step 2 $\implies$ $I/mR = \{\overline{a_1}, \ldots, \overline{a_n}\}, a_i \in I$ $\implies$ $I = (m, a_1, \ldots, a_n)$ finitely generated. So $R$ is Noetherian. $\square$

**Example.** $R = \mathbb{Z}[i\sqrt{5}]$
This is the integral closure of $\mathbb{Z}$ in $\mathbb{Q}(i\sqrt{5})$ so is a Dedekind domain.
But this is not a UFD ($2 \cdot 3 = (1 + i\sqrt{5})(1 - i\sqrt{5})$).

Summary:




**Proposition 11.** *Let $R$ be a Dedekind domain.*

1. *If $P \trianglelefteq R$ maximal, $Q \trianglelefteq R$, then*
   *$Q$ is $P$ primary $\iff$ $Q = P^k$ for some (unique) $k \in \mathbb{N}$*

2. *$0 \neq I \trianglelefteq R \implies I = P_1^{k_1} \cdots P_n^{k_n}$ ($k_i \in \mathbb{N}_{>0}, P_1, \dots, P_n$ distinct maximal ideals). This decomposition is unique up to order, and $Ass(I) = \{P_1, \dots, P_n\}$.*

*Proof.* 1. $(\Leftarrow)$ : true in any ring
$(\Rightarrow)$ : $Q$ $P$-primary $\implies$ consider $R \to R_P$. Then $Q^e \trianglelefteq R_p$ is nonzero.
$R_P$ is a DVR by Theorem 7, so $Q^e = (P^e)^k$ by Corollary 8.

Easy to check: $(P^e)^k = (P^k)^e$.
Contract: $Q = P^k$ (both are $P$-primary: localize at $S = R \setminus P$)
If $P^k = P^{k'}$, then $(P^e)^k = (P^e)^{k'}$ $\implies$ by Corollary 8, $k = k'$ (the inclusions are strict).

2. $R$ Noetherian $\implies I = Q_1 \cap \cdots \cap Q_n$ primary decomposition.
$\sqrt{Q_i} = P_i \implies P_1, \dots, P_n$ are distinct and maximal (since $R$ has dimension 1).
Then there are no embedded primes. So $Q_1, \dots, Q_n$ are unique. Then by part 1, $Q_i = P_i^{k_i}$ for some unique $k_i$.
$I = P_1^{k_1} \cap \cdots \cap P_n^{k_n} = P_1^{k_1} \cdots P_n^{k_n}$ since they are pairwise coprime.

□

**Example.** In $\mathbb{Z}[i\sqrt{5}]$, 2 is not prime but is irreducible. So 2 does not have a prime factorization.

Recall that $2 \mid 6$ and $6 = (1 + i\sqrt{5})(1 - i\sqrt{5})$.

$(2) \subsetneq (1 + i\sqrt{5}, 2)$, and the ideal on the right is maximal.

One can show: $(2) = (2, 1 + i\sqrt{5})^2$.

**Lemma 12.** *Let $R$ be a Dedekind domain.*

1. *If $P_1, \dots, P_n$ are distinct maximal ideals of $R$, $k_1, l_i \in \mathbb{N}$, then* $P_1^{k_1} \cdots P_n^{k_n} \subseteq P_1^{l_1} \cdots P_n^{l_n} \iff l_i \leq k_i$ *for all* $i$.
2. $a \in R \setminus \{0\} \implies (a) = P_1^{v_1(a)} \cdots P_n^{v_n(a)}$ *where* $\{P_1, \dots, P_n\} = Ass((a))$ *and* $v_i$ *is the valuation on* $R_{P_i}$
3. $a \in R \setminus \{0\}$, $P$ *maximal* $\implies P \notin Ass((a)) \implies v_p(a) = 0$.

*Proof.* Exercise. □

**Proposition 13.** *$I, J$ nonzero ideals in Dedekind domain, $I = P_1^{k_1} \cdots P_n^{k_n}$ and $J = P_1^{l_1} \cdots P_n^{l_n}$. Then*

$$I + J = P_1^{m_1} \cdots P_n^{m_n}, \quad m_i = \min\{l_i, k_i\}$$
$$I \cap J = P_1^{m_1} \cdots P_n^{m_n}, \quad m_i = \max\{l_i, k_i\}$$
$$IJ = P_1^{m_1} \cdots P_n^{m_n}, \quad m_i = l_i + k_i$$

*In particular $IJ = (I \cap J)(I + J)$.*

*Proof.* Exercise □

**Proposition 14.** *$R$ Dedekind domain, $I \trianglelefteq R$. $0 \neq a \in I \implies \exists b \in I$ so that $(a, b) = I$. In particular, every ideal in $R$ is generated by two elements.*

*Proof.* Let $0 \neq (a) \subseteq I$. Write $(a) = P_1^{k_1} \cdots P_n^{k_n}$ and $I = P_1^{l_1} \cdots P_n^{l_n}$ where $P_1, \dots, P_n$ are distinct maximal ideals.

By Lemma 12, $l_i \leq k_i$. For each $i$, there exists some

$$b_i \in (P_1^{l_1+1} \cdots P_i^{l_i} \cdots P_n^{l_n}) \setminus (P_1^{l_1+1} \cdots P_i^{l_i+1} \cdots P_n^{l_n}) \neq \varnothing$$

by uniqueness in Prop 11.
Notice that $b_i \in I$ for each $i$.
$b_i \in P_j^{l_j+1}$ for all $j \neq i$.
But $b_i \notin P_i^{l_i+1}$ (by Prop 13, this would force it to be in $P_1^{l_1+1} \cdots P_i^{l_i+1} \cdots P_n^{l_n}$).

Let $b := b_1 + \cdots + b_n \in I$. Then $b \notin P_i^{l_i+1}$ for all $i$.

We claim: $I = (a, b)$.
Know: $(a, b) = (a) + (b)$ and Prop 13 $\implies$ the prime factorization of $(a, b)$ contains only the maximal ideals $P_1, \dots, P_n$. So $(a, b) = P_1^{m_1} \cdots P_n^{m_n}$ for some $m_i$'s.

By Lemma 12, since $(a, b) \subseteq I$, $m_i \geq l_i$ for all $i$. But also $m_i \leq l_i$ for all $i$, since $b \notin P_i^{l_i+1}$ for all $i$. So $(a, b) = P_1^{l_1} \cdots P_n^{l_n} = I$. □

### 2.3.4 Fractional ideals, Cartier divisors, and the ideal class group

**Definition.** Let $R$ be a domain and $K = \text{Quot}(R)$. A **fractional ideal** of $R$ is an $R$-submodule $I$ of $K$ such that $aI \subseteq R$ for some $a \in R \setminus \{0\}$.

For $a_1, \ldots, a_n \in K$, $(a_1, \ldots, a_n) := Ra_1 + \cdots + Ra_n \subseteq K$ is the **fractional ideal generated by** $a_1, \ldots, a_n$

**Example.**

1. If $I \subseteq R$ then $I$ is a fractional ideal $\iff I \trianglelefteq R$.
2. $(1/2) = \frac{1}{2}\mathbb{Z} \subseteq \mathbb{Q}$ is a fractional ideal of $\mathbb{Z}$
3. $\mathbb{Z}_{(2)} \subseteq \mathbb{Q}$ is not a fractional ideal. This is a $\mathbb{Z}$-submodule, but you cannot multiply by a single element to get it into $\mathbb{Z}$

**Note.** Let $R$ be a domain and $K = \text{Quot}(R)$.

1. If $R$ is Noetherian, then every fractional ideal of $R$ is generated by finitely many elements of $K$
2. Operations on ideals extend to fractional ideals, e.g.

$$IJ = \{\sum_{i=1}^{n} a_i b_i \mid n \in \mathbb{N}, a_i \in I, b_i \in J\}$$

$$I :_K J = \{a \in K \mid aJ \subseteq I\}$$

**Definition.** Let $I \subseteq K$ be an $R$-submodule.
We say $I$ is an **invertible ideal** or **Cartier divisor** if $IJ = R$ for some $R$-submodule $J \subseteq K$.
$I$ is **principal** if $I = (a)$ for some $a \in K$.

**Lemma 15.** *Let $R$ be a domain and $K = Quot(R)$. Let $I \subseteq K$ be an $R$-submodule.*

1. *If $I$ is invertible with $IJ = R$ then $J = R :_K I$.*
2. *If $I$ is nonzero principal then $I$ is invertible*
3. *If $I$ is invertible then $I$ is a fractional ideal of $R$*

*Proof.*
1. Say $IJ = R$, Then $R = IJ \subseteq I \cdot (R :_K I) \subseteq R) \implies IJ = I \cdot (R :_K I)$. Multiply both sides by $J$: $J = R :_K I$.
2. $I = (a), a \neq 0 \implies a \in K^\times \implies a^{-1} \in K \implies (a) \cdot (a^{-1}) = R \implies I$ is invertible.

3. $I$ invertible $\implies I(R :_K I) = R$ by (1).
   Want to find $a \in R$ such that $aI \subseteq R$.

$$1 = \sum_{i=1}^{n} a_i b_i \text{ for } a_i \in I, b_i \in R :_K I \text{ for some suitable } n.$$

   If $b \in I$, then $b = b\sum_{i=1}^n a_i b_i = b\sum_{i=1}^n a_i(b_i b)$ and $b_i b \in R$ since $b \in I$ and $b_i \in R :_K I$. Let $a$ be the product of the denominators of $a_i$. Then $ab \in R$ and since $b$ was arbitrary, $aI \subseteq R$. So $I$ is fractional.

$\square$

**Example.**

1. Let $R$ be a PID and $I$ a fractional ideal of $R$. Then $aI \trianglelefteq R$ for some $a$.
   $aI = (b)$ for some $b$ since $R$ is a PID. So $I = (a/b)$ is principal.

   Thus all fractional ideals of a PID are principal. Then by Lemma 15, in a PID, $I$ nonzero principal $\iff$ $I$ invertible $\iff$ $I$ fractional.

2. $I = (x, y) \trianglelefteq \mathbb{R}[x, y]$, $K = \mathbb{R}(x, y)$.

$$R :_K I = \{f \in \mathbb{R}[x, y] \mid xf \in \mathbb{R}[x, y] \text{ and } yf \in \mathbb{R}[x, y]\} = \mathbb{R}[x, y].$$

   But $I \cdot (R :_K I) = (x, y)\mathbb{R}[x, y] = (x, y) \subsetneq \mathbb{R}[x, y]$. So by Lemma 15, $I$ is not invertible.

**Proposition 16.** *In a Dedekind domain, every nonzero fractional ideal is invertible.*

*Proof.* $R$ Dedekind domain, $I \neq 0$ fractional ideal of $R$.

Suppose for contradiction that $I$ is not invertible. So $I(R :_K I)$ is a proper ideal of $R$. Then $I(R :_K I)$ is contained in some maximal ideal $P$. Extend to $R_P$:

$$I^e(R_P :_K I^e) \subseteq P^e \subsetneq R_P \quad \text{(exercise)}$$

By Lemma 15, $I^e$ is not invertible. $R_P$ is a DVR so is a PID $\rightarrow\leftarrow$ . $\square$

**Note.** Let $R$ be a domain and $K = \mathrm{Quot}(R)$.

1. The invertible ideals of $R$ form an abelian group under multiplication with $I^{-1} := (R :_K I)$.

2. The nonzero principal ideals form a (normal) subgroup

**Definition.** The group of invertible ideals of a domain $R$ is called the **ideal group** or **group of Cartier divisors** $\mathrm{Div}(R)$.
The principal invertible ideals form a normal subgroup $\mathrm{Prin}(R) \trianglelefteq \mathrm{Div}(R)$, and

$$\mathrm{Pic}(R) := \mathrm{Div}(R)/\mathrm{Prin}(R)$$

is the **ideal class group** or the **Picard group** of $R$.

**Example.** If $R$ is a PID, $\mathrm{Pic}(R) = 1$.

**Proposition 17.** *If $I$ is an invertible ideal in a Dedekind domain, then $I = P_1^{k_1} \cdots P_n^{k_n}$ for suitable distinct maximal ideal of $R$ and $k_i \in \mathbb{Z}$, unique up to order.*

*Proof.* $aI \subseteq R$ for some $a \in R \setminus \{0\}$ since invertible implies fractional, by Lemma 15.

$(a) = P_1^{s_1} \cdots P_n^{s_n}$
$I = (a)^{-1} aI = P_1^{r_1 - s_1} \cdots P_n^{r_n - s_n}$.

Uniqueness is inherited from Prop 11. □

**Note.** $R$ Dedekind means we have a map

$$\mathrm{Div}(R) \to \text{free abelian groups on } \mathrm{Specmax}(R),$$

$$P_1^{k_1} \cdots P_n^{k_n} \mapsto (\varphi : \mathrm{Specmax}(R) \to \mathbb{Z}, \quad P_i \mapsto k_i)$$

**Proposition 18.** *Let $R$ be a Dedekind domain. Then TFAE:*

1. *$R$ is a PID*
2. *$R$ is a UFD*
3. $\mathrm{Pic}(R) = 1$

*Proof.* (1) $\Rightarrow$ (2): we showed this last semester.
(2) $\Rightarrow$ (3):
If $P \trianglelefteq R$ is maximal, then $P$ is also a minimal nonzero prime ideal $\dim R = 1$), so $b \in P \implies b = p_1 \cdots p_s$ by unique factorization $\implies p_i \in P$ for some $i$.

$0 \subsetneq (p_i) \subseteq P \implies (p_i) = P \implies P$ is principal. So $\mathrm{Pic}(R) = 1$. (3) $\Rightarrow$ (1):
If $\mathrm{Pic}(R) = 1$ then every invertible ideal is principal. But every nonzero ideal is invertible, by $R$ being a Dedekind domain. So every ideal is principal. □

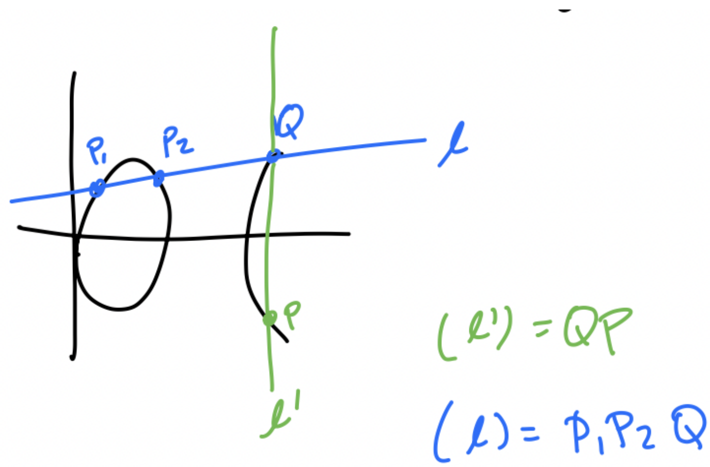
**Example.** $\mathcal{X} = V(y^2 - x(x-1)(x-\lambda)) \subseteq \mathbb{A}^2_{\mathbb{C}}$.
$(\lambda \in \mathbb{C} \setminus \{0\})$

There is no point for which both terms of the Jacobian of $f$ are 0, so $\mathcal{X}$ is smooth.

Take $\ell \in k[\mathcal{X}] \setminus \{0\}$ a "general" linear $\ell$.
$\ell$ vanishes to order 1 at 3 points.
By Lemma 12b, $(\ell) = P_1 P_2 P_3$.

Define $\varphi : \mathcal{X} \to \mathrm{Pic}(R), a \mapsto I(a) + \mathrm{Prin}(R)$.
Fact: This is injective and $\varphi(\mathcal{X}) = \mathrm{Pic}(R) \setminus \{(\overline{1})\}$.
Extend $\varphi$ to $\mathcal{X} \cup \{\infty\} \to \mathrm{Pic}(R)$. This extension will be bijective.

$\overline{P_1 P_2} = \overline{QP} = 1$ in the Picard group, so $\overline{P_1 P_2} = \overline{P}$.




### 2.3.5 Projective modules over PIDs

**Theorem 19.** *Let $R$ be a PID. If $M$ is a free $R$-module and $N \leq M$, then $N$ is free. In particular, every projective $R$-module is free.*

*Proof.* Let $M$ be the free $R$-module on the set $K$ (i.e. $M = F_K$).
So $F$ has basis $\{x_K \mid k \in K\} \subseteq M$.

We may well-order $K$ i.e. equip $K$ with a total order such that every nonempty subset has a minimal element. Write 0 for the minimal element in $K$.

Define $F_0 := \{0\}$.
For $k \in K$, define
$$F_k := \bigoplus_{i<k} Rx_i.$$
Let
$$\overline{F_k} := \bigoplus_{i \leq k} Rx_i = F_k \oplus Rx_k.$$
$(\overline{F_0} := Rx_0)$
Note:
$$M = \bigcup_{k \in K} \overline{F_k}.$$
Let $a \in N \cap \overline{F_k}$. Then $a = b + rx_k$ where $b \in F_k$, $r \in R$, and both are unique.
Define
$$\varphi_k : N \cap \overline{F_k} \to R, \quad a \mapsto r$$
From this we get a short exact sequence
$$0 \to N \cap F_k \to N \cap \overline{F_k} \xrightarrow{\varphi_k} \mathrm{Im}(\varphi_k) \to 0$$
Now $\mathrm{Im}(\varphi_k)$ is an ideal of $R$ and $R$ is a PID, so $\mathrm{Im}(\varphi_k)$ is free, hence the sequence splits.

Then $N \cap \overline{F_k} = (N \cap F_k) \oplus C_k$ for some $C_k \cong \mathrm{Im}(\varphi_k)$.

We claim: $N = \bigoplus C_k$. This would show that $N$ is free because the $C_k$'s are free.
Proof of claim:

1. First we'll show that $N = \sum C_k =: C$. It is clear that $C \leq N$. Suppose for contradiction that $C \subsetneq N$. For $a \in N$, let $\mu(a) := \min\{k \in K \mid a \in \overline{F_k}\}$.
   Let $J := \{\mu(a) \mid a \in N \setminus C\}$.
   By our assumption, $J \neq \varnothing$, so it has a minimal element $j$.

   $j = \mu(y)$ for some $y \in N \setminus C$.

   Then $y \in N \cap \overline{F_j} = (N \cap F_j) \oplus C_j$ i.e. $y = b + c$ where $b \in N \cap F_j$ and $c \in C_j$.
   Thus $b = y - c$ so $b \in N \setminus C$.
   So $\mu(b) < \mu(y) = j \rightarrow\leftarrow$ (contradicts minimality of $j$)

2. We claim the sum is direct:
   Suppose $c_1 + \cdots + c_n = 0$ for some $c_i \in C_{k_i}$, $k_1 < \cdots < k_n$. Suppose $k_n$ is minimal among all such relations. Then

$$-c_n = c_1 + \cdots + c_{n-1} \in (N \cap F_{k_n}) \cap C_{k_n} = \{0\} \implies c_n = 0 \rightarrow\leftarrow$$

   This contradicts the minimality of $k_n$.

□

**Note.** You can use this same argument to show that if $R$ is a ring in which every ideal is projective as an $R$-module, then every submodule of a projective module is a direct sum of ideals.

Our current argument shows this but for the statement where the second "projective" is replaced by "free."

# 3 Fields and Galois Theory

## 3.1 Fields and Field Extensions

### 3.1.1 Prime Fields

Notation: $k_1 \leq k_2$ denotes fields. We also write $k_2/k_1$ ($k_2$ over $k_1$)
In this case, $k_2$ is a **field extension** of $k_1$, and $k_1$ is a **subfield** of $k_2$.

**Proposition 1.** *Let $L$ be a field and $I \neq \varnothing$ an index set. If $L_i \leq L$ for all $i \in I$, then $\bigcap_{i \in I} L_i \leq L$. That is, the intersection of arbitrarily many subfields is a subfield.*

**Definition.** Let $I \neq \varnothing$, $L$ be a field, and $L_i \leq L$ for $i \in I$.
Let $\mathcal{N} := \{N \leq L \mid L_i \leq N \text{ for all } i \in I\}$. Then $\bigcap_{N \in \mathcal{N}} N$ is the **compositum** of the $L_i$.

**Definition.** The **prime field**

$$\bigcap_{N \leq L} N$$

of $L$ is the smallest subfield contained in $L$.

**Theorem 2.** *The set of subfields of a given field forms a* ***lattice*** *with respect to intersection and composition.*

**Definition.** A ring homomorphism $\varphi : k_1 \to k_2$ between fields is called a **homomorphism of fields**. If $\varphi$ is bijective, it is an **isomorphism**. A map $\varphi : K \to K$ is an **endomorphism**, and an **automorphism** if bijective.

**Note.** Every nontrivial homomorphism of fields must be injective (because the kernel is a subfield). So if $\varphi : k_1 \to k_2$ is a homomorphism of fields, $k_1 \cong \varphi(k_1) \leq k_2$ so we may view $k_1$ as a subfield of $k_2$.

**Theorem 3.** *The prime field of a field is either* $\mathbb{Q}$ *or* $\mathbb{F}_p$*,* $p$ *prime.*

*Proof.* Let $K$ be a field. The map $\varphi : \mathbb{Z} \to K$ defined by $n \mapsto 1 + \cdots + 1$ ($n$ times) is a ring homomorphism.

Its kernel $\ker \varphi$ is a principal ideal $(m)$.

Since $K$ is a domain, either $m = 0$ (making $\varphi$ injective and hence extend to $\mathbb{Q}$) or $m = p$ for some prime $p$ (making $\varphi(K) \cong \mathbb{F}_p$). □

**Definition.** The **characteristic** of a field $k$ is defined as

$$\operatorname{char}(k) := \begin{cases} 0 & \text{if the prime field is } \mathbb{Q} \\ p & \text{if the prime field is } \mathbb{F}_p \end{cases}$$

**Example.**

1. $\operatorname{char}(\mathbb{C}) = \operatorname{char}(\mathbb{R}) = \operatorname{char}(\mathbb{Q}) = 0$
2. $\operatorname{char}(\mathbb{F}_p) = p = \operatorname{char}(\mathbb{F}_{p^n}) = \operatorname{char}(\mathbb{F}_p(t))$

### 3.1.2 Algebraic and Transcendental Elements

**Definition.** Let $L \geq K$ be fields.

$x \in L$ is **algebraic** over $K$ if $f(x) = 0$ for some non-zero $f \in K[T]$.

The monic polynomial $f_x$ of minimal degree with this property is the **minimal polynomial** of $x$ over $K$.

If $x$ is not algebraic, it is **transcendental** over $K$.

**Definition.** If $L \geq K, x \in L$, the subfield of $L$ generated by $x$ over $K$ is

$$K(x) := \bigcap_{K \leq E \leq L, x \in E} E$$

For a set $B \subseteq L$, $K(B)$ is the compositum of all $K(x)$ for $x \in B$.

**Proposition 4.** *Let $L \geq K$ and $x \in L$. Define $\varphi : K[T] \to L$ by $T \mapsto x$.*

1. *$\varphi$ is injective $\implies$ $x$ is transcendental and $K(x) \cong K(T)$ (rational function field)*
2. *$\varphi$ is not injective $\implies$ $x$ is algebraic. Then $\ker \varphi = (f_x)$ and $K(x) \cong K[T]/(f_x)$.*

**Example.** Algebraic:

1. $i \in \mathbb{C}$, $f_i := T^2 + 1 \in \mathbb{R}[T]$ minimal polynomial over $\mathbb{R}$ (or $\mathbb{Q}$)
2. $\eta = e^{2\pi i/k}$ if $k$ is prime, $f_\eta = \frac{T^k - 1}{T - 1}$ over $\mathbb{R}$ or $\mathbb{Q}$

Transcendental:

1. $\pi, e$
2. $\mathbb{C}$ has infinitely many transcendentals over $\mathbb{Q}$

#### 3.1.3 Finite Field Extensions

**Note.** Any field extension $L \geq K$ allows $L$ to be viewed as a vector space over $K$

**Definition.** Let $L \geq K$ be a field extension.

The **degree** of $L$ over $K$ is $[L : K] := \dim_K L$

The extension $L/K$ is **finite** if $[L : K] < \infty$

For $x \in L$, $[K(x) : K]$ is the **degree of $x$ over $K$**

**Example.**

1. $[\mathbb{C} : \mathbb{R}] = 2, [\mathbb{R} : \mathbb{Q}] = \infty$
   $[K(x) : K] = \infty$ when $x$ is transcendental
2. $[L : K] = 1 \iff L = K$

**Theorem 5.** *Let $M \geq L \geq K$. $M/K$ is finite $\iff$ $M/L$ and $L/K$ are finite.*
*In this case, $[M : K] = [M : L][L : K]$.*

### 3.1.4 Algebraic Extensions

Recall: $L/K$ is algebraic if every $x \in L$ is algebraic over $K$ (transcendental otherwise)

**Theorem 6.** *Let $L \geq K$ be a field extension.*

1. *If $L/K$ is finite, then it is algebraic*
2. *If $x \in L$ is algebraic over $K$, $[K(x) : K] = \deg f_x$*

*Proof.* 1. For $x \in L$, the set $\{1, x, \ldots, x^n\}$ is linearly dependent (by assumption) where $n = [L : K] < \infty$. So there exists a non-zero $f \in K[T]$ such that $f(x) = 0$.

2. $K(x) \cong K[T]/(f_x) \cong \bigoplus_{i=1}^{\deg f_x - 1} K \cdot \overline{T}^i$ as $K$-vector spaces

□

**Note.** Algebraic extensions need not be finite. For example

$$\overline{\mathbb{Q}} = \{c \in \mathbb{C} \mid c \text{ is algebraic over } \mathbb{Q}\}$$

is not finite over $\mathbb{Q}$.

**Theorem 7.** *Let $L/K$ be algebraic. The following are equivalent:*

1. *$L/K$ is finite*
2. *There exists finite $B \subseteq L$ such that $L = K(B)$ ($L$ is finitely generated over $K$)*

*Proof.* $(\Rightarrow)$ : Take a basis.

$(\Leftarrow)$ : Let $B = \{x_1, \ldots, x_m\}, K(B) = L$. Induction on $m$:
If $m = 0$ then $L = K$
If $m \geq 1$, $L = K(x_1, \ldots, x_m)/K(x_1, \ldots, x_{m-1})$. Let $E = K(x_1, \ldots, x_{m-1})$,

$L = E(x_m)$ is finite over $E$. $E$ is finite over $K$ by induction. So $L/K$ is finite by Theorem 5. □

**Theorem 8.** *$M \geq L \geq K$.*
*$M/K$ is algebraic $\iff$ $M/L$ and $L/K$ are algebraic.*

*Proof.* $(\Rightarrow)$ : immediate.
$(\Leftarrow)$ : Let $x \in M$. Then $x$ is algebraic over $L \implies \sum_{i=0}^{n} a_i x^i = 0$ for some $n \in \mathbb{N} a_i \in L$.
$K' := K(a_0, \ldots, a_n)$. Then $x$ is algebraic over $K'$ (subfield of $L$).

Note: each $a_i$ is algebraic over $K$ since $L/K$ is. So by Theorem 7, $K'/K$ is finite.
Theorem 6 $\implies$ $K'(x)/K'$ is finite. So $K'(x)/K$ is algebraic by Theorem 6, hence $x$ is algebraic over $K$. □

**Theorem 9.** *$L \geq K$.*
*The set $\tilde{K} := \{x \in L \mid x \text{ is algebraic over } K\}$ is a subfield of $L$.*

*Proof.* $0 \neq x, y \in \tilde{K} \implies xy, x + y, x^{-1}y, \in E = K(x, y)$. But $E/K$ is finite so $E/K$ is algebraic, meaning $xy, x + y, x^{-1}y, \in \tilde{K}$. □

$\tilde{K}$ is called the **algebraic closure** of $K$ in $L$ (relative algebraic closure)

### 3.1.5 Transcendental Extensions

Recall: $L \geq K$ field extension $\implies \exists B \subseteq L$ transcendence basis so that $L/K(B)$ is algebraic and $B$ is algebraically independent.

If a finite $B$ exists, its length is uniquely determined and is called **transcendence degree**.

**Theorem 10.** *For $M \geq L \geq K$, $tr\ deg(M/K) = tr\ deg(M/L) + tr\ deg(L/K)$.*

## 3.2 Splitting Fields and Algebraic Closure

### 3.2.1 Stem Fields

**Definition.** $L/K$ is simple if $L = K(x)$ for some $x \in L$.

**Theorem 1** (Kronecker)**.** *If $K$ is a field and $f \in K[T]$ is irreducible of degree $n$, there exists a field extension $L/K$ with $[L : K] = n$ such that $L = K(x)$ for some $x \in L$ with $f(x) = 0$.*

*Proof.* Take $L := K[T]/(f)$. This is a field since $f$ is irreducible. $x := T + (f)$ works. $\square$

**Definition.** The field $L$ in Theorem 1 is called a **stem field** for $f$.

**Corollary 2.** *If $f \in K[T]$ is irreducible, all stem fields of $f$ are isomorphic.*

*Proof.* Say $L_1, L_2$ are stem fields of $f$, $L_1 = K(x_1)$ and $L_2 = K(x_2)$. Then $f$ irreducible means since $f_{x_1}, f_{x_2}$ both divide $f$, they must equal $f$ (up to nonzero constant). Then

$$L_1 = K[T]/(f_{x_1}) = K[T]/(f_{x_2}) = L_2$$

$\square$

**Example.** If $f(T) = T^3 - 2 \in \mathbb{Q}[T]$, if $a$ is a root of $f$, the stem fields are $\mathbb{Q}(a), \mathbb{Q}(\zeta a), \mathbb{Q}(\zeta^2 a)$ where $\zeta^3 = 1$ and $\zeta \neq 1$.

**Theorem 3.** *Let $K_1, K_2$ be fields. Let $\varphi : K_1 \to K_2$ be an isomorphism. Then we have a map $\tilde{\varphi} : K_1[T] \to K_2[T]$ induced by $\varphi$ which sends $T \mapsto T$ and such that $\tilde{\varphi}|_{K_1} = \varphi$.*
*Given $f \in K_1[T]$ irreducible, $f_2 := \tilde{\varphi}(f) \in K_2[T]$,*

$$L_1 := K_1(x_1), \quad f(x_1) = 0, \quad L_2 := K_2(x_2), \quad f_2(x_2) = 0$$

*there exists a unique isomorphism $\phi : L_1 \to L_2$ with $\phi|_{K_1} = \varphi$, $\quad \phi(x_1) = x_2$.*

$$\begin{tikzcd}
K_1[T] \arrow[rr, "\tilde{\varphi}"] \arrow[dr, "\pi_1"] & & K_2[T] \arrow[dr, "\pi_2"] & \\
& L_1 \arrow[rr, dashed, "\phi"] & & L_2 \\
K_1 \arrow[uu, hook] \arrow[ur, hook] \arrow[rr, "\varphi"'] & & K_2 \arrow[uu, hook] \arrow[ur, hook] &
\end{tikzcd}$$

*Proof.* Let $\pi_i : K_i[T] \to L_i, T \mapsto k_i$.
$\ker \pi_i = (f_i) = f_i K_i[T]$ (note $f_2$ is also irreducible)

Let $\psi := \pi_2 \circ \tilde{\varphi} : K_1[T] \to L_2$.
$\ker \psi \supseteq (f_1)$ but $f_1$ is irreducible, so $\ker \psi = (f_1)$. Hence there exists $\phi : L_1 \to L_2$ with $\phi \circ \pi_1 = \psi$.
The map is surjective and injective by construction, so $\phi$ is an isomorphism.

**Uniqueness: exercise.** □

### 3.2.2 Splitting Fields

**Definition.** Let $K$ be a field, $f \in K[T] \setminus K$. Then $N/K$ is a **splitting field** for $f \in K[T]$ if $f$ splits into linear factors $f = x_0 \prod (T - x_i)$ in $N[T]$ $(x_i \in N)$ and $N = K(x_1, \ldots, x_n)$.

If $S \subseteq K[T]$ is a set of nonconstant polynomials, then $N$ is a **splitting field for** $S$ if every element of $S$ splits into linear factors over $N$ and $N$ is generated by the zeroes of these polynomials.

**Example.** Say $f = T^3 - 2$. Let $a = 2^{\frac{1}{3}}$. We get:

$$\begin{tikzcd}
& \mathbb{Q}(a,\zeta) \arrow[dl, "2"'] \arrow[d, "2"] \arrow[dr, "2"] & \\
\mathbb{Q}(a) \arrow[dr, "3"'] & \mathbb{Q}(\zeta a) \arrow[d, "3"] & \mathbb{Q}(\zeta^2 a) \arrow[dl, "3"] \\
& \mathbb{Q} &
\end{tikzcd}$$

$\mathbb{Q}(a, \zeta)$ is the splitting field, and the three middle fields are the stem fields.
The minimal polynomial for $\zeta$ is $\frac{T^3-1}{T-1}$.

**Theorem 4.** *Every $f \in K[T] \setminus K$ has a splitting field that is a finite extension of $K$.*

*Proof.* $\deg f = n \geq 1$. Induction on $n$:
$n = 1$✓
$n > 1$: Wlog $f$ does not split over $K$, $f = gh$, $g$ irreducible.

By Kronecker, there exists $L_1/K$ stem field for $g$.
$L_1 = K_1(x_1), x_1 \in L_1, g(x_1) = 0 \implies f = (T - x_1) \cdot f_1 \in L_1[T]$
$\deg f_1 = n - 1$. $f_1$ has a splitting field $L/L_1$ by induction.

$L/K$ is a splitting field of $f$ over $K$. □

**Corollary 5.** *$S \subseteq K[T] \setminus K$ finite. There exists a splitting field for $S$ which is finite over $K$.*

**Theorem 6.** *If $\varphi : K_1 \to K_2$ is an isomorphism of fields and $\tilde{\varphi} : K_1[T] \to K_2[T]$ the induced isomorphism on polynomial rings, if $f_1 \in K_1[T] \setminus K_1$, $L_1$ is the splitting field of $f_1$ over $K$, and $L_2$ is the splitting field of $f_2 = \tilde{\varphi}(f_1)$ over $K_2$, then there exists an isomorphism $\phi : L_1 \to L_2$ so that $\phi|_{K_1} = \varphi$.*

*Proof.* Induction on $n = [L_1 : K_1]$:

If $n = 1$, then all zeros of $f_1$ are in $K_1$ and map bijectively to zeros of $f_2$ in $K_2$

If $n > 1$, then $f_1$ has an irreducible factor $g_1 \in K_1[T]$ of degree $\geq 2$
$g_2 := \tilde{\varphi}(g_1)$ irreducible factor of $f_2$
$x_1 \in L_1, g_1(x_1) = 0$
$x_2 \in L_2, g_2(x_2) = 0$

Theorem 3 $\implies \exists \varphi^* : K_1(x_1) \to K_2(x_2), x_1 \mapsto x_2$ with $\varphi^*|_{K_1} = \varphi$.

$L_1$ is a splitting field for $f_1$ over $K_1(x_1)$.
$L_2$ is a splitting field for $f_2$ over $K_2(x_2)$.
Both have degree $\leq n - 1$. By induction, there exists $\phi : L_1 \to L_2$ an isomorphism with $\phi|_{K_1(x_1)} = \varphi^*$. This proves the claim. $\square$

**Corollary 7.** *Every polynomial has a splitting field which is determined uniquely up to isomorphism.*

### 3.2.3 Construction of an algebraic closure

**Definition.** $L$ is **algebraically closed** if every nonconstant polynomial with coefficients in $L$ has a root in $L$.
$L$ is an **algebraic closure** of a field $K$ if $L$ is algebraically closed and $L$ is algebraic over $K$.

**Example.** $\mathbb{C}$ is an algebraic closure of $\mathbb{R}$ but is not an algebraic closure of $\mathbb{Q}$.

**Proposition 8.** *Let $K$ be a field. TFAE:*

1. *$K$ is algebraically closed*
2. *Every $f \in K[x]$ splits into linear factors*
3. *Every irreducible polynomial has degree 1*
4. *$K$ has no proper algebraic extensions*

*Proof.* (1) $\Rightarrow$ (2): By induction
(2) $\Rightarrow$ (3): Clear
(3) $\Rightarrow$ (4): The minimal polynomial of an element that is algebraic over $K$ must have degree 1
(4) $\Rightarrow$ (1): If $f \in K[X]$ then $f$ has a stem field which is algebraic over $K$, hence equals $K$. So $f$ has a root in $K$. $\square$

**Proposition 9.** *Let $L/K$ be a field extension, $L$ algebraically closed. Then the algebraic closure of $K$in $L$ is an algebraic closure of $K$.*

*Proof.* Let $\tilde{K} \leq L$ be the relative algebraic closure. Let $f \in \tilde{K}[X]$. Then $f$ has a root $x$ in $L$.
$\tilde{K}(x)/\tilde{K}$ is algebraic $\implies$ $\tilde{K}(x)/K$ is algebraic $\implies$ $x \in \tilde{K}$ $\implies$ $\tilde{K}$ is algebraically closed. □

**Algebraic closure of $K$ in $L$**$= \{a \in L \mid a \text{ is algebraic over } K\} = \tilde{K}$

**Theorem 10.** *Every field has an algebraic closure.*

*Proof.* Let $K$ be a field. Let $\mathcal{X} = (X_f)_{f\in I}$. Let $I = \{f \in K[X] \mid \deg f \geq 1\}$.
Let $K[\mathcal{X}]$ be the polynomial ring in the variables $X_f$.
Let
$$A := (f(X_f) \mid f \in I) \trianglelefteq K[\mathcal{X}].$$
Want to show: $A$ is a proper ideal.
Assume for contradiction that $A = K[\mathcal{X}]$. Then
$$1 = \sum_{i=1}^{n} g_i f_i(X_{f_i})$$
where $g_i \in K[\mathcal{X}], f_i \in I$ and $n$ is some suitable positive integer.

Let $L/K$ be a splitting field for $\{f_1, \dots, f_n\}$. Then there exists some $x_i \in L$, a zero of $f_i$, for each $i$. Plug in $x_i$ for $X_{f_i}$. Plug in 0 for all other $X_f$'s. We get $1 = 0$ in $L$. $\rightarrow\leftarrow$

So $A \subsetneq K[\mathcal{X}]$. Then there exists some $Q \trianglelefteq K[\mathcal{X}], A \subseteq Q$ ($Q$ is maximal).

Let $E_1 = K[\mathcal{X}]/Q$. Then
$$K \hookrightarrow K[\mathcal{X}] \to K[\mathcal{X}]/Q = E_1$$
is a homomorphism, so has to be injective. So without loss of generality, view $K$ as a subfield of $E_1$.
By construction, every nonconstant polynomial $f \in K[X]$ has a zero in $E_1$.

Inductively, get $K = E_0 \leq E_1 \leq \cdots$
Every nonconstant polynomial in $E_i[X]$ has a zero in $E_{i+1}$. Take
$$E := \bigcup_{i=1}^{\infty} E_i.$$
□

**Theorem 11.** *Let $K$ be a field, $\tilde{K}$ algebraically closed. Let $\varphi : K \to \tilde{K}$ be a homomorphism. Then if $L/K$ is algebraic, there exists some $\phi : L \to \tilde{K}$ so that $\phi|_K = \varphi$.*

*Proof.* Exercise. □

**Corollary 12.** *An algebraic closure of a field is determined uniquely up to isomorphism.*

*Proof.* Let $K$ be a field, $\overline{K}$, $\overline{K'}$ two algebraic closures. If $\varphi : K \hookrightarrow \overline{K'}$ is an embedding, by Theorem 11, we get $\phi : \overline{K} \to \overline{K'}$ an extension of $\varphi$. So $\phi(\overline{K})$ is an algebraically closed subfield of $\overline{K'}$ and $\phi(\overline{K})/\overline{K}$ is algebraic.

So $\overline{K'}/\phi(\overline{K})$ is algebraic, meaning $\phi$ is surjective, hence an isomorphism. □

**Corollary 13.** *Let $K$ be a field, $f \in K[X]$, $\overline{K}$ the algebraic closure of $K$. Then there exists a unique splitting field of $f$ contained in $\overline{K}$.*

## 3.3 Normal Extensions

### 3.3.1 Automorphisms of extensions

**Definition.** Let $M/K$, $L/K$ be field extensions. Let $\varphi : L \to M$ be a homomorphism.
We say $\varphi$ is a **$K$-homomorphism** if $\varphi|_K = \mathrm{id}_K$.
$\mathrm{Aut}(L/K)$ is the set of $K$-automorphisms of $L$ = **automorphism group of $L$ over $K$**

**Note.** $\mathrm{Aut}(L/K)$ is a subgroup of $\mathrm{Aut}(L)$.

**Example.** $\mathrm{Aut}(\mathbb{C}/\mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$ (generated by complex conjugation)

**Theorem 1.** *Let $f \in K[X]$ be irreducible with stem field $L$. Then $|\mathrm{Aut}(L/K)| = |\{y \in L \mid f(y) = 0\}| \leq [L : K]$.*

*Proof.* $L = K(x)$ for some zero $x$ of $f$.
$\varphi \in \mathrm{Aut}(L/K) \implies \varphi(f)(x) = \varphi(f(x)) = 0$.
So $\varphi$ must send $x$ to another root of $f$. On the other hand, if $y \in L$ and $f(y) = 0$, then Theorem 2.3 $\implies \exists \varphi : L \to L, \varphi|_K = \mathrm{id}_K, \varphi(x) = y$.
Thus $\mathrm{Aut}(L/K)$ is in bijection with $\{y \in L \mid f(y) = 0\}$.

$\deg f = [L : K] \implies$ gives last claim (this is by Theorem 2.6) □

**Example.**

1. $K = \mathbb{Q}, f = X^4 - 2, L = \mathbb{Q}(x)$ stem field.
   $f = (X - 2^{\frac{1}{4}})(X + 2^{\frac{1}{4}})(X^2 + 2^{\frac{1}{2}}) \in L[X]$
   $\mathrm{Aut}(L/K) = \mathbb{Z}/2\mathbb{Z}$ (generated by sending the root you choose to its negative)

2. If we looked at $K = \mathbb{Q}(i)$, $f$ as above, we would get something different.

**Proposition 2.** *Let $K$ be a field, $f \in K[X]$ so that $f$ has $n$ pairwise distinct roots, $N/K$ splitting field of $f$. Then there exists $\pi : \mathrm{Aut}(N/K) \to S_n$ injective, induced from the action of $\mathrm{Aut}(N/K)$ on the zeros of $f$.*

*Proof.* Iterated application of Theorem 2.3 shows injectivity. □

**Example.** If $N/\mathbb{Q}$ is the splitting field of $f = X^4 - 2 \in \mathbb{Q}[X]$, then the roots are $a, -a, ia, -ia$ and the automorphism group contains

$$\varphi : a \mapsto ia, \quad \pi(\varphi) = (1234)$$

and

$$\psi : i \mapsto -i, \quad \pi(\psi) = (24)$$

so $D_4 \leq \pi(\operatorname{Aut}(N/K)) \leq S_4$. And since e.g. $(123) \notin \pi(\operatorname{Aut}(N/K))$ we know $\operatorname{Aut}(N/K) \cong D_4$.

Note that in this example, $[N : K] = 8$. A natural question is: is it always true that $|\operatorname{Aut}(N/K)| \leq [N : K]$?

**Theorem 3** (Dedekind). *Let $H$ be a monoid, $K$ a field. If $\sigma_1, \ldots, \sigma_n : H \to K^\times$ are homomorphisms of monoids, then TFAE:*

1. *$\sigma_1, \ldots, \sigma_n$ are pairwise disjoint*
2. *$\sigma_1, \ldots, \sigma_n$ are linearly independent as maps $H \to K$*

*Proof.* (2) $\Rightarrow$ (1) is clear.
(1) $\Rightarrow$ (2):
Suppose $\sum_{i=1}^n a_i \sigma_i = 0$ for $a_i \in K$ not all zero.

Without loss of generality assume this relation is of minimal length (number of nonzero $a_i$).
Without loss of generality assume $a_1 = 1, a_2 \neq 0$.
$\sigma_1 \neq \sigma_2 \implies \exists h \in H$ such that $\sigma_1(h) \neq \sigma_2(h) \implies$

$$\sum_{i=1}^n a_i \sigma_i(h) \sigma_i(g) = \sum_{i=1}^n a_i \sigma_i(hg) = 0 \text{ for all } g \in H.$$

But also

$$\sigma_1(h) \sum_{i=1}^n a_i \sigma_i(g) = 0 \text{ for all } g \in H.$$

Take the difference of these to get

$$\sum_{i=2}^n (\sigma_i(h) - \sigma_1(h)) a_i \sigma_i(g) = 0.$$

Note that we constructed these coefficients of $\sigma_i(g)$ to be nonzero for all $i = 2, \ldots, n$. But then

$$\sum_{i=2}^n (\sigma_i(h) - \sigma_1(h)) \sigma_i = 0$$

is a shorter nontrivial relation. $\rightarrow\leftarrow$ $\square$

**Corollary 4.** *If $L_1$ and $L_2$ are extensions of a field $K$ and $[L_1 : K] < \infty$ then there are at most $[L_1 : K]$ different $K$-homomorphisms $L_1 \to L_2$. In particular, $|\operatorname{Aut}(L_1/K)| \leq [L_1 : K]$.*

*Proof.* Let $H = L_1 \setminus \{0\}$ in Theorem 3. Then the number of homomorphisms is bounded by the dimension of $\operatorname{Hom}_K(L_1, L_2)$ as an $L_2$-vector space.

$\dim_{L_2}(\operatorname{Hom}_K(L_1, L_2)) = \dim_K L_1 = [L_1 : K]$. □

### 3.3.2 Normality

**Definition.** Let $N/K$ be algebraic. We say $N/K$ is **normal** if every irreducible polynomial in $K[X]$ which has a root in $N$ splits into linear factors in $N$.

**Theorem 5.** *Let $K$ be a field and $\overline{K}$ the algebraic closure of $K$. Suppose $N/K$ is algebraic. Then TFAE:*

1. *$N/K$ is normal*
2. *Every $K$-homomorphism $\varphi : N \to \overline{K}$ is a $K$-automorphism of $N$.*
3. *$N$ is the splitting field of a set of irreducible polynomials.*

*Proof.* (1) ⇒ (2): $N/K$ is normal. Say $\varphi : N \to \overline{K}$ is a $K$-homomorphism. $x \in N$ has minimal polynomial $f_x \implies f_x$ has zeros $x_1, \ldots, x_n \in N$ and $\deg f_x = n$.

$L := K(x_1, \ldots, x_n) \geq N \implies L/K$ is finite and $\varphi(L) \subseteq L$ because $\varphi$ is a $K$-homomorphism. Then $\varphi|_L : L \to L$ must be surjective ($L$ is a finite-dimensional vector space). So there exists $y \in L$ such that $\varphi(y) = x$. Then $\varphi$ is surjective. So $\varphi$ is an automorphism of $N/K$.
(2) ⇒ (3):
$N/K$ algebraic $\implies$ for every $x \in N$ there exists a minimal polynomial $f_x$. Let $y \in \overline{K}$ be any zero of $f_x$. By Theorem 2.3, there exists $\varphi : K(x) \to K(y), x \mapsto y$ so that $\varphi|_K = \operatorname{id}_K$.

Then Theorem 2.6 implies that there exists a $K$-homomorphism $\phi : N \to \overline{K}, \phi|_N = \varphi$. So $\phi \in \operatorname{Aut})(N/K) \implies y \in N \implies N$ contains the splitting field for $f_x \implies N$ is normal and $N$ is the splitting field for $\{f_x \mid x \in N\}$.

We've now shown 1 ⇒ 2, 2 ⇒ 1, 2 ⇒ 3. Remains to show:
(3) ⇒ (2):
Let $\varphi : N \to \overline{K}$ be a $K$-homomorphism. Then $f(\varphi(x)) = 0$ for any $f \in K[X]$ and $x \in N$ so that $f(x) = 0$. Thus $\varphi(N) \leq N$.
As in the first part, conclude that $\varphi \in \operatorname{Aut}(N/K)$. □

**Corollary 6.** *$N/K$ is a finite normal extension $\iff$ $N$ is the splitting field of a polynomial with coefficients in $K$.*

### 3.3.3 Finite Fields

**Proposition 7.** *If $K$ is a finite field, then $|K| = p^n$ where $p$ is prime and $n \in \mathbb{N} \setminus \{0\}$.*

*Proof.* $|K| < \infty \implies K$ has prime field $\mathbb{F}_p$ for some prime $p$.

$K$ is a vector space over $\mathbb{F}_p$ of some dimension $n < \infty$. □

**Proposition 8.** *If $p$ is a prime and $K$ is a field of characteristic $p$, then $\varphi : K \to K, a \mapsto a^p$ is an endomorphism of $K$.*

*Proof.* $\varphi(1) = 1$ and $\varphi$ is multiplicative. To show that $\varphi$ is also additive,

$$\varphi(a+b) = (a+b)^p = \sum_{i=0}^{p} \binom{p}{i} a^i b^{p-i} b^{p-i} = a^p + b^p = \varphi(a) + \varphi(b).$$

□

**Definition.** $\varphi$ as in Proposition 8 is called the **Frobenius endomorphism**.

**Theorem 9.** *If $K$ is a finite field, $K^\times$ is cyclic.*

*Proof.* We proved last semester that since $K^\times$ is finite and abelian, $K$ is a direct product of finite groups of prime power order.

Suffices to show: every Sylow subgroup is cyclic.
Let $U$ be a $q$-Sylow of $K^\times$. Then there exists some $k \in \mathbb{N} \setminus \{0\}$ such that $u^{q^k} = 1$ for all $u \in U$. Thus $|U| \leq q^k$ since all elements are now roots of $x^{q^k} - 1$. So without loss of generality choose $k$ minimal.

Then pick $u \in U$ of order $q^k \implies U = \langle u \rangle$ is cyclic. □

**Theorem 10.** *If $p$ is a prime and $n \in \mathbb{N} \setminus \{0\}$, then there exists a field $K$ with $|K| = p^n$, unique up to isomorphism. It is the splitting field of $f(X) = X^q - X$ over $\mathbb{F}_p$, where $q = p^n$.*

*Proof.* Take $N/\mathbb{F}_p$ splitting field of $f$. Let $K = \{y \in N \mid f(y) = 0\}$. By Proposition 8, $K$ is a field.

$K$ contains $\mathbb{F}_p$ and all roots of $f$. And $K \subseteq N$. So $K = N$.
As in Theorem 9, $|K| = q$.

For uniqueness, if $|L| = q$ then by Theorem 9, $L^\times$ is cyclic of order $q - 1$ so $a^q = a$ for all $a \in L$. Thus $L$ is a splitting field of $f$, so $L \cong N$. □

## 3.4 Resultants

### 3.4.1 The resultant as a linear combination

**Definition.** Let $f = \sum_{i=0}^{n} a_i X^i$ and $g = \sum_{j=0}^{m} b_j X^j, a_n \neq 0, b_m \neq 0$.
The **resultant** of $f$ and $g$ is

$$\operatorname{Res}(f,g) := \det \begin{pmatrix} a_n & a_{n-1} & \cdots & a_0 & & & & \\ & a_n & a_{n-1} & \cdots & a_0 & & & \\ & & \ddots & & & & \ddots & \\ & & & a_n & \cdots & \cdots & a_0 & \\ b_m & \cdots & b_0 & & & & & \\ & \ddots & & \ddots & & & & \\ & & b_m & \cdots & \cdots & b_0 & & \end{pmatrix} \begin{matrix} \left.\vphantom{\begin{matrix} a \\ a \\ a \\ a \end{matrix}}\right\} m \\ \left.\vphantom{\begin{matrix} a \\ a \\ a \end{matrix}}\right\} n \end{matrix}$$

Set $R(f,g) = 0$ if $f = 0$ or $g = 0$.

Note: if $f, g$ are constant, $R(f,g) = 0$.

**Theorem 1.** *$f, g \in K[X] \setminus K \implies \exists u, v \in K[X]$ so that $R(f,g) = uf + vg$,* $\deg u < \deg g$, $\deg v < \deg f$.

*Proof.* Let $f, g$ be as above. Without loss of generality assume $m \neq 0 \neq n$.

$$X^{m-1} f = a_n X^{m+n-1} + a_{n-1} X^{m+n-2} + \cdots + a_0 X^{m-1}$$
$$X^{m-2} f = a_n X^{m+n-2} + a_{n-1} X^{m+n-3} + \cdots + a_1 X^{m-1} + a_0 X^{m-2}$$
$$\vdots$$
$$f = a_n X^n + \cdots + a_0$$
$$X^{n-1} g = b_m X^{m+n-1} + \cdots$$
$$\vdots$$
$$g = \cdots$$

This is a linear system: If

$$c = \begin{pmatrix} X^{m-1} f \\ \vdots \\ f \\ X^{n-1} g \\ \vdots \\ g \end{pmatrix}$$

then

$$c = c_{m+n-1} X^{m+n-1} + \cdots + c_0 X^0 = C \begin{pmatrix} X^{m+n-1} \\ \vdots \\ X^0 \end{pmatrix}$$

where $C = (c_{m+n-1}, \dots, c_0)$ is as in the definition of $R(f,g)$.

By Cramer's rule,

$$\det(C)X^i = \det(c_{m+n-1}, \dots, c_{i+1}, c, c_{i-1}, \dots, c_0)$$

Then for $i = 0$, $\det C = \det(c_{m+n-1}, \dots, c_1, c)$. So performing Laplace on the last column, $R(f,g)$ is a linear combination of $f$ and $g$ with coefficients of degree bounded by $m-1$ and $n-1$, respectively. $\square$

**Corollary 2.** *Let $f, g$ be as in the definition ($a_n \neq 0$ and $b_m \neq 0$). Then*
*$R(f,g) = 0 \iff f$ and $g$ have a common factor.*

*Proof.* $(\Leftarrow)$ : Suppose $f = f_1 h, g = g_1 h$. Say $\deg f_1 < \deg f$ and $\deg g_1 < \deg g$ ($h$ is not a constant).
Then $g_1 f - f_1 g = 0$.
$\deg g_1 < \deg g \implies \deg g_1 \leq m-1$.
$\deg f_1 < \deg g \implies \deg f_1 \leq n-1$.
$\implies X^{m-1}f, \dots, Xf, f, X^{n-1}g, \dots, g$ are $K$-linearly dependent. So there exists $a \in K^{m+n}$, $a \neq 0$ such that

$$a \cdot C \begin{pmatrix} X^{n+m-1} \\ \vdots \\ X^0 \end{pmatrix} = 0.$$

Thus $\det C = R(f,g) = 0$ since $X^{m+n-1}, \dots, X^0$ are linearly independent.
$(\Rightarrow)$ : $R(f,g) = 0 \implies uf = vg$ for some $u, v \in K[X]$ with $\deg u < \deg g$ and $\deg v < \deg f$. Without loss of generality $u, v$ are coprime. For degree reasons, not all prime factors of $f$ can divide $v$ (with appropriate multiplicities), so we have the claim. $\square$

**Example.** $f = X^2 - 4X + 3 = (X-1)(X-3)$
$g = 2X - 2 = 2(X-1)$
$R(f,g) = 0$

$$\det \begin{pmatrix} 1 & -4 & 3 \\ 2 & -2 & 0 \\ 0 & 2 & -2 \end{pmatrix} = 1(4) - 2(8-6) = 0.$$

### 3.4.2 Representation in zeros

**Theorem 3.**

$$f = a_n \prod_{i=1}^{n} (X - \alpha_i), \quad g = b_m \prod_{j=1}^{m} (X - \beta_j) \in K[X] \setminus K$$

$$\implies R(f,g) = a_n^m b_m^n \prod_{i=1}^{n} \prod_{j=1}^{m} (\alpha_i - \beta_j)$$

**Corollary 4.**

$$R(f,g) = a_n \prod_{i=1}^{n} g(\alpha_i) = b_m \prod_{j=1}^{m} f(\beta_j)$$

**Example.** If $f = (X-1)(X-3)$ and $g = X-2$ then

$$R(f,g) = \det \begin{pmatrix} 1 & -4 & 3 \\ 1 & -2 & 0 \\ 0 & 1 & -2 \end{pmatrix} = -1.$$

And

$$R(f,g) = (2-1)(2-3) = (1-2)(3-2) = -1.$$

**Corollary 5.** *$K$ field, $\overline{K}$ algebraic closure, $x \in \overline{K}$, $y \in \overline{K} \setminus \{0\}$ with minimal polynomials $f_x, f_y$. Then*

1. *$x + y$ is a zero of $R(f_x(X-Y), f_y(Y))$*
2. *$\frac{x}{y}$ is a zero of $R(f_x(X \cdot Y), f_y(Y))$*

*where the resultant is taken with respect to $y$.*

*Proof.* Exercise. $\square$

**Example.** $x = 2, y = \sqrt{5}$.
Find the polynomial in $\mathbb{Q}[X]$ that has $2+\sqrt{5}$ as a root.
$f_x = X - 2$ and $f_x(X-Y) = X - Y - 2$
$f_y = X^2 - 5$
So

$$R(f_x(X-Y), f_y(Y)) = R(X-Y-2, Y^2-5) = \det \begin{pmatrix} -1 & X-2 & 0 \\ 0 & -1 & X-2 \\ 1 & 0 & -5 \end{pmatrix} = (X-2)^2 - 5 = X^2 - 4X - 1.$$

*Proof.* (Of Theorem 3)
If $a \in K$ and $f, g \in K[X]$, $\deg f = n$ and $\deg g = m$, then $R(af, g) = a^m R(f,g)$ and $R(f, ag) = a^n R(f,g)$.
So without loss of generality, assume $a_n = b_m = 1$.
View the zeros of $f, g$ as variables. $f, g \in K[A_1, \ldots, A_n, B_1, \ldots, B_m][X]$.
$R(f,g) \in K[A_1, \ldots, A_n, B_1, \ldots, B_m]$.
By Corollary 2, $R(f,g) = 0$ when $A_i = B_j$ i.e. $A_i - B_j$ divides $R(f,g)$ for all $i, j$. Thus

$$S := \prod_i \prod_j (A_i - B_j) \text{ divides } R(f,g).$$

$\deg R(f,g) \leq n \cdot m$ as a polynomial in the $A_i$'s, so if $R(f,g) = S \cdot T$, $\deg T = 0$ in the $A_i$'s.
A similar argument in the $B$ variables forces $T \in K$.

Letting $A_i = 0$, $B_j = 1$ for all $i, j$, we get

$$(-1)^{nm} = R(X^n, (X-1)^m) = T \cdot S = T \cdot (-1)^{nm} \implies T = 1.$$

$\square$

### 3.4.3 The discriminant of a polynomial

**Definition.** $K$ field, $f \in K[X] \setminus K$, $\deg f = n$. $x_1, \dots, x_n \in \overline{K}$ roots of $f$. Then

$$D(f) = \prod_{i<j} (x_i - x_j)^2$$

is the **discriminant** of $f$.

**Example.** $f = X^2 + aX + b \in K[X]$ has roots $\frac{a}{2} \pm \sqrt{\frac{a^2}{4} - b}$ (assume $K$ has characteristic $\neq 2$)
Then $D(f) = a^2 - 4b$.

**Theorem 6.** *$f \in K[X]$ monic,* $\deg f = n \geq 1$. *Then*

$$D(f) = (-1)^{\frac{n(n+1)}{2}} R(f, f').$$

*In particular, $D(f) \in K$.*

*Proof.* By Corollary 4, $R(f, f') = \prod_{i=1}^n f'(x_i)$ where $f = \prod_{i=1}^n (X - x_i)$ in $\overline{K}[X]$.

$$\prod_{i=1}^n f'(x_i) = \prod_{i=1}^n \sum_{k=1}^n \prod_{j \neq k} (x_i - x_j) = \prod_{i=1}^n \prod_{i \neq j} (x_i - x_j) = \prod_{i \neq j} (x_i - x_j) =$$

$$\prod_{i<j} (x_i - x_j)^2 (-1)^{\frac{n(n-1)}{2}} = (-1)^{\frac{n(n-1)}{2}} D(f).$$

□

**Example.** $f = X^2 + aX + b$, $f' = 2X + a$, $n = 2$
$\frac{n(n-1)}{2} = 1$

$$R(f, f') = \det \begin{pmatrix} 1 & a & b \\ 2 & a & 0 \\ 0 & 2 & a \end{pmatrix} = a^2 - 2a^2 + 4b = D(f).$$

## 3.5 Separable extensions

### 3.5.1 Separable polynomials

**Definition.** Let $K$ be a field. $f \in K[X] \setminus K$ is **separable** if it has only simple roots (in an algebraic closure $\overline{K}$).
If $L/K$ is algebraic, $x \in L$ is **separable over** $K$ if its minimal polyomial is.
$L/K$ is **separable** if every $x \in L$ is separable over $K$.
Otherwise, $L/K$ (resp. $x, f_x$) is **inseparable**.

**Proposition 1.** *Let $K$ be a field, $f \in K[X] \setminus K$. TFAE:*

1. *$f$ is inseparable*
2. *$D(f) = 0$*
3. *$f$ and $f'$ have a common factor or $f' = 0$*

*Proof.* (1) $\iff$ (2) by definition of discriminant
(2) $\iff$ (3): $D(f) = 0 \iff R(f, f') = 0$ so wlog $\deg f \geq 2$: then if $f' = 0$ or $f$ and $f'$ have a common factor, $R(f, f') = 0$
Conversely, if $R(f, f') = 0$ and $f' \neq 0$ then $f$ and $f'$ have a common factor (Corollary 4.2) □

**Note.** If $f \in K[X]$ has a double/multiple root and char$K = 0$ then $f$ is not irreducible.

If char$K > 0$ then $f$ may be irreducible and inseparable. e.g. $f = X^p - t, K = \mathbb{F}_p(t)$ has a $p$-fold root:
$f = (X - x_1) \cdots (X - x_p) \in N[X]$ where $N$ is a splitting field.
$x_1 \neq 0 \implies \zeta_i = \frac{x_i}{x_1}$ satisfies $\zeta_i^p = 1$.
But then $\zeta_i = 1$ since Frobenius is injective.
So $f = (X - x_1)^p$.

But $f$ is irreducible: if $f = g \cdot h$ for $g, h \in K[X] \setminus K$ monic, then $g = (X - x_1)^s, h = (X - x_1)^r$ for some $r, s < p$.Thus $x_1^s \in K$.
$ms + np = 1$ since $\gcd(s, p) = 1$. So $x_1 = x_1^{ms+np} = (x_1^s)^m (x_1^p)^n \in K \rightarrow\leftarrow$

**Theorem 2.** *$K$ field, $f \in K[X] \setminus K$ irreducible.*

1. *If* char$K = 0$ *then $f$ is separable.*
2. *If* char$K = p > 0$ *then $f$ is inseparable $\iff f' = 0$.*
   *In this case, $f$ is a polynomial in $X^p$ i.e. $\exists g \in K[X]$ such that $f = g(X^p)$.*

*Proof.*
1. $\deg f' = \deg f - 1 \implies$ since $f$ is irreducible, $f, f'$ have no common factor $\implies f$ is separable, by Prop 1.
2. $f$ inseparable $\iff$ since $f$ is irreducible, $f'(x) = 0 \iff$ exponents are all multiples of $p$.

□

So all algebraic extensions of a characteristic 0 field are separable.

### 3.5.2 Perfect fields

**Definition.** Let $K$ be a field. $K$ is **perfect** if every irreducible polynomial with coefficients in $K$ is separable.

**Example.** $K = \mathbb{F}_p(t)$ is not perfect.

**Theorem 3.** 1. *Every field of characteristic 0 is perfect.*

2. $\mathrm{char}K = p > 0$. *Then $K$ is perfect $\iff$ the Frobenius endomorphism is an automorphism of $K$ (surjectivity)*

*Proof.* (1) we already knew.
For (2), let $\phi$ be the Frobenius endomorphism. We will show: $\phi$ is not surjective $\iff$ $K$ is not perfect.

($\Leftarrow$):
$K$ not perfect $\implies$ $\exists f \in K[X]$ inseparable and irreducible $\implies$ by Theorem 2, $f = \sum_{i=0}^{n} a_i X^{p^i}$.

Suppose for contradiction that $\phi$ is surjective. Then $a_i = b_i^p$ for some $b_i \in K$. So $f = \sum b_i^p X^{p^i} = (\sum b_i X^i)^p \rightarrow\leftarrow$ (not irreducible)

($\Rightarrow$):
Suppose $\phi$ is not surjective. Then there exists $t \in K \setminus K^p \implies f := X^p - t$ does not split over $K$.

Take $L/K$ splitting field, $y \in L/K$ a zero of $f$ (so $y^p = t$). Then $f = (X-y)^p$ (as in the example) is inseparable.

$y$ has some minimal polynomial $f_y$, which is irreducible. $f_y \mid f \implies f_y$ is inseparable (not linear) and irreducible. Thus $K$ is not perfect. □

**Corollary 4.** *Finite fields are perfect.*

### 3.5.3 The separable degree

**Definition.** $L/K$ algebraic extension, $\overline{K}$ algebraic closure of $K$.

$$[L:K]_s = \left|\{\varphi : L \to \overline{K} \text{ K-homomorphisms}\}\right|$$

is the **separable degree of $L$ over $K$**

**Example.** $L \geq K = \mathbb{F}_p(t)$ splitting field of $f = X^p - t$
$L$ is normal $\implies \left|\{\varphi : L \to \overline{K} \text{ K-homomorphisms}\}\right| = |\mathrm{Aut}(L/K)|$
$\varphi \in \mathrm{Aut}(L/K) \implies \varphi$ maps zeros to zeros $\implies \varphi = \mathrm{id}_L$. So $[L:K]_s = 1$ (i.e. $L/K$ is not separable)

**Lemma 5.** *$K$ field, $\overline{K}$ algebraic closure, $f \in K[X] \setminus K$ irreducible with stem field $L \implies$ $[L:K]_s = \left|\{y \in \overline{K} \mid f(y) = 0\}\right|$.*

*Proof.* Theorem 2.3. □

**Proposition 6.** *$K$ field, $f \in K[X]\setminus K$ irreducible with stem field $L$. Then $[L:K]_s \leq [L:K]$ with equality when $f$ is separable.*

*Proof.* Lemma 5. □

**Theorem 7.** *$M \geq L \geq K$ tower of algebraic extensions $\implies [M : K]_s = [M : L]_s \cdot [L : K]_s$.*

*Proof.* Pick an algebraic closure $\overline{K}$ containing $M$. Without loss of generality $M/L$ is simple (the general case follows from induction).
Then $M$ is a stem field of an irreducible $f \in L[X]$. $\varphi : M \to \overline{K}$ $K$-homomorphism $\implies \varphi|_L : L \to \overline{K}$ is a $K$-homomorphism. Hence, we want to count extensions of a given $\psi : L \to \overline{K}$ to $M$.

$M/L$ is simple so by Theorem 2.3, there are exactly $|\{y \in \overline{K} \mid f(y) = 0\}|$ extensions.
Lemma 5 $\implies |\{y \in \overline{K} \mid f(y) = 0\}| = [M : L]_s$. □

**Corollary 8.** *An extension generated by a separable element is separable.*

*Proof.* $K$ field, $K(x)$ simple extension with $x$ separable over $K$. Let $y \in K(x)$. Then by Proposition 6, $[K(x) : K] = [K(x) : K]_s$, which, by Theorem 7, $= [K(x) : K(y)]_s \cdot [K(y) : K]_s = [K(x) : K(y)] \cdot [K(y) : K]_s$.

We know: $[K(x) : K] = [K(x) : K(y)] \cdot [K(y) : K]$, thus $[K(y) : K]_s = [K(y) : K]$. Then by Proposition 6, $K(y)/K$ is separable. □

### 3.5.4 Primitive element theorem

**Definition.** If $L = K(x)$ is a simple extension, we say $x$ is a **primitive element** for $L/K$.

**Theorem 9** (Primitive Element Theorem)**.** *If $L = K(y, z)/K$ is finite and $z$ is separable over $K$, then $L = K(x)$ for some $x \in L$ i.e. $L$ admits a primitive element.*
*In particular, every finite separable extension is a simple separable extension.*

*Proof.* If $|K| < \infty$, then $|L| < \infty$ and $L \setminus \{0\}$ is cyclic, so we can just take a generator.

So without loss of generality assume $|K| = \infty$. Let $f_y, f_z$ be minimal polynomials of $y, z$ over $K$ respectively. Let $N$ be a splitting field of $f_y \cdot f_z$. In $N[X]$,

$$f_y = \prod_{i=1}^{n}(x - y_i) \quad f_z = \prod_{j=1}^{m}(x - z_j)$$

where $y_i \in N$, $y_1 = y$, $z_j \in N$, $z_j = z$, and $z_1, \ldots, z_m$ are distinct ($f_z$ was separable).

Let $a \in K$. If $y_i + az_j = y + az$ for some $i, j \neq 1$ then

$$a = \frac{y_i - y}{z - z_j}$$

$\implies a$ is in a finite list.

$|K| = \infty \implies \exists a \in K$ for which $y_i + az_j \neq y + az$ for all $i, j \neq 1$.

Let $x := y + az$. Claim: $L = K(x)$.
$y = x - az \implies f_y(x - az) = 0 \implies z$ is a zero of $f_y(x - a \cdot X) =: h \in K(x)[X] \implies z$ is a zero of $\gcd(f_z, h)$ in $K(x)[X]$.
If $j \neq 1$,

$$h(z_j) = f_y(x - az_j) = f_y(y + az - az_j) \neq 0.$$

Thus $\gcd(f_z, h) = X - z \in K(x)[X] \implies z \in K(x) \implies y = x - az\imath K(x) \implies K(x) = L$.
The last claim follows by induction. □

**Example.** $K = \mathbb{F}_p(t, u)$
$L$ splitting field of $(X^p - t)(X^p - u) \in K[X]$
Exercise: if $L = K(y, z)$ where $y^p = t$ and $z^p = u$ then $[L : K] = p^2$.
Then $x \in L \implies x \in K[y, z] \implies$

$$x = \sum_{i=0}^{p-1} \sum_{j=0}^{p-1} a_{ij} y^i z^j, \quad a_{ij} \in K.$$

$$x^p = \sum_{i=0}^{p-1} \sum_{j=0}^{p-1} a_{ij}^p y^{ip} z^{jp} = \sum_{i=0}^{p-1} \sum_{j=0}^{p-1} a_{ij}^p t^i u^j \in K$$

$\implies [K(x) : K] \leq p$ for all $x \in L \implies$ no primitive element exists for $L/K$.

**Theorem 10.** *$L/K$ finite $\implies [L : K]_s \leq [L : K]$ with equality exactly when $L/K$ is separable.*

*Proof.* $[L : K]_s \leq [L : K]$ from Corollary 3.4 (Dedekind's theorem)
$(\Leftarrow)$ : $L/K$ finite separable $\implies L/K$ simple separable (Theorem 9) $\implies [L : K] = [L : K]$ by Prop 6.
$(\Rightarrow)$ : $[L : K]_s = [L : K]$
Let $L = K(y_1, \dots, y_n), y_i \in L$.
Let $L_i = K(y_1, \dots, y_i)$. We get a tower $K = L_0 \subseteq \cdots \subseteq L_n = L \implies [L_i : L_{i-1}]_s = [L_i : L_{i-1}]$ for all $i$.

By Proposition 6, $L_i/L_{i-1}$ is stem field of separable polynomial, hence is separable. Thus $y_i$ is separable over $K \implies$ by Theorem 9, $L_2 = K(y_2, y_1)$ is a simple extension $\implies L_2/K$ is separable by Proposition 6/

Induction gives: $L/K$ is separable. □

**Corollary 11.** *$L/K$ algebraic, $y_1, \dots, y_n \in L$ separable over $K \implies K(y_1, \dots, y_n)/K$ separable.*

**Corollary 12.** *$K$ field, $N$ splitting field of a separable polynomial with coefficients in $K$. Then*

$$|\mathrm{Aut}(N/K)| = [N : K].$$

**Example.** $N/\mathbb{F}_p$ splitting field of $X^{p^n} - X$ separable $\implies |N| = p^n \implies |\mathrm{Aut}(N/\mathbb{F}_p)| = [N : \mathbb{F}_p] = n$.

$\phi$ Frobenius automorphism $\implies \phi \in \mathrm{Aut}(N/\mathbb{F}_p) \implies \langle\phi\rangle \leq \mathrm{Aut}(N/\mathbb{F}_p)$.

Let $m = |\phi|$.

$m$ is minimal such thar $\phi^m = 1$ i.e. such that $a^{p^m} = a$ for all $a \in N \implies m = n \implies \mathrm{Aut}(N/\mathbb{F}_p) = \langle\phi\rangle$ cyclic group of order $n$.

### 3.5.5 Separable and perfect closure

**Definition.** $K$ field, $\overline{K}$ algebraic closure
$K^s := \{a \in \overline{K} \mid a \text{ separable over } K\}$ is called the **separable closure of $K$ (in $\overline{K}$)**

**Proposition 13.** *1. $K^s \leq \overline{K}$*

*2. Every separable polynomial in $K[X]$ splits into linear factors in $K^s[X]$*

*3. $K^s/K$ is normal*

*Proof.* Exercise □

Assume until the end of this section that char$K = p > 0$.

**Definition.** Let $L/K$ be algebraic. Then $L$ is **purely inseparable** if no element in $L/K$ is separable over $K$.

**Note.**

1. $L/K$ is purely inseparable $\iff$ $L/K$ has no intermediate field extensions which are separable over $K$.

2. If $L/K$ is finite ($L \neq K$), then $L/K$ is purely inseparable $\iff [L : K]_s = 1$.

**Proposition 14.** *$L/K$ algebraic extension. TFAE:*

*1. $L/K$ is purely inseparable and is perfect*

*2. L is perfect and minimal with this property (i.e. if $E/K$ is an extension with E perfect, then there is a $K$-embedding $L \hookrightarrow E$)*

*If L satisfies these conditions, then the embedding given by (2) is unique. Any extension satisfying the conditions is uniquely isomorphic to*

$$K_p := \{a \in \overline{K} \mid a^{p^n} \in K \text{ for some } n \geq 0\}.$$

*Proof.* $K_p/K$ is a subfield of $\overline{K}$ and is perfect by definition. Moreover, $K_p/K$ is purely inseparable $\implies K_p$ satisfies (1).

If $E/K$ is perfect, then there exists an embedding $\overline{K} \hookrightarrow E$ extending $K \hookrightarrow E$.
If $a \in K_p$ then $a^{p^n} \in K$ some $n \implies a^{p^n} \in E = E^p \implies a \in E$ ($p$-th roots are unique).

So $K_p \leq E$ i.e. $K_p$ satisfies (2).

If $L/K$ is algebraic and satisfies (2), then since $K_p$ is perfect, $L \hookrightarrow K_p$. Now $K_p/L$ is purely inseparable but $L$ is perfect $\implies K_p = L$.

Similarly, if $L/K$ satisfies (1), then $L$ is perfect so $K_p \hookrightarrow L$ by the first part of the proof $\implies L/K_p$ is purely inseparable, $K_p$ is perfect $\implies L = K_p$. □

**Definition.** A extension as in Proposition 14 is called a **perfect closure of** $K$. Notation: $K^{p^{-\infty}}$.

**Example.** $L = K(a)$, $f_a$ minimal polynomial.
Exercise: $f_a = g(X^{p^n})$ for some $n$, some irreducible and separable $g \implies$
$g$ is the minimal polynomial of $a^{p^n}$.

So we have $K \leq K(a^{p^n}) \leq K(a) = L$ where the first extension is separable and the second extension is purely inseparable.

## 3.6 The Galois Correspondence

### 3.6.1 Galois extensions

**Definition.** A extension $L/K$ is a **Galois extension** if it is normal and separable.
The Galois group is $\mathrm{Gal}(L/K) = \mathrm{Aut}(L/K)$.

**Lemma 1.** *A finite extension is a Galois extension $\iff$ it is the splitting field of a separable polynomial.*

**Theorem 2.** *A finite extension is Galois $\iff$ $|\mathrm{Aut}(L/K)| = [L : K]$.*

*Proof.* ($\Rightarrow$): $L/K$ finite and Galois $\implies$ by Lemma 1, $L$ is the splitting field of a separable polynomial in $K[X] \implies$ by Corollary 5.12, we get the claim.
($\Leftarrow$): $|\mathrm{Aut}(L/K)| \leq |\{\varphi : L \to \overline{K}\}| \leq [L : K]$, meaning $|\mathrm{Aut}(L/K)| \leq [L : K]_s \leq [L : K]$ so $|\mathrm{Aut}(L/K)| = [L : K] \implies L/K$ is normal and separable (Theorem 5.10) □

### 3.6.2 Fixed Fields

**Lemma 3.** *$N$ field, $G \leq \mathrm{Aut}(N)$.*
*Then $N^G := \{n \in N \mid \forall \sigma \in G, \sigma(n) = n\}$ is a field.*

*Proof.* Exercise. □

**Lemma 4** (Artin)**.** *$N$ field, $G \le \mathrm{Aut}(N)$, $G$ finite $\implies [N : N^G] = |G|$.*

*Proof.* $|G| \le |\mathrm{Aut}(N/N^G)| \le [N : N^G]$.
Say $G = \{\sigma_1, \dots, \sigma_n\}$. Let $y_1, \dots, y_{n+1} \in N$.
Want to show: these satisfy a nontrivial $N^G$ relation.
Consider

$$\sum_{i=1}^{n+1} \sigma_j^{-1}(y_i) a_i = 0 \text{ for } j = 1, \dots, n.$$

This is a linear system in indeterminates $a_1, \dots, a_{n+1}$. Then there exists a solution $(z_1, \dots, z_{n+1}) \neq (0, \dots, 0) \in N^{n+1}$. Without loss of generality, assume $z_1 \neq 0$. By Dedekind, $\sigma_1, \dots, \sigma_n$ are linearly independent. In particular, $\sum_{j=1}^n \sigma_j$ is not the zero map.

After multiplying $(z_1, \dots, z_{n+1})$ with an element of $N$, we may assume $\sum_{j=1}^n \sigma_j(z_1) \neq 0$. Apply $\sigma_j$ to the $j$-th equation and add:

$$0 = \sum_{j=1}^n \sigma_j \left( \sum_{i=1}^{n+1} \sigma_j^{-1}(y_i) z_i \right)$$

$$0 = \sum_{j=1}^n \sum_{i=1}^{n+1} y_i \sigma_j(z_i) = \sum_{i=1}^{n+1} y_i \left( \sum_{j=1}^n \sigma_j(z_i) \right)$$

Notice that $\sum_{j=1}^n \sigma_j(z_i) \in N^G$.

Since the coefficient of $y_i$ is $\sum \sigma_j(z_1) \neq 0$, $[N : N^G] \le |G| \implies [N : N^G] = |G|$. ∎

**Theorem 5.** *$L/K$ finite, $G \le \mathrm{Aut}(L/K)$. TFAE:*

1. *$L/K$ is Galois with group $G$*
2. *$L^G = K$*

*Proof.* (1) ⇒ (2): $L/K$ Galois with group $G$, i.e. $G = \mathrm{Aut}(L/K)$.
Suppose there exists $x \in L^G \setminus K$. Let $f_x$ be the minimal polynomial for $x$ over $K$.
$\deg f_x \ge 2$ and $f_x$ is separable. $f_x$ has a root $x' \neq x$ in $L$, since $L$ is normal. By Theorem 2.3, there exists a $K$-homomorphism $K(x) \to K(x')$ sending $x$ to $x'$.
Since $L$ is normal, this extends to an automorphism $\varphi$ of $L$. But $\varphi(x) \neq x \rightarrow\leftarrow$ So $x \in L^G$.

(2) ⇒ (1): $G \le \mathrm{Aut}(L/K)$, $L^G = K$.
$|\mathrm{Aut}(L/K)| = |\mathrm{Aut}(L/L^G)| = [L : L^G] = [L : K]$ (using Artin's lemma), so $L/K$ is Galois by Theorem 2. ∎

### 3.6.3 The Galois correspondence theorem

$L/K$ Galois extension $\implies$ the set of intermediate fields forms a lattice
Similarly (last semester) the subgroups of $\mathrm{Gal}(L/K)$ forms a lattice

**Definition.** Let $V, V'$ be lattices. An **anti-homomorphism** of lattices is a map $\Phi : V \to V'$ such that

- $\Phi(v_1 \wedge v_2) = \Phi(v_1) \vee \Phi(v_2)$
- $\Phi(v_1 \vee v_2) = \Phi(v_1) \wedge \Phi(v_2)$

for all $v_1, v_2 \in V$. If $\Phi$ is also bijective, it is an **anti-isomorphism**.

**Theorem 6** (Galois Correspondence)**.** *$L/K$ finite Galois extension, $G = \mathrm{Gal}(L/K)$*

1. *$H \le G \implies |H| = [L : L^H]$ and $[G : H] = [L^H : K]$*
2. *$H \mapsto L^H$ defines an anti-isomorphism of lattices $\{\{e\} \le H \le G\} \to \{K \le E \le L\}$ with inverse $E \mapsto \mathrm{Gal}(L/E)$*
3. *$H_1, H_2 \le G$ are conjugate in $G \iff L^{H_1}$ and $L^{H_2}$ are $K$-isomorphic*
4. *$H \trianglelefteq G \iff L^H/K$ is Galois. In this case, $\mathrm{Gal}(L^H/K) \cong G/H$.*

**Note.** In particular, there are only finitely many intermediate fields.

**Example.** $K = \mathbb{Q}$,
$L$ splitting field of $f := X^3 - 2$,
$\zeta \in L$ so that $1 \ne \zeta, \zeta^3 = 1$
$L = \mathbb{Q}(\zeta, \sqrt[3]{2})$
$\mathrm{Gal}(L/K) \cong S_3$
Subgroup lattice:

$$\begin{tikzcd}
 & \{e\} \arrow[dl, no head] \arrow[d, no head] \arrow[dr, no head] \arrow[drr, no head] & & \\
\langle(123)\rangle = A_3 \arrow[dr, no head] & \langle(23)\rangle \arrow[d, no head] & \langle(12)\rangle \arrow[dl, no head] & \langle(13)\rangle \arrow[dll, no head] \\
 & S_3 & &
\end{tikzcd}
\qquad
\begin{tikzcd}
 & L \arrow[dl, no head] \arrow[d, no head] \arrow[dr, no head] \arrow[drr, no head] & & \\
M \arrow[dr, no head] & L_1 \arrow[d, no head] & L_2 \arrow[dl, no head] & L_3 \arrow[dll, no head] \\
 & K = \mathbb{Q} & &
\end{tikzcd}$$

$M = L^{A_3}/\mathbb{Q}$ Galois with group $S_3/A_3 \cong \mathbb{Z}/2\mathbb{Z}$.

Zeros of $f$: $\alpha_1 = \sqrt[3]{2}$, $\alpha_2 = \zeta\sqrt[3]{2}$, $\alpha_3 = \zeta^2\sqrt[3]{2}$

$L_1 = L^{\langle(23)\rangle} = \mathbb{Q}(\alpha_1) = \mathbb{Q}(\sqrt[3]{2})$.
Similarly for $L_2$ and $L_3$. Also $\langle(12)\rangle, \langle(13)\rangle$, and $\langle(23)\rangle$ are conjugate, so $L_1$, $L_2$, $L_3$ are $K$-isomorphic.

$D(f) = -R(f, f') = -108 = -27 \cdot 4 = 6^2 \cdot (-3)$ and also equals $\prod_{i<j}(\alpha_i - \alpha_j)^2 \in L^2$.

So $L$ contains a square root of $-3$. $E := \mathbb{Q}(\sqrt{-3}) \implies E \leq L^{A_3} = M \implies$ for degree reasons, $E = M$.

Can also find $\sqrt{-3}$ explicitly: Let $\alpha = \alpha_1$.

$$-108 = D(f) = ((\alpha - \zeta\alpha)(\alpha - \zeta^2\alpha)(\zeta\alpha - \zeta^2\alpha))^2 = (\alpha - \zeta\alpha)^2 \implies$$

$$\frac{2(1-\zeta)^3}{6} = \frac{(\alpha - \zeta\alpha)^3}{6} = \sqrt{-3} \implies M = \mathbb{Q}(\zeta).$$

*Proof.* (Of Theorem 6)

1. $H \leq G \implies E = L^H$, field by Lemma 3.
   Theorem 5 $\implies L/E$ is Galois with group $H$.
   By Theorem 2, $|H| = |\mathrm{Gal}(L/E)| = [L : E]$ and $|G| = [L : K]$ so the claim follows by Lagrange's theorem

2. $L \geq E \geq K$
   $L$ Galois $\implies L$ is splitting field of separable $f \in K[X]$
   View $f \in E[X]$. Then $L/E$ is Galois.
   Let $H = \mathrm{Gal}(L/E)$. $E = L^H$ by Theorem 5 and $\mathrm{Gal}(L/L^H) = \mathrm{Gal}(L/E) = H$.
   Similarly, $L^{\mathrm{Gal}(L/E)} = L^H = E$.
   Both maps are bijective.
   $H_1 \leq H_2 \leq G \implies L^{H_2} \leq L^{H_1}$ so this map is inclusion-reversing. Conversely, $L_2 \leq L_1 \leq L \implies$ every $\sigma \in \mathrm{Gal}(L/L_1)$ is also in $\mathrm{Gal}(L/L_2)$ so $\mathrm{Gal}(L/L_1) \leq \mathrm{Gal}(L/L_2)$.

   If $H_1, H_2 \leq G$, then $\Phi(H_1 \cap H_2) = L^{H_1 \cap H_2} \geq L^{H_1}L^{H_2}$ since $H_1 \cap H_2 \subseteq H_1$ and $H_1 \cap H_2 \subseteq H_2$. On the other hand, if $E \leq L$ is arbitrary so that $L^{H_1} \leq E$ and $L^{H_2} \leq E$, then $H := \mathrm{Gal}(L/E) \implies H \leq H_1$ and $H \leq H_2$ so $H \leq H_1 \cap H_2$. Thus $L^{H_1 \cap H_2} \leq E = L^H$.
   Hence $L^{H_1 \cap H_2} = L^{H_1}L^{H_2} = \Phi(H_1) \vee \Phi(H_2)$.

   Similar argument for $\Phi(H_1 \vee H_2) = \Phi(H_1) \cap \Phi(H_2)$.

3. See notes

4. By (3), $H \trianglelefteq G \iff E := L^H$ is stable under all $\sigma \in G$.
   ($\Rightarrow$): $H \trianglelefteq G$, $\psi : E = L^H \to \overline{K}$. $L$ normal $\implies \exists$ extension $\tilde{\psi} : L \to \overline{K}$, $\tilde{\psi} \in \mathrm{Gal}(L/K)$ i.e. $\psi \in \mathrm{Aut}(E/K)$ as noted above.
   So $E$ is normal and separable $\implies E/K$ is Galois.
   ($\Leftarrow$): $E/K$ Galois $\implies E/K$ is normal. $\sigma \in G \implies \sigma|_E : E \to L$ is an automorphism of $E \implies H \trianglelefteq G$ (again by the remark)
   We get $\pi : G = \mathrm{Gal}(L/K) \to \mathrm{Gal}(E/K)$ by restriction. $\ker \pi = \mathrm{Gal}(L/E) = H$. So $G/H \cong \mathrm{Gal}(E/K)$.

$\square$

### 3.6.4 Base change of Galois extensions

**Theorem 7.** *$N/K$ finite Galois, $L/K$ arbitrary with $N \leq \overline{L}$. $M := NL \leq \overline{L} \implies M/L$ is Galois with group* $\mathrm{Gal}(M/L) \cong \mathrm{Gal}(N/N \cap L)$.

*Proof.* $N/K$ Galois $\implies \exists f \in K[X]$ separable, splitting field $N$.

$M/L$ is splitting field of $f \in L[X] \implies$ Galois.

$N$ normal $\implies \varphi : \mathrm{Gal}(M/L) \to \mathrm{Gal}(N/N \cap L)$ by restriction.
Claim: $\varphi$ is injective.
If $y_1, \ldots, y_n$ are the zeros of $f$ and $\sigma \in \mathrm{Gal}(M/L)$ with $\sigma|_N = \mathrm{id}_N$, then $\sigma(y_i) = y_i$ for all $i$.
So $\sigma = \mathrm{id}_M$ since $M = L(y_1, \ldots, y_n) \implies \varphi$ is injective.

$N^{\varphi(\mathrm{Gal}(M/L))} = N \cap L \implies$ by Galois correspondence, $\varphi(\mathrm{Gal}(M/L)) = \mathrm{Gal}(N/N \cap L)$. □

Important special case:
$L/K$ finite Galois, $N \cap L = K$

**Corollary 8.** *$[M : L]$ divides $[N : K]$ with $M, L, N, K$ as in Theorem 7.*

### 3.6.5 The Galois group of a polynomial

$N_1, N_2$ splitting fields of separable $f \in K[X] \implies$ (exercise) $\mathrm{Gal}(N_1/K) \cong \mathrm{Gal}(N_2/K)$.

**Definition.** $K$ field, $f \in K[X]$ with splitting field $N$, $y_1, \ldots, y_n$ zeros of $f$ in $N$
$\pi : \mathrm{Gal}(N/K) \to S_n$ by permutation of $y_1, \ldots, y_n$
$\pi(\mathrm{Gal}(N/K)) =: \mathrm{Gal}(f) \leq S_n$ is the **Galois group of** $f$

**Proposition 9.** *$K$ field, $f \in K[X]$ separable with splitting field $N$.*

1. *$\pi : \mathrm{Gal}(N/K) \to S_n$ is faithful (injective)*
2. *$\mathrm{Gal}(f)$ acts transitively on $\{1, \ldots, n\} \iff f$ is irreducible*
3. *If $\mathrm{char} K \neq 2$, $\mathrm{Gal}(f) \leq A_n \iff D(f) \in K^2$*

*Proof.*

1. We have seen this
2. $y_1, \ldots, y_n \in N$. Without loss of generality, $f = \prod_{i=1}^{n}(X - y_i) \in N[X]$ (monic)
   Suppose $\mathrm{Gal}(f) \leq S_n$ is transitive. $f_1$ minimal polynomial of $y_1$ over $K \implies f_1 \mid f$
   $\sigma_i : \mathrm{Gal}(N/K)$ such that $\sigma_i(y_i) = y_i$ for $i = 2, \ldots, n$
   $f_1(y_i) = f_1(\sigma_i(y_1)) = \sigma_i(f_1(y_1)) = 0 \implies f_1 = f \implies f$ is irreducible.

   Conversely, if $f$ is irreducible, for each $i = 1, \ldots, n$ there exists a $K$-homomorphism $\varphi_i : K(y_1) \to K(y_i)$ so that $\varphi(y_1) = y_i$. We can extend $\varphi_i$ to $\sigma_i : N \to \overline{K}$, $\sigma_i \in \mathrm{Gal}(N/K)$ since $N$ is normal $\implies \mathrm{Gal}(N/K)$ acts transitively on $y_1, \ldots, y_n \implies \mathrm{Gal}(f)$ acts transitively on $\{1, \ldots, n\}$

3. $\sqrt{D(f)} \in N$
   $\sqrt{D(f)} \in K \iff \sigma(\sqrt{D(f)}) = \sqrt{D(f)}$ for all $\sigma \in \mathrm{Gal}(f) \iff \mathrm{sign}(\sigma(\sqrt{D(f)}) = \mathrm{sign}(\sqrt{D(f)})$ for all $\sigma \in \mathrm{Gal}(f) \iff \mathrm{sign}\sigma = 1$ for all $\sigma \in \mathrm{Gal}(f) \iff \mathrm{Gal}(f) \leq A_n$.

$\square$

**Example.** $f = X^4 + 1 \in \mathbb{Q}[X]$
$N$ splitting field $\implies \mathrm{Gal}(f) \leq S_4$.
$f = (X-\alpha)(X-i\alpha)(X+\alpha)(X+i\alpha) \in \mathbb{Q}[\alpha][X]$ where $\alpha^2 = i$. So a stem field is a splitting field $\implies |\mathrm{Gal}(f)| = 4$. Then $\mathrm{Gal}(f)$ is either $\mathbb{Z}/4\mathbb{Z}$ or $V_4$. Easy to see that it is $V_4$.

## 3.7 Cyclotomic extensions

### 3.7.1 Roots of unity

**Proposition 1.** *Let $p$ be either a prime or 0. Let $K$ be a field of characteristic $p$. Let $n = p^r m$ where $p \nmid m$. Let $N$ be a splitting field of $X^n - 1 =: f_n \in K[X]$. Then $N/K$ is Galois and the zeros of $f_n$ in $N$ form a subgroup of $(N^\times, \cdot)$ isomorphic to $\mathbb{Z}/m\mathbb{Z}$.*

*Proof.* Let $n = p^r m$ and $p = \mathrm{char}K$
$f_n = (X^m - 1)^{p^r} = f_m^{p^r} \implies N$ is a splitting field of $f_m$ and is separable, so $N/K$ is Galois.

Let $\mu$ = set of zeros of $f_n$. So $|\mu| = m$. If $\zeta_1, \zeta_2 \in \mu$ then $\zeta_1\zeta_2 \in \mu$, $\zeta_1^{-1}1\mu$ so $\mu \leq N^\times$. And so $\mu$ is cyclic as in the proof of Theorem 3.9. $\square$

**Definition.** Let $K$ be a field and $N/K$ a splitting field of $f_n = X^n - 1$. The roots of $f_n$ in $N$ are called the **$n$-th roots of unity**.
$\mu_{n,K}$ := set of $n$-th roots of unity in $N$
$\zeta \in \mu_{n,K}$ is **primitive** if $\langle\zeta\rangle = \mu_{n,K}$.
$\Phi_n := \prod_{\zeta \text{ primitive}}(X - \zeta)$ is the **$n$-th cyclotomic polynomial**

**Proposition 2.** *$K$ field of characteristic $p \geq 0$, $p \nmid n$.*

1. *$X^n - 1 = \prod_{m|n} \Phi_m$.*
2. *$\Phi_n \in \mathbb{Z}[X]$ or $\Phi_n \in \mathbb{F}_p[X]$*
3. *$\deg \Phi_n = \varphi(n)$ Euler $\varphi$-function ($\varphi(n) := |\{a \in \{1, \ldots, n\} \mid \gcd(a,n) = 1\}|$)*

*Proof.* By Proposition 1, $|\mu_{m,K}| = m$ for all $m \mid n$ (since $p \nmid n$)

1. $\zeta \in \mu_{n,K}$, $|\zeta| = m \implies m \mid n$ and $\zeta \in \mu_{m,K}$ is primitive
   Conversely, if $\zeta \in \mu_{m,K}$ is primitive, then $|\zeta| = m$ and $m \mid n$ so
   $$X^n - 1 = \prod_{\zeta \in \mu_{m,K}} (X - \zeta) = \prod_{m|n} \prod_{\zeta \in \mu_{n,K} \text{ and primitive}} (X - \zeta) = \prod_{m|n} \Phi_m$$
2. Will prove by induction on $n$ ($n = 1$ is clear)
   $X^n - 1 = \Phi_n \cdot g$ where $g = \prod_{m|n, m\neq n} \Phi_m$ by (1).
   So by induction $g \in \mathbb{Z}[X]$ or $g \in \mathbb{F}_p[X]$.
   But $g, \Phi_n$ are monic. Gauss' Lemma $\implies \Phi_n \in \mathbb{Z}[X]$ or $\Phi_n \in \mathbb{F}_p[X]$.

3. Clear.

□

### 3.7.2 Cyclotomic fields

$\mu_n := \mu_{n,\mathbb{Q}}$

**Definition.** $\mathbb{Q}(\mu_n)$ is the **$n$-th cyclotomic field**

**Theorem 3.** *1. $\Phi_n \in \mathbb{Q}[X]$ is irreducible*

*2.* $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong ((\mathbb{Z}/n\mathbb{Z})^\times, \cdot)$

*Proof.* 1. Notes

2. Take $\sigma \in \mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$, $\zeta_n$ primitive. Then $\sigma(\zeta_n)$ is primitive, so $\sigma(\zeta_n) = \zeta_n^{v(\sigma)}$ for $v(\sigma) \in (\mathbb{Z}/n\mathbb{Z})^\times$ and $\gcd(v(\sigma), n) = 1$ so $v(\sigma) \in (\mathbb{Z}/n\mathbb{Z})^\times$. This is a homomorphism. Surjective by part 1 (transitive). Injective by being stem field.

□

**Definition.** $N/K$ is a **cyclic extension** (resp. abelian, solvable) if $N/K$ is Galois with group that is cyclic (resp. abelian, solvable).

We just saw: $\mathbb{Q}(\mu_n)/\mathbb{Q}$ is abelian, and if $n$ is prime, cyclic.

**Theorem** (Kronecker-Weber)**.** *$N/\mathbb{Q}$ abelian $\implies N \leq \mathbb{Q}(\zeta_n)$ some $n$.*

### 3.7.3 Wedderburn's Theorem

Recall: A **skew field** is a set $S$ with operations $\cdot, +$ which satisfy all the field axioms except possibly commutativity of $\cdot$.
$C(S) = \{s \in S \mid sx = xs \text{ for all } x \in S\}$ is the **center** of $S$

**Theorem 4.** *A finite skew field is a field.*

*Proof.* Let $S$ be a skew field, $|S| < \infty$. $K := C(S) \implies K$ is a finite field, $\mathrm{char}K = p > 0$. $|K| = p^r = q$ for some $r$. $|S| = q^n$ some $n$. We want: $n = 1$.

Let $x_1, \ldots, x_m$ be representatives of conjugacy classes in $S^\times$.
Let $S_i^\times := C_{S^\times}(x_i)$ be the centralizer of $x_i$ in $S^\times$.
Now

$$q^n - 1 = |S^\times| = \sum_{i=1}^{m} [S^\times : S_i^\times]$$

If $S_i := S_i \cup \{0\}$ is an abelian group under addition, then each $S_i$ is a skew field. Thus $|S_i| = q^{n_i}$ where $n_i = \dim_K(S_i) < n$ (since $x_i \notin K$). Thus $\frac{X^n - 1}{X^{n_i} - 1}$ is a polynomial for all $i$ such that $x_i \notin K$ and $q^n - 1 = (q-1) + \sum_{x_i \notin K} \frac{q^n - 1}{q^{n_i} - 1}$.

By Proposition 2, $\Phi_n \mid X^n - 1$ and is irreducible, so $\Phi_n \mid \frac{X^n-1}{X^{n_i}-1}$ for $i$ such that $x_i \notin K$, meaning
$$\prod_{\gcd(i,n)=1, 1\leq i\leq n} (X - \zeta_n^i) = \Phi_n(q) \text{ divides } q-1.$$
Complex absolute value $|q - \zeta_n^i| \geq ||q| - |\zeta_n^i|| = |q-1| = q-1$ so $n = 1$ (i.e. there is only one factor). $\square$

## 3.8 Computation of Galois Groups

### 3.8.1 Dedekind's reduction criterion

Let $f$ be monic separable over $K$, $N$ a splitting field for $f$, $x_1, \ldots, x_n$ roots of $f$ in $N$.
$\hat{K} := K[t_1, \ldots, t_n]$, $\hat{N} := N[t_1, \ldots, t_n]$.
$\sigma \in S_n$ acts on $\hat{N}$ by permutation of $t_i$, $\sigma \mapsto \sigma * a$.

**Lemma 1.** *Let*
$$s := \prod_{\sigma \in S_n} \left( X - \sigma * \left( \sum_{i=1}^{n} x_i t_i \right) \right)$$

1. $s \in \hat{K}[X]$
2. $h \in \hat{K}[X]$ *irreducible factor of* $s \implies \mathrm{Gal}(f)$ *is in* $S_n$ *conjugate to* $G := \{\tau \in S_n \mid \tau * h = h\}$

**Theorem 2.** $f \in \mathbb{Z}[X]$ *separable monic,* $p$ *prime,* $p \nmid D(f)$

1. $\overline{f} \in \mathbb{F}_p[X]$ *is separable (coefficientwise reduction)*
2. *There exists an injective homomorphism* $\mathrm{Gal}(\overline{f}) \hookrightarrow \mathrm{Gal}(f)$ *(inclusion after renumbering zeros)*

*Proof.*

1. $p \nmid D(f) \implies D(\overline{f}) = D(f) \pmod p \neq 0 \implies \overline{f}$ is separable.
2. $s, \overline{s}$ as in Lemma 1 for $f, \overline{f}$
   From the proof of Lemma 1, $s \in \mathbb{Z}[t_1, \ldots, t_n][X]$, $\overline{s} = s \pmod p$
   Let $h$ be an irreducible factor of $s$ in $\hat{\mathbb{Q}}[X]$. Without loss of generality, $h \in \mathbb{Z}[t_1, \ldots, t_n][X]$, $\overline{h} := h \pmod p$.
   Let $\overline{g}$ be an irreducible factor of $\overline{h}$.
   $\overline{g}$ is also a factor of $\overline{s}$.

   By the Lemma, $\mathrm{Gal}(\overline{f}) = \{\sigma \in S_n \mid \sigma * \overline{g} = \overline{g}\}$ and $\mathrm{Gal}(f) = \{\sigma \in S_n \mid \sigma * h = h\}$.

   Suppose for contradiction that $\sigma \in S_n$ satisfies $\sigma * \overline{g} = \overline{g}$ but $h_1 := \sigma * h \neq h$.
   Then $\sigma * s = s \implies h_1$ is a factor of $s$.
   So $s = hh_1q$ (UFD) $\implies \overline{s} = \overline{hh_1}\overline{q}$.

   Now $\sigma * \overline{h} = \overline{h_1}$, and $\overline{g} \mid \overline{h}$. So $\overline{g} \mid \sigma * \overline{g}$ and $\sigma * \overline{g} \mid \overline{h_1} \implies \overline{g} \mid \sigma * \overline{h}$.
   So $s$ has a multiple factor $\rightarrow\leftarrow$ .

$\square$

**Corollary 3** (Dedekind)**.** *If $f \in \mathbb{Z}[X]$ is monic and separable, $p \nmid D(f)$ and $\overline{f} = \overline{f_1} \cdots \overline{f_r} \in \mathbb{F}_p[X]$, $\deg \overline{f_i} = n_i$ and $\overline{f_i}$ are irreducible, then $\mathrm{Gal}(f)$ contains a permutation of cycle type $(n_1, \ldots, n_r)$.*

**Example.**

1. $f = X^4 + X - 1 \in \mathbb{Z}[X]$
   $D(f) = \pm 229$
   mod 2 $\overline{f}$ is irreducible degree 4, so $\mathrm{Gal}(f)$ contains a 4-cycle
   mod 3 $\overline{f} = (X^3 + X^2 + X + 2)(X - 1)$ and $X^3 + X^2 + X + 2$ is irreducible in $\mathbb{Z}/3\mathbb{Z}[X]$ so $\mathrm{Gal}(f)$ contains a 3-cycle
   mod 23 $\overline{f} = (X^2 + 14X + 2)(X + 4)(X_5)$ and $X^2 + 14X + 2$ is irreducible in $\mathbb{Z}/23\mathbb{Z}[X]$ so $\mathrm{Gal}(f)$ contains a transposition.

   Conclude $\mathrm{Gal}(f) = S_4$.

2. $f = X^4 - 7X^2 + 3X - 1 \in \mathbb{Z}[X]$
   $D(f) = 33489 = 183^2$ so $\mathrm{Gal}(f) \leq A_4$.
   mod 2 $\overline{f} = (X + 1)(X^3 + X^2 + 1)$ so $\mathrm{Gal}(f)$ contains a 3-cycle
   mod 3 $\overline{f} = (X^2 + 1)^2$ so $\mathrm{Gal}(f)$ contains a (2,2)-cycle
   Conclude $\mathrm{Gal}(f) = A_4$.

## 3.9 Extensions solvable by radicals

### 3.9.1 Radical extensions

**Definition.** $L/K$ is a **radical extension** if there is a chain of subfields $K = K_0 \subseteq K_1 \subsetneq \cdots \subsetneq K_r = L$ so that $K_i = K_{i-1}(x_i)$ and $x_i^{n_i} \in K_{i-1}$ for some $n_i \in \mathbb{N} \setminus \{0\}$.
It is a **2-radical extension** if some splitting field of $f$ is contained in a radical extension of $K$.
A polynomial $f$ is **solvable by radicals** if some splitting field of $f$ is contained in a radical extension of $K$.

**Proposition 1.** *$K$ field of characteristic $p \geq 0$, $p \nmid n$, assume $K$ contains a primitive $n$-th root of unity. Then for all $a \in K$, $X^n - a \in K[X]$ has a stem field which is Galois over $K$ with group $\mathbb{Z}/d\mathbb{Z}$ for some $d \mid n$.*

**Proposition 2.** *$K$ field, $p$ is a prime, $p \neq \mathrm{char} K$. Suppose $K$ contains a primitive $p$-th root of unity. Then if $N/K$ is cyclic of degree $p$, $N = K(x)$ for some $x$ such that $x^p \in K$.*

*Proof.* Let $\zeta$ be a primitive $p$-th root of unity in $K$.
$\mathrm{Gal}(N/K) \cong \mathbb{Z}/p\mathbb{Z} = \langle \sigma \rangle$.
By the primitive element theorem, $N = K(y)$ some $y \in N$.
Let $y_i = \sigma^{i-1}(y)$.
Let $z_i = y_1 + \zeta^i y_2 + \cdots + \zeta^{(p-1)i} y_p$.
$\sigma(z_i) = \zeta^{-i} z_i$ for all $i$.
So $\sigma(z_i^p) = z_i^p$ i.e. $z_i^p \in K$ for all $i$.

We claim: one of the $z_i$ generates $N/K$.
$y_1, \ldots, y_p$ are solutions to the linear system

$$\begin{aligned} x_1 + \zeta x_2 + \cdots &= z_1 \\ x_1 + \zeta^2 x_2 + \cdots &= z_2 \\ &\vdots \end{aligned}$$

The determinant of the coefficient matrix is $\prod_{i>j}(\zeta^i - \zeta^j) \neq 0$. So $y_i \in K(z_1, \ldots, z_n)$.
$K(z_i) \neq K$ for some $i \implies K(z_i) = N$. $\square$